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Standard Plate Count: A Quantitative Test

In determining the total numbers of bacteria in water, we are faced with the same problems that are encountered with soil. Water organisms have great variability in physiological needs, and no single medium, pH, or temperature is ideal for all types. Despite the fact that only small numbers of organisms in water will grow on nutrient media, the **standard plate count** can perform an important function in water testing. Probably its most important use is to give us a tool to reveal the effectiveness of various stages in the purification of water. Plate counts made of water before and after storage, for example, can tell us how effective holding is in reducing bacterial numbers.

In this exercise, various samples of water will be evaluated by routine standard plate count procedures. Since different dilution procedures are required for different types of water, two methods are given.

TAP WATER PROCEDURE

If the water is of low bacterial count, such as in the case of tap water, use the following method.

Materials:

1.0 ml pipettes
2 tryptone glucose extract agar pours (TGEA)
2 sterile Petri plates
Quebec colony counter and hand counters
water samples

1. Liquefy two tubes of TGEA and cool to 45° C.
2. After shaking the sample of water 25 times transfer 1 ml of water to each of the two sterile Petri plates.

3. Pour the medium into the dishes, rotate sufficiently to get good mixing of medium and water, and let cool.
4. Incubate at 35° C for 24 hours.
5. Count the colonies of both plates on the Quebec colony counter and record your average count of the two plates on the Laboratory Report.

SURFACE WATER PROCEDURE

If the water is likely to have a high bacterial count, as in the case of surface water, proceed as follows:

Materials:

1 bottle (75 ml) of tryptone glucose extract agar (TGEA)
6 sterile Petri plates
2 water blanks (99 ml)
1.0 ml pipettes

1. Liquefy a bottle of TGEA medium and cool to 45° C.
2. After shaking your water sample 25 times, produce two water blanks with dilutions of 1:100 and 1:1000. See Exercise 23.
3. Distribute aliquots from these blanks to six Petri dishes, which will provide you with two plates each of 1:100, 1:1000, and 1:10,000 dilutions.
4. Pour one-sixth of the TGEA medium into each plate and rotate sufficiently to get even mixing of the water and medium.
5. Incubate at 35° C for 24 hours.
6. Select the pair of plates that has 30 to 300 colonies on each plate and count all the colonies on both plates. Record the average count for the two plates on the second portion of Laboratory Report 64, 65.

PART 12 Microbiology of Milk and Food Products

Milk and food provide excellent growth media for bacteria when suitable temperatures exist. This is in direct contrast to natural waters, which lack the essential nutrients for pathogens. The introduction of a few pathogens into food or milk products becomes a much more serious problem because of the ability of these substances to support tremendous increases in bacterial numbers. Many milk-borne epidemics of human diseases have been spread by contamination of milk by soiled hands of dairy workers, unsanitary utensils, flies, and polluted water supplies. The same thing can be said for improper handling of foods in the home, restaurants, hospitals, and other institutions.

We learned in Part 11 that bacteriological testing of water is primarily qualitative—emphasis being placed on the presence or absence of coliforms as indicators of sewage. Bacteriological testing of milk and food may also be performed in this same manner, using similar media and procedures to detect the presence of coliforms. However, most testing by public health authorities is quantitative. Although the presence of small numbers of bacteria in these substances does not necessarily mean that pathogens are lacking, low counts do reflect better care in handling of food and milk than is true when high counts are present.

Standardized testing procedures for milk products are outlined by the American Public Health Association in *Standard Methods for the Examination of Dairy Products*. The procedures in Exercises 66, 67, and 67 are excerpts from that publication. Copies of the book may be available in the laboratory as well as in the library.

Exercises 69, 70, and 71 pertain to bacterial counts in dried fruit and meats, as well as to spoilage of canned vegetables and meats. Since bacterial counts in foods are performed with some of the techniques you have learned in previous exercises, you will have an opportunity to apply some of those skills here. Exercises 72 and 73 pertain to fermentation methods used in the production of wine and yogurt.

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Standard Plate Count of Milk

The bacterial count in milk is the most reliable indication we have of its sanitary quality. It is for this reason that the American Public Health Association recognizes the standard plate count as the official method in its *Milk Ordinance and Code*. Although human pathogens may not be present in a high count, it may indicate a diseased udder, unsanitary handling of milk, or unfavorable storage temperatures. In general, therefore, a high count means that there is a greater likelihood of disease transmission. On the other hand, it is necessary to avoid the wrong interpretation of low plate counts, since it is possible to have pathogens such as the brucellosis and tuberculosis organisms when counts are within acceptable numbers. Routine examination and testing of animals act as safeguards against the latter situation.

In this exercise, standard plate counts will be made of two samples of milk: a supposedly good sample and one of known poor quality. *Odd-numbered students will work with the high-quality milk and even-numbered students will test the poor-quality sample.* A modification of the procedures in Exercise 23 will be used.

HIGH-QUALITY MILK

Materials:

- milk sample
- 1 sterile water blank (99 ml)
- 4 sterile Petri plates
- 1.1 ml dilution pipettes
- 1 bottle of TGEA (40 ml)
- Quebec colony counter
- mechanical hand counter

1. Following the procedures used in Exercise 23, pour four plates with dilutions of 1:1, 1:10, 1:100, and 1:1000. Before starting the dilution procedures, shake the milk sample 25 times in the customary manner.
2. Incubate the plates at 35° C for 24 hours and count the colonies on the plate that has between 30 and 300 colonies.
3. Record your results on the first portion of Laboratory Report 66, 67.

POOR-QUALITY MILK

Materials:

- milk sample
- 3 sterile water blanks (99 ml)
- 4 sterile Petri plates
- 1.1 ml dilution pipettes
- 1 bottle TGEA (50 ml)
- Quebec colony counter
- mechanical hand counter

1. Following the procedures used in Exercise 23, pour four plates with dilutions of 1:10,000, 1:100,000, 1:1,000,000, and 1:10,000,000. Before starting the dilutions, shake the milk sample 25 times in the customary manner.
2. Incubate the plates at 35° C for 24 hours and count the colonies on the plate that has between 30 and 300 colonies.
3. Record your results on the first portion of Laboratory Report 66, 67.

Direct Microscopic Count of Organisms in Milk: The Breed Count

67

When it is necessary to determine milk quality in a much shorter time than is possible with a standard plate count, one can make a **direct microscopic count** on a slide. This is accomplished by staining a measured amount of milk that has been spread over an area one square centimeter on a slide. The slide is examined under oil and all of the organisms in an entire microscopic field are counted. To increase accuracy, several fields are counted to get average field counts. Before the field counts can be translated into organisms per milliliter, however, it is necessary to calculate the field area.

High-quality milk will have very few organisms per field, necessitating the examination of many fields. A slide made of poor-quality milk, on the other hand, will reveal large numbers of bacteria per field, thus requiring the examination of fewer fields. An experienced technician can determine, usually within 15 minutes, whether or not the milk is of acceptable quality.

In addition to being much faster than the SPC, the direct microscopic count has two other distinct advantages. First of all, it will reveal the presence of bacteria that do not form colonies on an agar plate at 35° C; thermophiles, psychrophiles, and dead bacteria would fall in this category. Secondly, the presence of excessive numbers of leukocytes and pus-forming streptococci on a slide will be evidence that the animal that produced the milk has an udder infection (mastitis).

In view of all these advantages, it is apparent that the direct microscopic count has real value in milk testing. It is widely used for testing raw milk in creamery receiving stations and for diagnosing the types of contamination and growth in pasteurized milk products.

In this exercise, samples of raw whole milk will be examined. Milk that has been separated, blended, homogenized, and pasteurized will lack leukocytes and normal flora.

SLIDE PREPARATION

There are several acceptable ways of spreading the milk onto the slide. Figure 67.1 illustrates a method using a guide card. The Breed slide used in figure 67.2

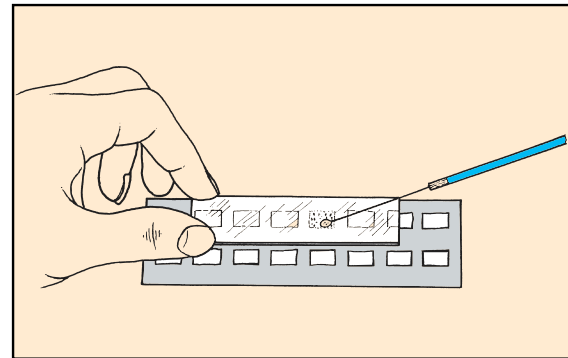


Figure 67.1 Using a guide card to spread milk sample over one square centimeter on a slide

has five one-centimeter areas that are surrounded by ground glass, obviating the need for a card. Proceed as follows:

Materials:

Breed slide or guide card
Breed pipettes (0.01 ml)
methylene blue, xylol, 95% alcohol
beaker of water and electric hot plate
samples of raw milk (poor and high quality)

1. Shake the milk sample 25 times to completely disperse the organisms and break up large clumps of bacteria.
2. Transfer 0.01 ml of milk to one square on the slide. The pipette may be filled by capillary action or by suction, depending on the type of pipette. The instructor will indicate which method to use. *Be sure to wipe off the outside tip of the pipette with tissue before touching the slide* to avoid getting more than 0.01 ml on the slide.
3. Allow the slide to air-dry and then place it over a beaker of boiling water for 5 minutes to steam-fix it.
4. Flood the slide with **xylol** to remove fat globules.
5. Remove the xylol from the slide by flooding the slide with **95% ethyl alcohol**.
6. Gently immerse the slide into a beaker of **distilled water** to remove the alcohol. Do not hold it under running water; the milk film will wash off.

Exercise 67 • Direct Microscopic Count of Organisms in Milk: The Breed Count

7. Stain the smear with **methylene blue** for 15 seconds and dip the slide again in water to remove the excess stain.
8. Decolorize the smear to pale blue with 95% alcohol and dip in distilled water to stop decolorization.
9. Allow the slide to completely air-dry before examination.

CALIBRATION OF MICROSCOPE (Microscope Factor [MF])

Before counting the organisms in each field it is necessary to know what part of a milliliter of milk is represented in that field. The relationship of the field to a milliliter is the **microscope factor** (MF). To calculate the MF, it is necessary to use a stage micrometer to measure the diameter of the oil immersion field. By applying the formula πr^2 to this measurement, the area is easily determined. With the amount of milk (0.01 ml) and the area of the slide (1 cm²), it is a simple matter to calculate the MF.

Materials:

stage micrometer

1. Place a stage micrometer on the microscope stage and bring it into focus under oil. Measure the diameter of the field, keeping in mind that each space is equivalent to 0.01 mm.
2. Calculate the area of the field in square millimeters, using the formula πr^2 ($\pi = 3.14$).
3. Convert the area of the field from square millimeters to square centimeters by dividing by 100.
4. Calculate the number of fields in one square centimeter by dividing one square centimeter by the area of the field in square centimeters.
5. To get the part of a milliliter that is represented in a single field (**microscope factor**), multiply the number of fields by 100. The value should be around 500,000. Therefore, a single field represents 1/500,000 of a ml of milk. Record your computations on the Laboratory Report.

EXAMINATION OF SLIDE

Two methods of counting the bacteria can be used: individual cells may be tallied or only clumps of bacteria may be counted. In both cases, the number per milliliter will be higher than a standard plate count, but a

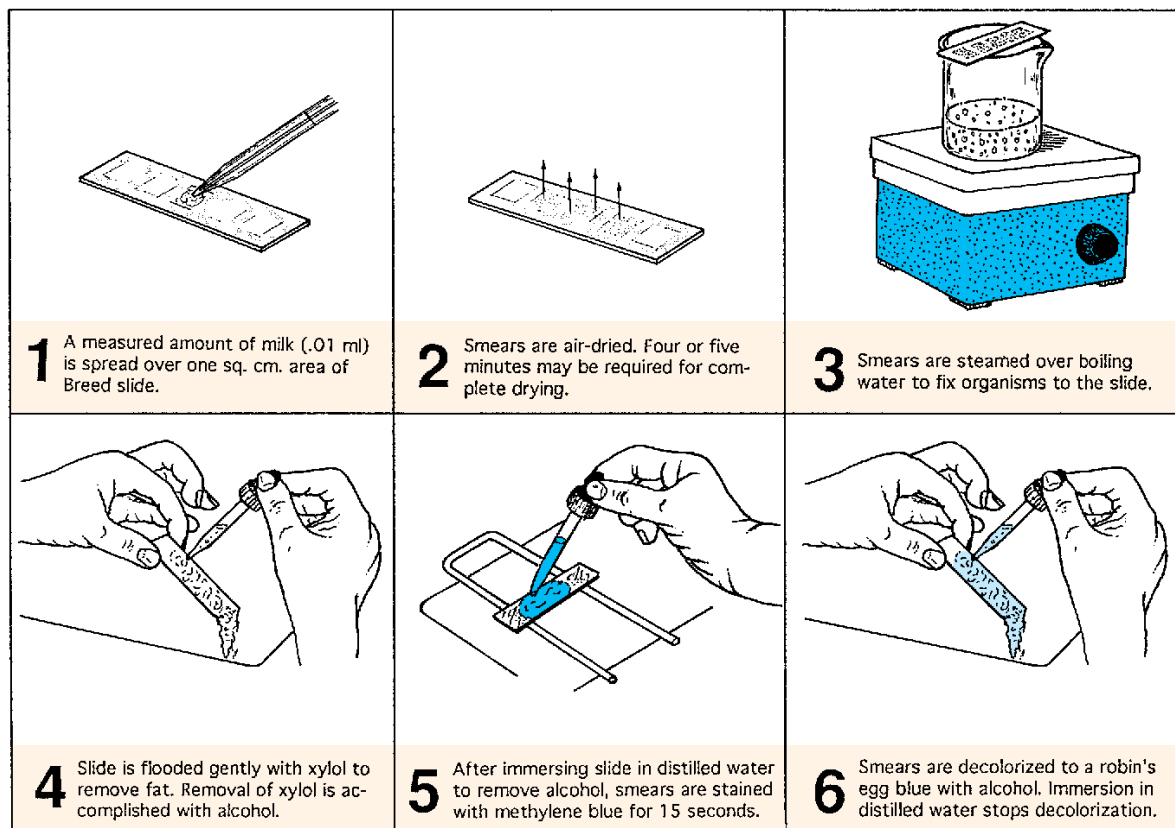


Figure 67.2 Procedure for making a stained slide of a raw milk sample

Direct Microscopic Count of Organisms in Milk: The Breed Count • Exercise 67

clump count will be closer to the SPC. Both methods will be used.

1. After the microscope has been calibrated, replace the stage micrometer with the stained slide. Examine it under oil immersion optics.
2. Count the individual cells in five fields and record your results on the Laboratory Report. A field is the entire area encompassed by the oil immersion lens. As you see leukocytes, record their numbers, also.
3. Count only clumps of bacteria in five fields, recording the numbers of leukocytes as well. Record the totals on the Laboratory Report.
4. Calculate the number of organisms, clumps, and body cells per milliliter using the microscope factor.

LABORATORY REPORT

Complete the last portion of Laboratory Report 66, 67.

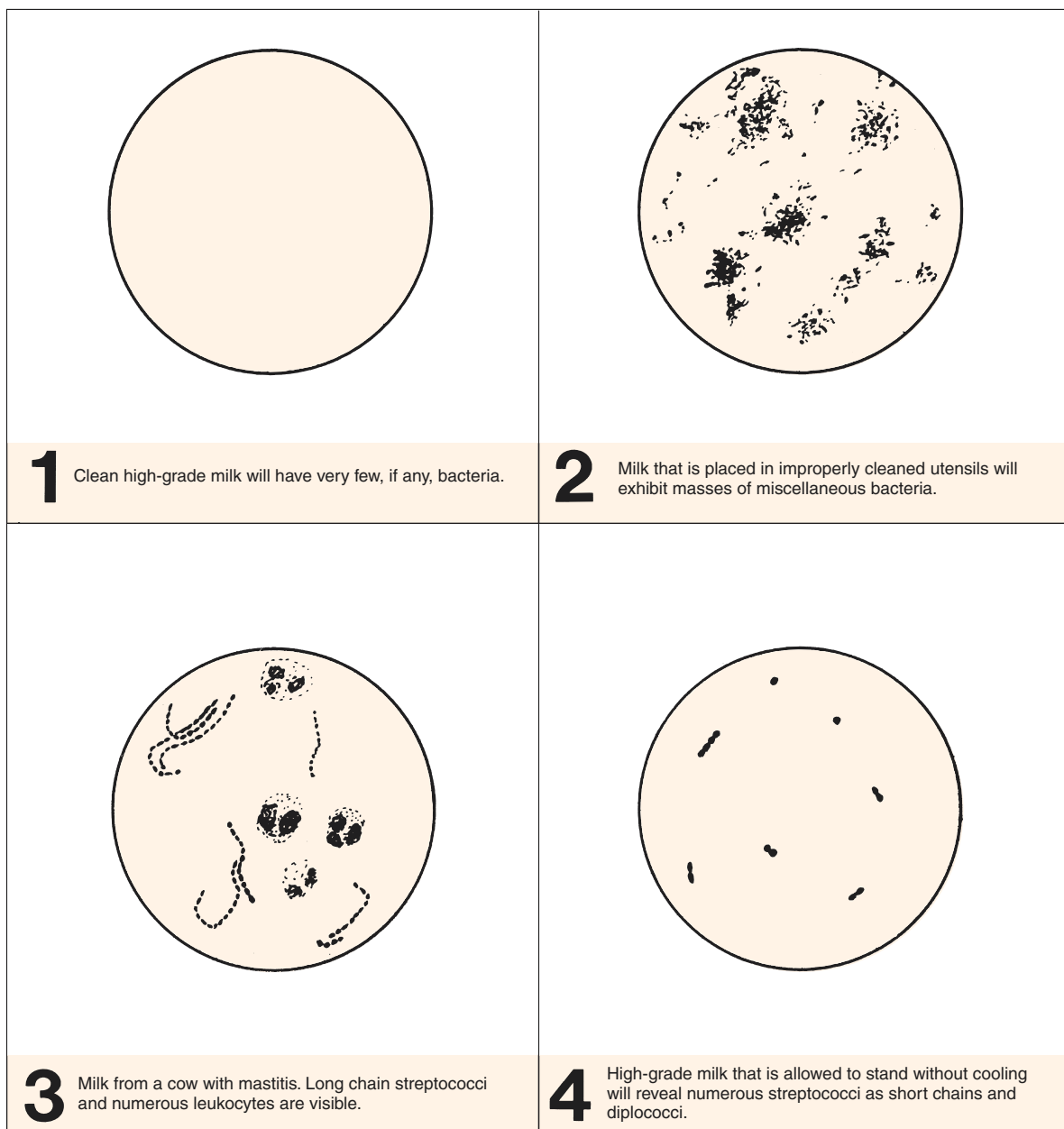


Figure 67.3 Microscopic fields of milk samples

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Reductase Test

Milk that contains large numbers of actively growing bacteria will have a lowered oxidation-reduction potential due to the exhaustion of dissolved oxygen by microorganisms. The fact that methylene blue loses its color (becomes reduced) in such an environment is the basis for the **reductase test**. In this test, 1 ml of methylene blue (1:25,000) is added to 10 ml of milk. The tube is sealed with a rubber stopper and slowly inverted three times to mix. It is placed in a water bath at 35° C and examined at intervals up to 6 hours. The time it takes for the methylene blue to become colorless is the **methylene blue reduction time** (MBRT). The shorter the MBRT, the lower the qual-

ity of milk. An MBRT of 6 hours is very good. Milk with an MBRT of 30 minutes is of very poor quality.

The validity of this test is based on the assumption that all bacteria in milk lower the oxidation-reduction potential at 35° C. Large numbers of psychrophiles, thermophiles, and thermodurics, which do not grow at this temperature, would not produce a positive test. Raw milk, however, will contain primarily *Streptococcus lactis* and *Escherichia coli*, which are strong reducers; thus, this test is suitable for screening raw milk at receiving stations. Its principal value is that less technical training of personnel is required for its performance.

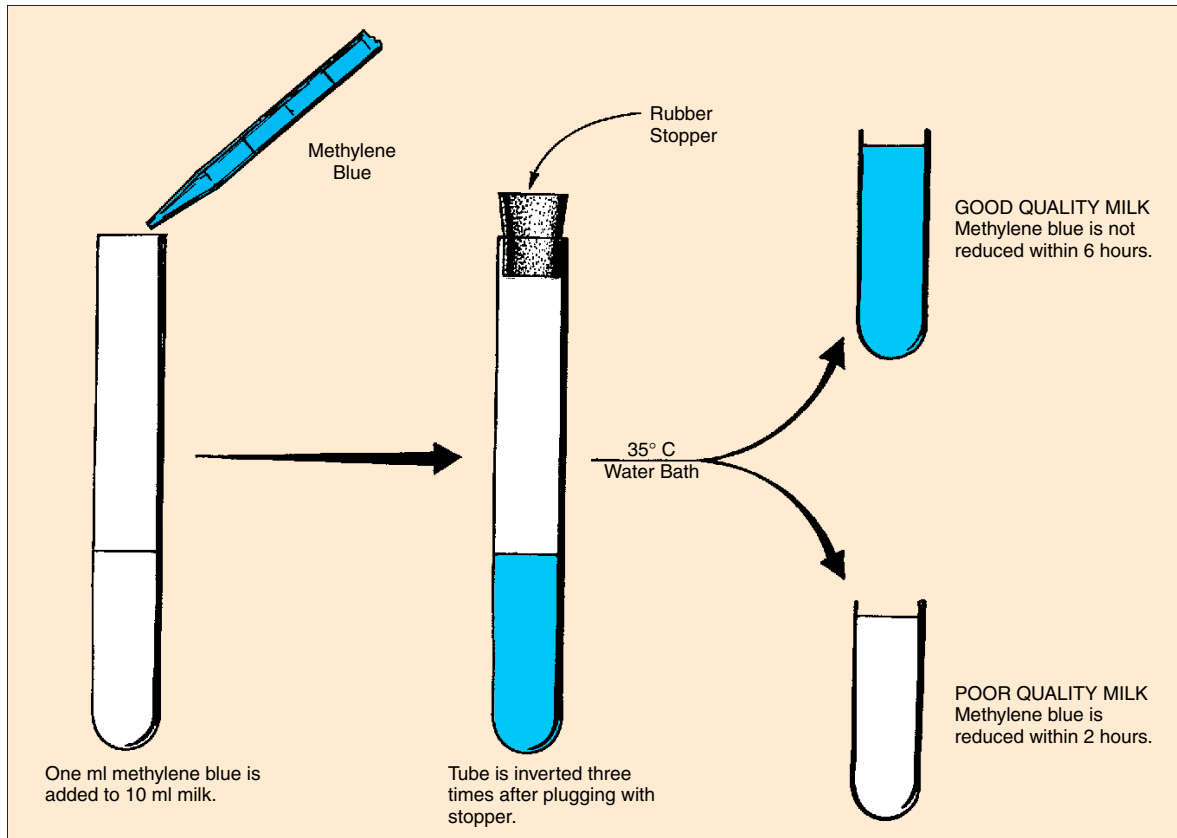


Figure 68.1 Procedure for testing raw milk with reductase test

The Reductase Test • Exercise 68

In this exercise, samples of low- and high-quality raw milk will be tested.

Materials:

2 sterile test tubes with rubber stoppers for each student
raw milk samples of low- and high-quality (samples A and B)
water bath set at 35° C
methylene blue (1:25,000)
10 ml pipettes
1 ml pipettes
gummed labels

1. Attach gummed labels with your name and type of milk to two test tubes. Each student will test a good-quality as well as a poor-quality milk.
2. Using separate 10 ml pipettes for each type of milk, transfer 10 ml to each test tube. To the milk in the tubes add 1 ml of methylene blue with a 1 ml pipette. Insert rubber stoppers and gently in-

vert three times to mix. Record your name and the time on the labels and place the tubes in the water bath, which is set at 35° C.

3. After 5 minutes incubation, remove the tubes from the bath and invert once to mix. This is the last time they should be mixed.
4. Carefully remove the tubes from the water bath 30 minutes later and every half hour until the end of the laboratory period. *When at least four-fifths of the tube has turned white*, the end point of reduction has taken place. Record this time on the Laboratory Report. The classification of milk quality is as follows:

Class 1: Excellent, not decolorized in 8 hours.

Class 2: Good, decolorized in less than 8 hours, but not less than 6 hours.

Class 3: Fair, decolorized in less than 6 hours, but not less than 2 hours.

Class 4: Poor, decolorized in less than 2 hours.

69

Bacterial Counts of Foods

The standard plate count, as well as the multiple tube test, can be used on foods much in the same manner that they are used on milk and water to determine total counts and the presence of coliforms. To get the organisms in suspension, however, a food blender is necessary.

In this exercise, samples of ground meat, dried fruit, and frozen food will be tested for total numbers of bacteria. This will not be a coliform count. The instructor will indicate the specific kinds of foods to be tested and make individual assignments. Figure 69.1 illustrates the general procedure.

Materials:

per student:

- 3 Petri plates
- 1 bottle (45 ml) of Plate Count agar or
Standard Methods agar
- 1 99 ml sterile water blank
- 2 1.1 ml dilution pipettes

per class:

- food blender
- sterile blender jars (one for each type of food)
- sterile weighing paper

180 ml sterile water blanks (one for each type of food)
samples of ground meat, dried fruit, and frozen vegetables, thawed 2 hours

1. Using aseptic techniques, weigh out on sterile weighing paper 20 grams of food to be tested.
2. Add the food and 180 ml of sterile water to a sterile mechanical blender jar. Blend the mixture for 5 minutes. This suspension will provide a 1:10 dilution.
3. With a 1.1 ml dilution pipette dispense from the blender 0.1 ml to plate I and 1.0 ml to the water blank. See figure 69.1.
4. Shake the water blank 25 times in an arc for 7 seconds with your elbow on the table as done in Exercise 23 (Bacterial Population Counts).
5. Using a fresh pipette, dispense 0.1 ml to plate III and 1.0 ml to plate II.
6. Pour agar (50° C) into the three plates and incubate them at 35° C for 24 hours.
7. Count the colonies on the best plate and record the results on the Laboratory Report.

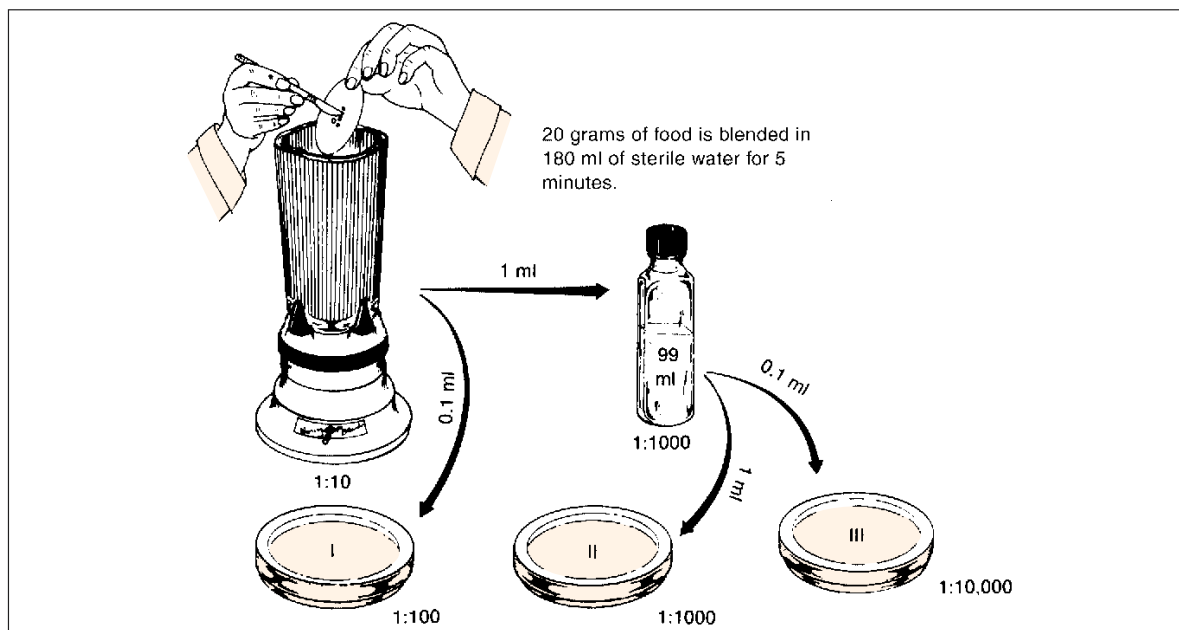


Figure 69.1 Dilution procedure for bacterial counts of food

Microbial Spoilage of Canned Food

70

Spoilage of heat-processed, commercially canned foods is confined almost entirely to the action of bacteria that produce heat-resistant endospores. Canning of foods normally involves heat exposure for long periods of time at temperatures that are adequate to kill spores of most bacteria. Particular concern is given to the processing of low-acid foods in which *Clostridium botulinum* can thrive to produce botulism food poisoning.

Spoilage occurs when the heat processing fails to meet accepted standards. This can occur for several reasons: (1) lack of knowledge on the part of the processor (usually the case in home canning); (2) carelessness in handling the raw materials before canning, resulting in an unacceptably high level of contamination that ordinary heat processing may be inadequate to control; (3) equipment malfunction that results in undetected underprocessing; and (4) defec-

tive containers that permit the entrance of organisms after the heat process.

Our concern here will be with the most common types of food spoilage caused by heat-resistant spore-forming bacteria. There are three types: “flat sour,” “T.A. spoilage,” and “stinker spoilage.”

Flat sour pertains to spoilage in which acids are formed with no gas production; result: sour food in cans that have flat ends. **T.A. spoilage** is caused by thermophilic anaerobes that produce acid and gases (CO_2 and H_2 , but not H_2S) in low-acid foods. Cans swell to various degrees, sometimes bursting. **Stinker spoilage** is due to spore-formers that produce hydrogen sulfide and blackening of the can and contents. Blackening is due to the reaction of H_2S with the iron in the can to form iron sulfide.

In this experiment you will have an opportunity to become familiar with some of the morphological and

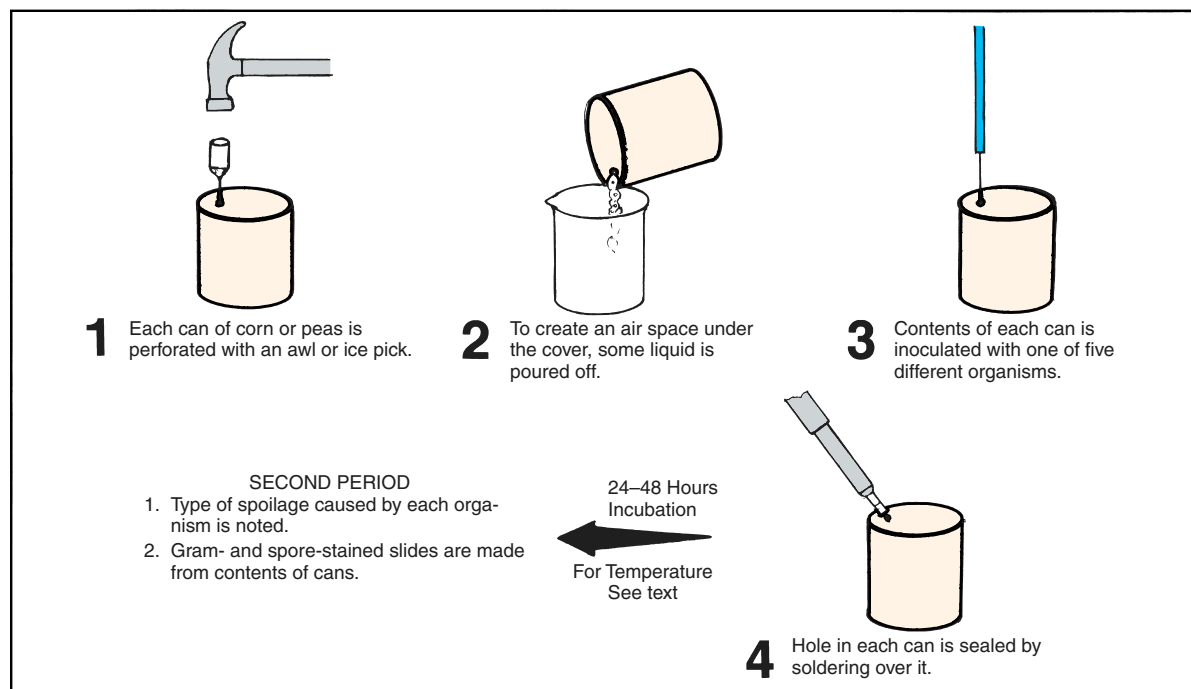


Figure 70.1 Canned food inoculation procedure

Exercise 70 • Microbial Spoilage of Canned Food

physiological characteristics of organisms that cause canned food spoilage, including both aerobic and anaerobic endospore formers of *Bacillus* and *Clostridium*, as well as a non-spore-forming bacterium.

Working as a single group, the entire class will inoculate 10 cans of vegetables (corn and peas) with five different organisms. Figure 70.1 illustrates the procedure. Note that the cans will be sealed with solder after inoculation and incubated at different temperatures. After incubation the cans will be opened so that stained microscope slides can be made to determine Gram reaction and presence of endospores. Your instructor will assign individual students or groups of students to inoculate one or more of the 10 cans. One can of corn and one can of peas will be inoculated with each of the organisms. Proceed as follows:

FIRST PERIOD
(Inoculations)**Materials:**

5 small cans of corn
5 small cans of peas
cultures of *B. stearothermophilus*,
B. coagulans, *C. sporogenes*,
C. thermosaccharolyticum, and *E. coli*
ice picks or awls
hammer
solder and soldering iron
plastic bags
gummed labels and rubber bands

1. Label the can or cans with the name of the organism that has been assigned to you. Use white gummed labels. In addition, place a similar label on one of the plastic bags to be used after sealing of the cans.
2. With an ice pick or awl, punch a small hole through a flat area in the top of each can. This can be done easily with the heel of your hand or a hammer, if available.
3. Pour off a small amount of the liquid from the can to leave an air space under the lid.
4. Use an inoculating needle to inoculate each can of corn or peas with the organism indicated on the label.
5. Take the cans up to the demonstration table where the instructor will seal the hole with solder.

6. After sealing, place each can in two plastic bags. Each bag must be closed separately with rubber bands, and the outer bag must have a label on it.
7. Incubation will be as follows till the next period:
 - **55° C**—*C. thermosaccharolyticum* and *B. stearothermophilus*
 - **37° C**—*C. sporogenes* and *B. coagulans*
 - **30° C** — *E. coli*

Note: If cans begin to swell during incubation, they should be placed in refrigerator.

SECOND PERIOD
(Interpretation)

After incubation, place the cans under a hood to open them. The odors of some of the cans will be very strong due to H₂S production.

Materials:

can opener, punch type
small plastic beakers
Parafilm
gram-staining kit
spore-staining kit

1. Open each can carefully with a punch-type can opener. If the can is swollen, hold an inverted plastic funnel over the can during perforation to minimize the effects of any explosive release of contents.
2. Remove about 10 ml of the liquid through the opening, pouring it into a small plastic beaker. Cover with Parafilm. This fluid will be used for making stained slides.
3. Return the cans of food to the plastic bags, reclose them, and dispose in a proper trash bin.
4. Prepare gram-stained and endospore-stained slides from your canned food extract as well as from the extracts of all the other cans. Examine under brightfield oil immersion.
5. Record your observations on the report sheet on the demonstration table. It will be duplicated and a copy will be made available to each student.

LABORATORY REPORT

Complete the first portion of Laboratory Report 70, 71.

Microbial Spoilage of Refrigerated Meat

71

Contamination of meats by microbes occurs during and after slaughter. Many contaminants come from the animal itself, others from utensils and equipment. The conditions for rapid microbial growth in freshly cut meats are very favorable, and spoilage can be expected to occur rather quickly unless steps are taken to prevent it. Although immediate refrigeration is essential after slaughter, it will not prevent spoilage indefinitely, or even for a long period of time under certain conditions. In time, cold-tolerant microbes will destroy the meat, even at low refrigerator temperatures.

Microorganisms that grow at temperatures between 5° and 0° C are classified as being either psychrophilic or psychrotrophic. The difference between the two groups is that **psychrophiles** seldom grow at temperatures above 22° C and **psychrotrophs** (psychrotolerants or low-temperature mesophiles) grow well above 25° C. While the optimum growth temperature range for psychrophiles is 15°–18° C, psychrotrophs have an optimum growth temperature range of 25°–30° C. It is the psychrotrophic microorganisms that cause most meat spoilage during refrigeration.

The majority of psychrophiles are gram-negative and include species of *Aeromonas*, *Alcaligenes*, *Cytophagia*, *Flavobacterium*, *Pseudomonas*, *Serratia*, and *Vibrio*. Gram-positive psychrophiles include species of *Arthrobacter*, *Bacillus*, *Clostridium*, and *Micrococcus*.

Psychrotrophs include a much broader spectrum of gram-positive and gram-negative rods, cocci, vibrios, spore-formers, and non-spore-formers. Typical genera are *Acinetobacter*, *Chromobacterium*, *Citrobacter*, *Corynebacterium*, *Enterobacter*, *Escherichia*, *Klebsiella*, *Lactobacillus*, *Moraxella*, *Staphylococcus*, and *Streptococcus*.

The widespread use of vacuum or modified atmospheric packaging of raw and processed meat has resulted in food spoilage due to facultative and obligate anaerobes, such as *Lactobacillus*, *Leuconostoc*, *Pediococcus*, and certain Enterobacteriaceae.

Although most of the previously mentioned psychrotrophic representatives are nonpathogens, there are significant pathogenic psychrotrophs such as

Aeromonas hydrophila, *Clostridium botulinum*, *Listeria monocytogenes*, *Vibrio cholera*, *Yersinia enterocolitica*, and some strains of *E. coli*.

In addition to bacterial spoilage of meat there are many yeasts and molds that are psychrophilic and psychrotrophic. Examples of psychrophilic yeasts are *Cryptococcus*, *Leucosporidium*, and *Torulopsis*. Psychrotrophic fungi include *Candida*, *Cryptococcus*, *Saccharomyces*, *Alternaria*, *Aspergillus*, *Cladosporium*, *Fusarium*, *Mucor*, *Penicillium*, and many more.

Our concern in this experiment will be to test one or more meat samples for the prevalence of psychrophilic-psychrotrophic organisms. To accomplish this, we will liquefy and dilute out a sample of ground meat so that it can be plated out and then incubated in a refrigerator for 2 weeks. After incubation, colony counts will be made to determine the number of organisms of this type that exist in a gram of the sample.

Figure 71.1 illustrates the overall procedure. Work in pairs to perform the experiment.

FIRST PERIOD

Materials:

at demonstration table:

ground meat and balance
sterile foil-wrapped scoopula
1 blank of phosphate buffered water (90 ml)
blender with sterile blender jar
sterile Petri dish or sterile filter paper

per pair of students:

4 large test tubes of sterile phosphate buffered water (9 ml each)
4 TSA plates
9 sterile 1 ml pipettes
L-shaped glass spreading rod
beaker of 95% ethyl alcohol

At Demonstration Table

1. With a sterile scoopula, weigh 10g of ground meat into a sterile Petri plate or onto a sterile piece of filter paper.

Exercise 71 • Microbial Spoilage of Refrigerated Meat

2. Pour 90 ml of sterile buffered water from water blank into a sterile blender jar and add the meat.
3. Blend the meat and water at moderate speed for **1 minute**.

Student Pair

1. Label the four water blanks 1 through 4.
2. Label the four Petri plates with their dilutions, as indicated in figure 71.1. Add your initials and date also.
3. Once blender suspension is ready, pipette 1 ml from jar to tube 1.
4. Using a fresh 1 ml pipette, mix the contents in tube 1 and transfer 1 ml to tube 2.
5. Repeat step 4 for tubes 3 and 4, *using fresh pipettes* for each tube.
6. Dispense 0.1 ml from each tube to their respective plates of TSA. Note that by using only 0.1 ml per plate you are increasing the dilution factor by 10 times in each plate.
7. Using a sterile L-shaped glass rod, spread the organisms on the agar surfaces. Sterilize the rod

each time by dipping in alcohol and flaming gently. Be sure to let rod cool completely each time.

8. Incubate the plates for 2 weeks in the back of the refrigerator (away from door-opening) where the temperature will remain between 0° and 5° C.

SECOND PERIOD

Materials:

Quebec colony counters
hand tally counters
gram-staining kit

1. After incubation, count the colonies on all the plates and calculate the number of psychrophiles and psychrotrophs per gram of meat.
2. Select a colony from one of the plates and prepare a gram-stained slide. Examine under oil immersion and record your observations on the Laboratory Report.

LABORATORY REPORT

Complete the last portion of Laboratory Report 70, 71.

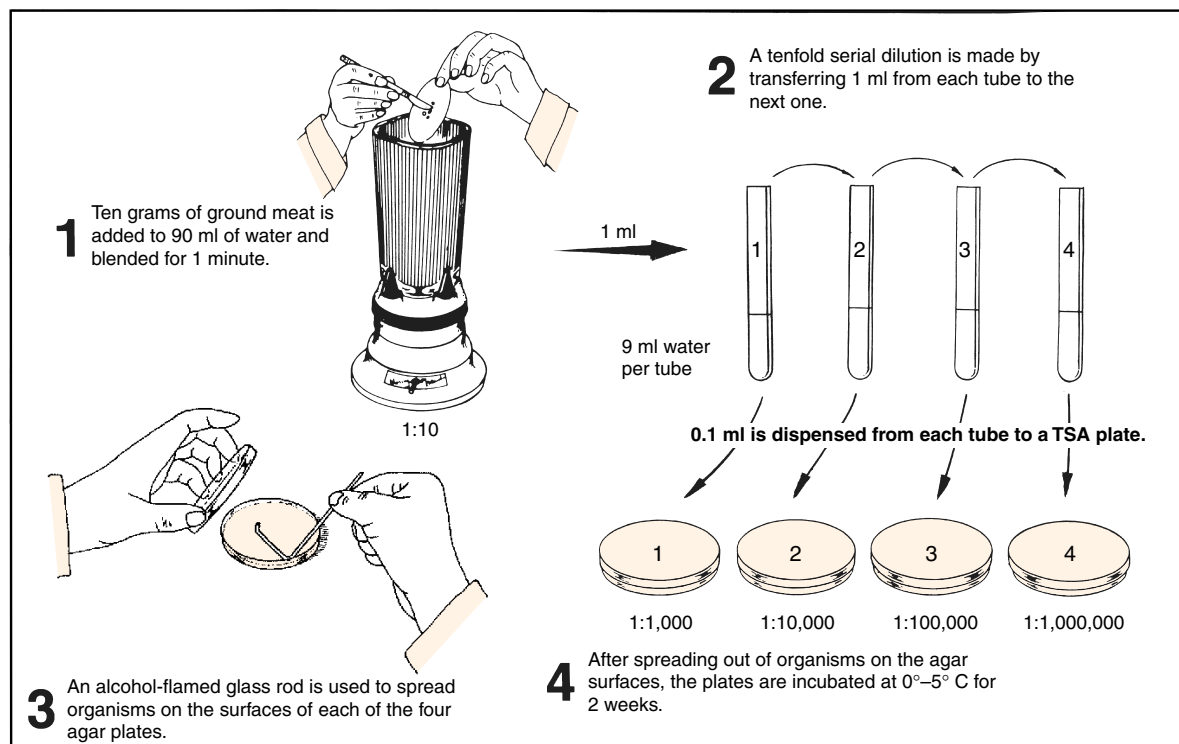


Figure 71.1 Dilution and inoculation procedure

Microbiology of Alcohol Fermentation

72

Fermented food and beverages are as old as civilization. Historical evidence indicates that beer and wine making were well established as long ago as 2000 B.C. An Assyrian tablet states that Noah took beer aboard the ark.

Beer, wine, vinegar, buttermilk, cottage cheese, sauerkraut, pickles, and yogurt are some of the more commonly known products of fermentation. Most of these foods and beverages are produced by different strains of yeasts (*Saccharomyces*) or bacteria (*Lactobacillus*, *Acetobacter*, etc.).

Fermentation is actually a means of food preservation because the acids formed and the reduced environment (anaerobiasis) hold back the growth of many spoilage microbes.

Wine is essentially fermented fruit juice in which alcoholic fermentation is carried out by *Saccharomyces cerevisiae* var. *ellipsoideus*. Although we usually associate wine with fermented grape juice, it may also be made from various berries, dandelions, rhubarb, etc. Three conditions are necessary: simple sugar, yeast, and anaerobic conditions. The reaction is as follows:



Commercially, wine is produced in two forms: red and white. To produce red wines, the distillers use red grapes with the skins left on during the initial stage of the fermentation process. For white wines either red or white grapes can be used, but the skins are discarded. White and red wines are fermented at 13° C (55° F) and 24° C (75° F), respectively.

In this exercise we will set up a grape juice fermentation experiment to learn about some of the characteristics of sugar fermentation to alcohol. Note in figure 72.1 that a balloon will be attached over the mouth of the fermentation flask to exclude oxygen uptake and to trap gases that might be produced. To detect the presence of hydrogen sulfide production we will tape a lead acetate test strip inside the neck of the flask. The pH of the substrate will also be monitored before and after the reaction to note any changes that occur.

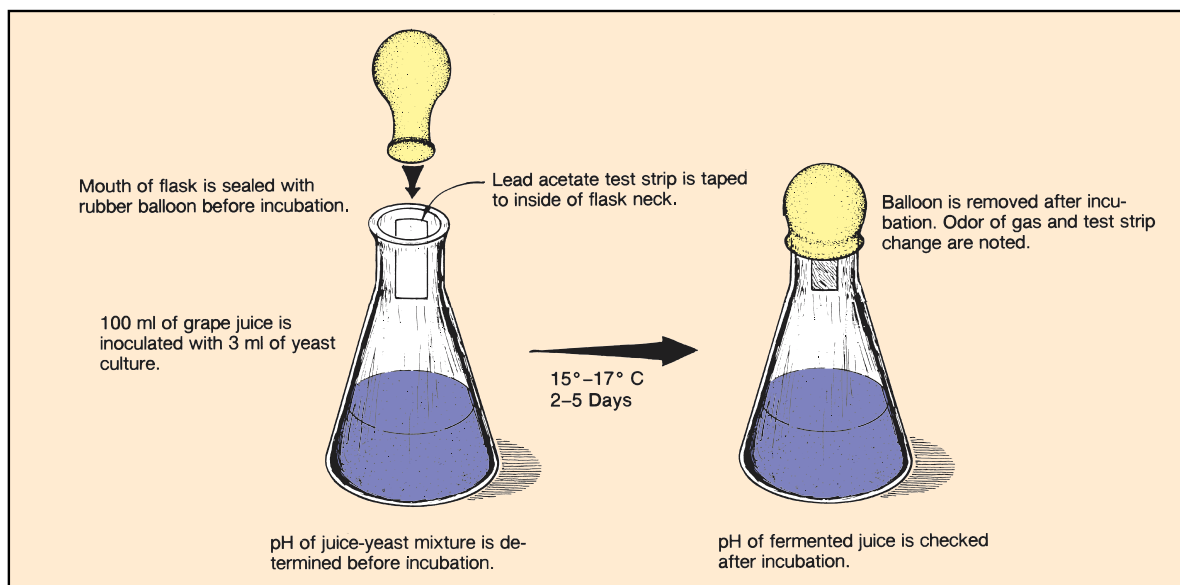


Figure 72.1 Alcohol fermentation setup

Exercise 72 • Microbiology of Alcohol Fermentation

FIRST PERIOD

Materials:

100 ml grape juice (no preservative)
bottle of juice culture of wine yeast
125 ml Erlenmeyer flask
1 10 ml pipette
balloon
hydrogen sulfide (lead acetate) test paper
tape
pH meter

1. Label an Erlenmeyer flask with your initials and date.
2. Add about 100 ml of grape juice to the flask (fermenter).
3. Determine the pH of the juice with a pH meter and record the pH on the Laboratory Report.
4. Agitate the container of yeast juice culture to suspend the culture, remove 5 ml with a pipette, and add it to the flask.
5. Attach a short strip of tape to a piece of lead-acetate test paper (3 cm long), and attach it to the inside surface of the neck of the flask. Make certain that neither the tape nor the test strip protrudes from the flask.

6. Cover the flask opening with a balloon.
7. Incubate at 15°–17° C for 2–5 days.

SECOND PERIOD

Materials:

pH meter

1. Remove the balloon and note the aroma of the flask contents. Describe the odor on the Laboratory Report.
2. Determine the pH and record it on the Laboratory Report.
3. Record any change in color of the lead-acetate-test strip on the Laboratory Report. If any H₂S is produced, the paper will darken due to the formation of lead sulfide as hydrogen sulfide reacts with the lead acetate.
4. Wash out the flask and return it to the drain rack.

LABORATORY REPORT

Complete the first portion of Laboratory Report 72, 73 by answering all the questions.

Microbiology of Yogurt Production

73

For centuries, people throughout the world have been producing fermented milk products using yeasts and lactic acid-producing bacteria. The **yo-gurt** of eastern central Europe, the **kefir** of the Cossacks, the **koumiss** of central Asia, and the **leben** of Egypt are just a few examples. In all of these fermented milks, lactobacilli act together with some other microorganisms to curdle and thicken milk, producing a distinctive flavor desired by the producer. Kefir of the Cossacks is made by charging milk with small cauliflower-like grains that contain *Streptococcus lactis*, *Saccharomyces delbrueckii*, and *Lactobacillus brevis*. As the grains swell in the milk they release the growing microorganisms to ferment the milk. The usual method for producing yogurt in large-scale production is to add pure cultures of *Streptococcus thermophilus* and *Lactobacillus bulgaricus* to pasteurized milk.

In this exercise you will produce a batch of yogurt from milk by using an inoculum from commercial yogurt. Gram-stained slides will be made from the finished product to determine the types of organisms that control the reaction. If proper safety measures are followed, the sample can be tasted.

Two slightly different ways of performing this experiment are provided here. Your instructor will indicate which method will be followed.

METHOD A (First Period)

Figure 73.1 illustrates the procedure for this method. Note that 4 g of powdered milk are added to 100 ml of whole milk. This mixture is then heated to boiling and cooled to 45°C. After cooling, the milk is inoculated with yogurt and incubated at 45° C for 24 hours. Proceed:

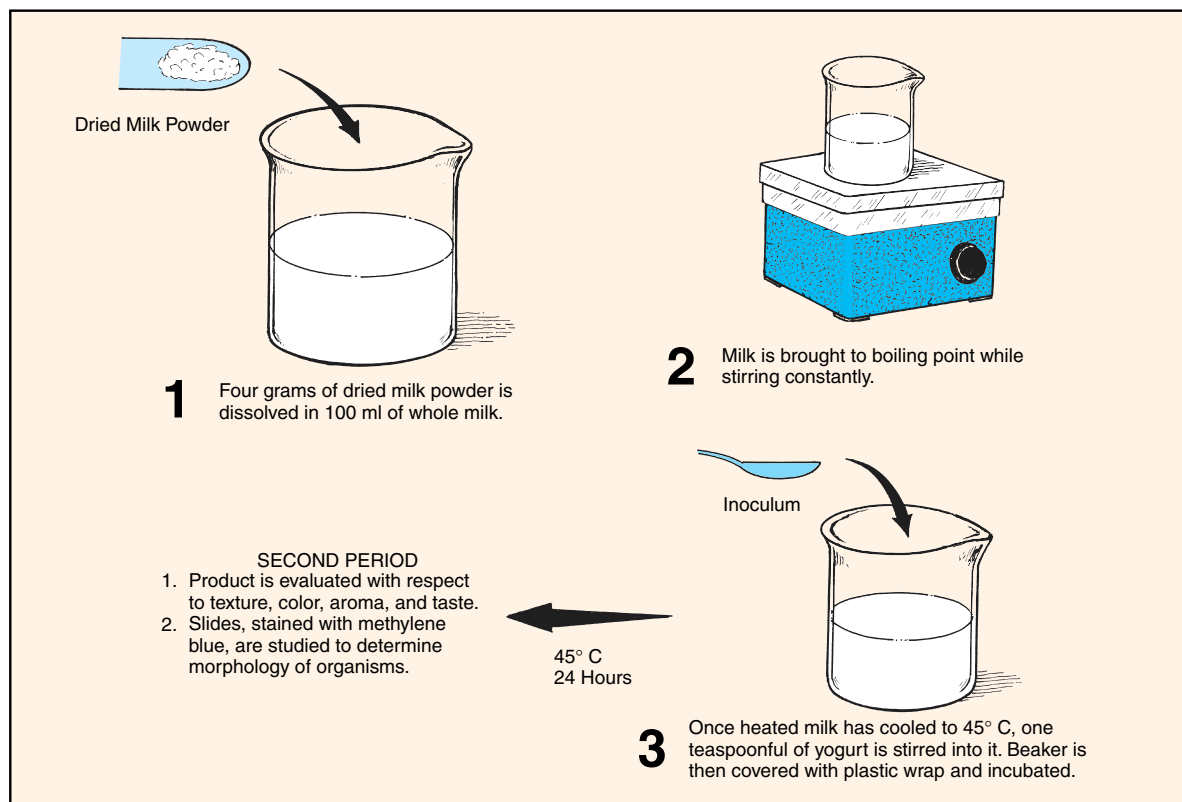


Figure 73.1 Yogurt production by Method A

Exercise 73 • Microbiology of Yogurt Production

Materials:

dried powdered milk
whole milk
commercial yogurt (with viable organisms)
small beaker, graduate, teaspoon, stirring rod
plastic wrap
filter paper (for weighing)

1. On a piece of filter paper weigh 4 grams of dried powdered milk.
2. To a beaker of 100 ml of whole milk add the powdered milk and stir thoroughly with sterile glass rod to dissolve.
3. Heat to boiling, while stirring constantly.
4. Cool to 45° C and inoculate with 1 teaspoon of the commercial yogurt. Stir. Be sure to check the label to make certain that product contains a live culture. Cover with plastic wrap.
5. Incubate at 45° C for 24 hours.

METHOD B (First Period)

Figure 73.2 illustrates a slightly different method of culturing yogurt, which, due to its simplicity, may be preferred. Note that no whole milk is used and provisions are made for producing a sample for tasting.

Materials:

small beaker, graduate, teaspoon, stirring rod
dried powdered milk
commercial yogurt (with viable organisms)
plastic wrap
filter paper for weighing

paper Dixie cup (5 oz size) and cover
electric hot plate or Bunsen burner and tripod

1. On a piece of filter paper weigh 25 grams of dried powdered milk.
2. Heat 100 ml of water in a beaker to boiling and cool to 45° C.
3. Add the 25 grams of powdered milk and 1 teaspoon of yogurt to the beaker of water. Mix the ingredients with a sterile glass rod.
4. Pour some of the mixture into a sterile Dixie cup and cover loosely. Cover the remainder in the beaker with plastic wrap.
5. Incubate at 45° C for 24 hours.

SECOND PERIOD (Both Methods)

1. Examine the product and record on the Laboratory Report the color, aroma, texture, and, if desired, the taste.

CAUTION

Refrain from working with other bacteria or doing other exercises while tasting the yogurt.

2. Make slide preparations of the yogurt culture. Fix and stain with methylene blue. Examine under oil immersion and record your results on Laboratory Report 72, 73.

LABORATORY REPORT

Complete the last portion of Laboratory Report 72, 73 by answering all the questions.

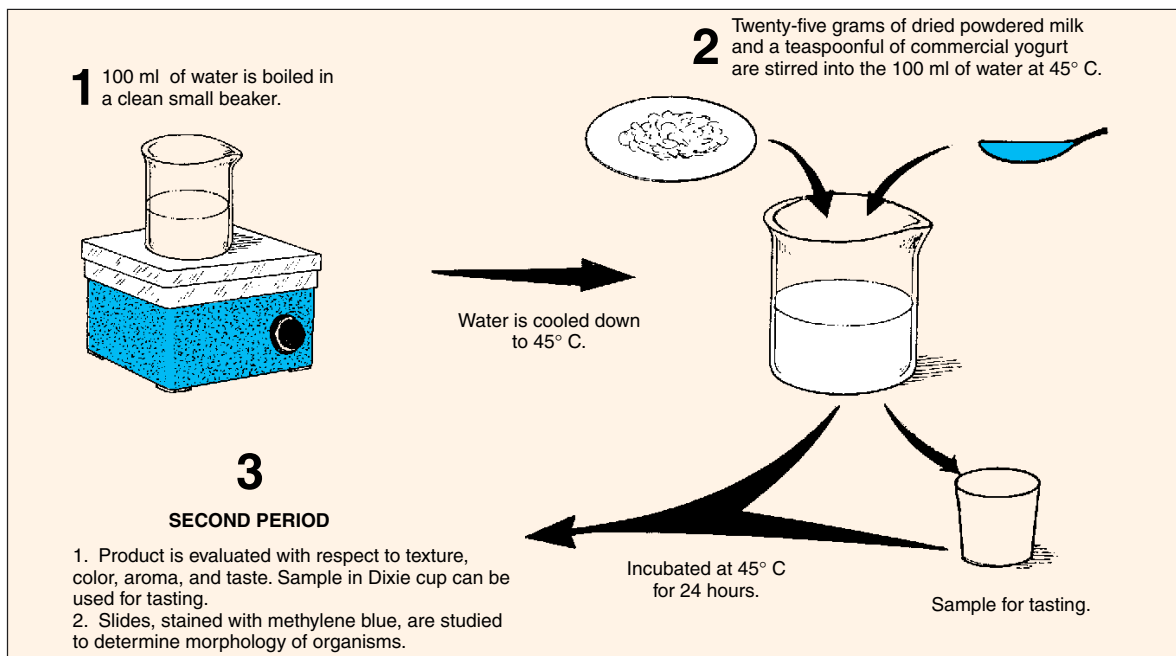


Figure 73.2 Yogurt production by Method B

PART 13 Bacterial Genetic Variations

Variations in bacteria that are due to environmental factors and that do not involve restructuring DNA are designated as **temporary variations**. Such variations may be morphological or physiological and disappear as soon as the environmental changes that brought them about disappear. For example, as a culture of *E. coli* becomes old and the nutrients within the tube become depleted, the new cells that form become so short that they appear coccoidal. Reinoculation of the organism into fresh media, however, results in the reappearance of distinct bacilli of characteristic length.

Variations in bacteria that involve alteration of the DNA macromolecule are designated as **permanent variations**. It is because they survive a large number of transfers that they are so named. Such variations are due to mutations. Variations of this type occur spontaneously. They also might be induced by physical and chemical methods. Some permanent variations also are caused by the transfer of DNA from one organism to another, either directly by conjugation or indirectly by phage. It is these permanent genetic variations that the three exercises of this unit represent.

Exercises 74 and 75 of this unit demonstrate how spontaneous mutations are constantly occurring in bacterial populations. The genetic change that occurs in these two exercises pertains to the development of bacterial resistance to streptomycin. In Exercise 76 we will study how chemically induced mutagenicity that causes back mutations is used in the Ames test to determine possible carcinogenicity of chemical compounds.

74

Mutant Isolation by Gradient Plate Method

An excellent way to determine the ability of organisms to produce mutants that are resistant to antibiotics is to grow them on a **gradient plate** of a particular antibiotic. Such a plate consists of two different wedgelike layers of media: a bottom layer of plain nutrient agar and a top layer of nutrient agar with the antibiotic. Since the antibiotic is only in the top layer, it tends to diffuse into the lower layer, producing a gradient of antibiotic concentration from low to high.

In this exercise we will make a gradient plate using streptomycin in the medium. *E. coli*, which is normally sensitive to this antibiotic, will be spread over the surface of the plate and incubated for 4 to 7 days. Any colonies that develop in the high concentration area will be streptomycin-resistant mutants.

PLATE PREPARATION

The gradient plate used in this experiment will have a high concentration of 100 mcg of streptomycin per milliliter of medium. This concentration is 10 times the strength used in sensitivity disks in the Kirby-Bauer test method. Prepare a gradient plate as follows:

Materials:

- 1 sterile Petri plate
- 2 nutrient agar pours (10 ml per tube)
- 1 tube of streptomycin solution (1%)
- 1 wood spacer ($\frac{1}{8}$ " $\times$ $\frac{1}{2}$ " $\times$ 2")
- 1 ml pipette

1. Liquefy two pours of nutrient agar and cool to 50° C.
2. With wood spacer under one edge of Petri plate (see figure 74.1), pour contents of one agar pour into plate. Let stand until solidification has occurred.
3. Remove the wood spacer from under the plate.
4. Pipette 0.1 ml of streptomycin into second agar pour, mix tube between palms, and pour contents over medium of plate that is now resting level on the table.
5. Label the low and high concentration areas on the bottom of the plate.

INOCULATION

The inoculation procedure is illustrated in figure 74.2. The technique involves spreading a measured amount

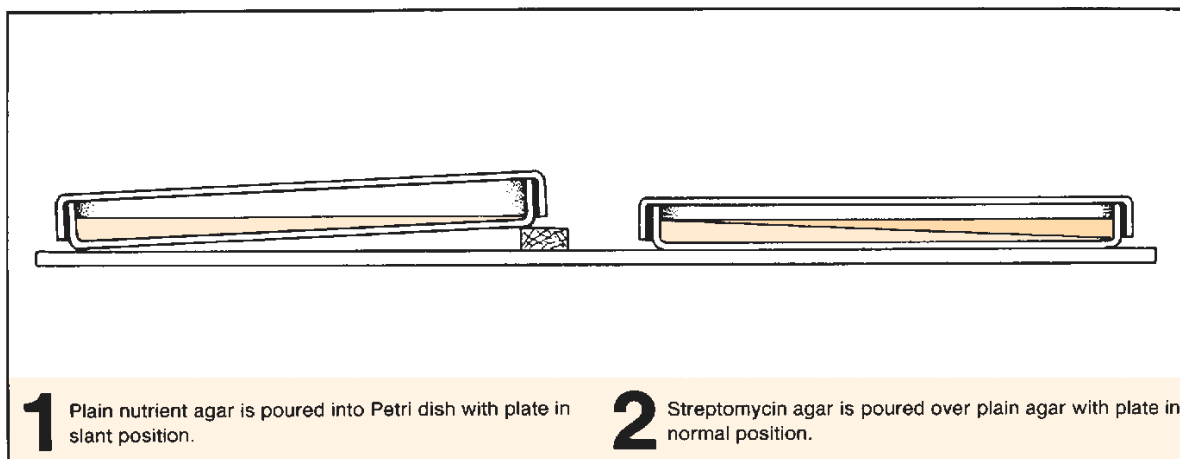


Figure 74.1 Procedure for pouring a gradient plate

Mutant Isolation by Gradient Plate Method • Exercise 74

of the culture on the surface of the medium with a glass bent rod to provide optimum distribution.

Materials:

- 1 beaker of 95% ethanol
- 1 glass rod spreader
- nutrient broth culture of *E. coli*
- 1 ml pipette

1. Pipette 0.1 ml of *E. coli* suspension onto surface of medium in Petri plate.
2. Sterilize glass spreading rod by dipping it in alcohol first and then passing it quickly through the flame of a Bunsen burner. Cool the rod by placing against sterile medium in plate before contacting organisms.
3. Spread the culture evenly over the surface with the glass rod.
4. Invert and incubate the plate at 37° C for 4 to 7 days in a closed cannister or plastic bag. Unless incubated in this manner, excessive dehydration might occur.

FIRST EVALUATION

After 4 to 7 days, look for colonies of *E. coli* in the area of high streptomycin concentration. Count the colonies that appear to be resistant mutants and record your count on the Laboratory Report.

Select a well-isolated colony in the high concentration area and, with a sterile loop, smear the colony over the surface of the medium toward the higher concentration portion of the plate. Do this with two or three colonies. Return the plate to the incubator for another 2 or 3 days.

FINAL EVALUATION

Examine the plate again to note what effect the spreading of the colonies had on their growth. Record your observations on Laboratory Report 74, 75.

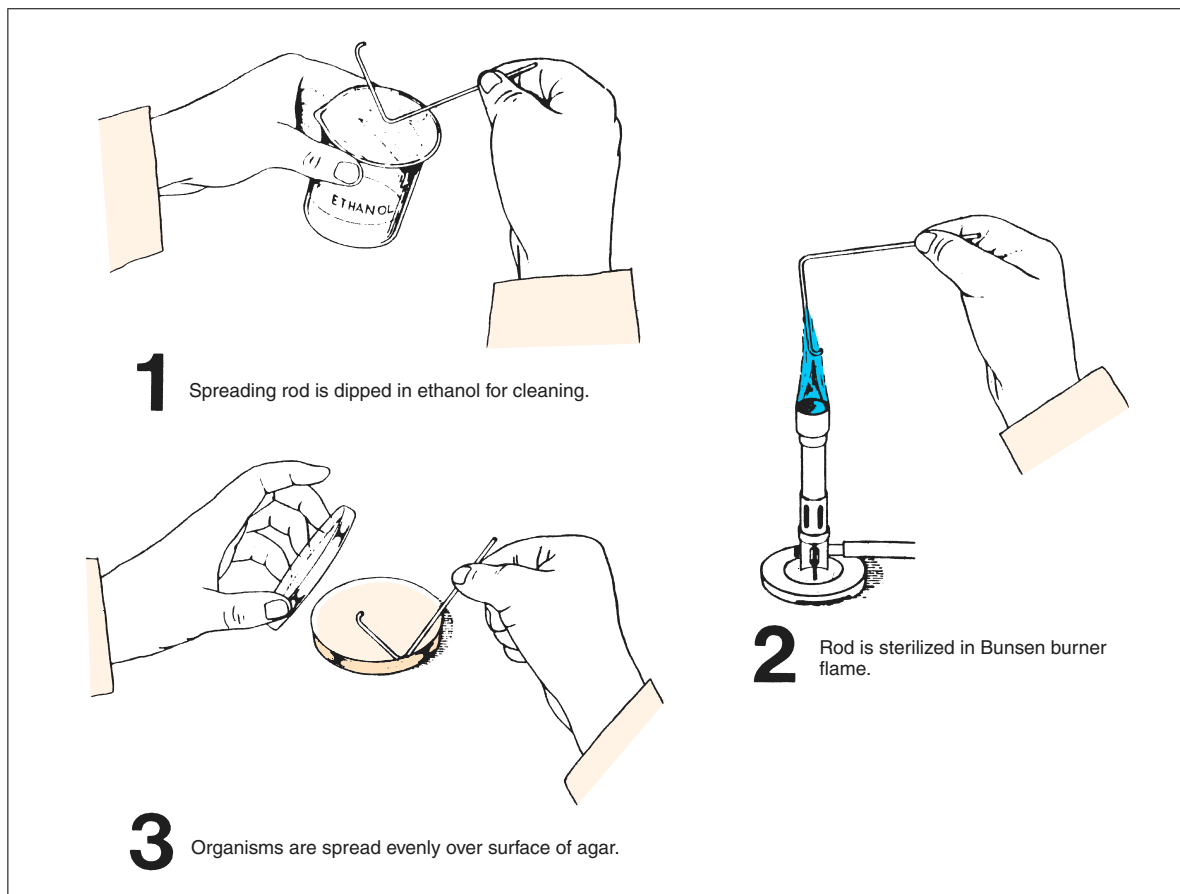


Figure 74.2 Procedure for spreading organisms on gradient plate

75

Mutant Isolation by Replica Plating

In the last exercise it was observed that *E. coli* could develop mutant strains that are streptomycin-resistant. If we had performed this experiment with other organisms and with other antibiotics, the results would have been quite similar. The question that logically develops in one's mind from this experiment is: What mechanism is involved here? Is a mutation of this sort induced by the antibiotic? Or does the mutation occur spontaneously and independently of the presence of the drug? If we could demonstrate the presence of a streptomycin-resistant mutant occurring on a medium that lacks streptomycin, then we could assume that the mutation occurs spontaneously.

To determine whether or not such a colony exists on a plain agar plate having 500 to 1,000 colonies could be a laborious task. One would have to transfer

organisms from each colony to a medium containing streptomycin. This is somewhat self-defeating, too, in light of the low incidence of mutations that occur. Many thousands of the transfers might have to be made to find the first mutant. Fortunately, we can resort to replica plating to make all the transfers in one step. Figure 75.1 illustrates the procedure. In this technique a velveteen-covered colony transfer device is used to make the transfers.

Note in figure 75.1 that organisms are first dispersed on nutrient agar with a glass spreading rod. After incubation, all colonies are transferred from the nutrient agar plate to two other plates: first to a nutrient agar plate and second to a streptomycin agar plate. After incubation, streptomycin-resistant strains are looked for on the streptomycin agar.

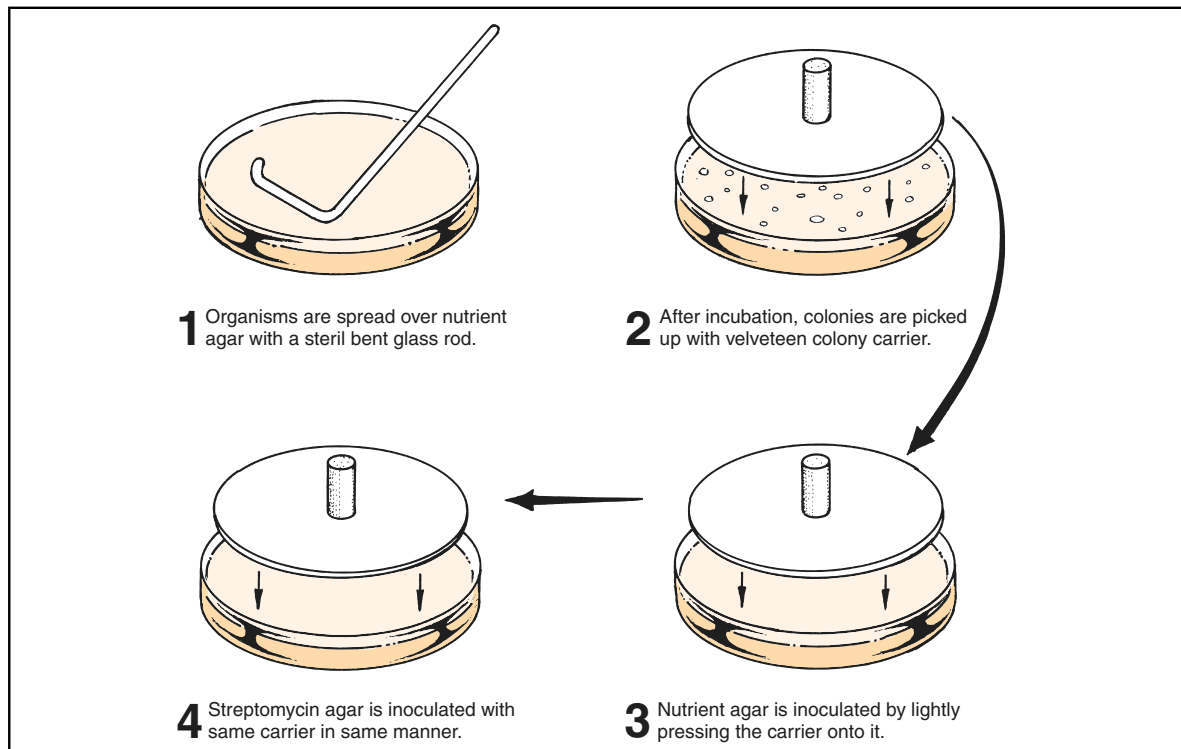


Figure 75.1 Replica plating technique

Mutant Isolation by Replica Plating • **Exercise 75**

FIRST PERIOD

Materials:

- 1 Petri plate of nutrient agar
 - 1 bent glass spreading rod
 - 1 ml serological pipette
 - beaker of 95% ethanol
 - Bunsen burner
 - broth culture of *E. coli*
1. Pipette 0.1 ml of *E. coli* from broth culture to surface of medium in Petri dish.
 2. With a sterile bent glass rod, spread the organisms over the plate following the routine shown in figure 75.1.
 3. Incubate this plate at 37° C for 24 hours.

SECOND PERIOD

Materials:

- 1 Petri plate culture of *E. coli* from previous period
- 1 Petri plate of nutrient agar per student

- 1 Petri plate of streptomycin agar
(100 micrograms of streptomycin per ml of medium)
- 1 sterile colony carrier per student

1. Carefully lower the sterile colony carrier onto the colonies of *E. coli* on the plate from the previous period.
2. Inoculate the plate of nutrient agar by lightly pressing the carrier onto the medium.
3. Now *without* returning the carrier to the original culture plate, inoculate the streptomycin agar in the same manner.
4. Incubate both plates at 37° C for 2 to 4 days in an enclosed cannister.

THIRD PERIOD

Materials:

- Quebec colony counter and hand counter
1. Examine both plates and record the information called for on Laboratory Report 74, 75.
 2. Tabulate the results of other members of the class.

76

Bacterial Mutagenicity and Carcinogenesis:
The Ames Test

The fact that carcinogenic compounds induce increased rates of mutation in bacteria has led to the use of bacteria for screening chemical compounds for possible carcinogenesis. The **Ames test**, developed by Bruce Ames at the University of California–Berkeley, has been widely used for this purpose.

The conventional way to determine whether a chemical substance is carcinogenic is to inject the material into animals and look for the development of tumors. If tumors develop, it is presumed that the substance can cause cancer. Although this method works well, it is costly, time-consuming, and cumbersome, especially if it is applied to all the industrial chemicals that have found their way into food and water supplies.

The Ames test serves as a screening test for the detection of carcinogenic compounds by testing the ability of chemical agents to induce bacterial mutations. Although most mutagenic agents are carcinogenic, some are not; however, the correlation between carcinogenesis and mutagenicity is high—around 83%. Once it has been determined that a specific agent is mutagenic, it can be used in animal tests to confirm its carcinogenic capability.

The standard way to test chemicals for mutagenesis has been to measure the rate of *back mutations* in strains of auxotrophic bacteria. In the Ames test a strain of *Salmonella typhimurium*, which is auxotrophic for histidine (unable to grow in the absence of histidine), is exposed to a chemical agent. After chemical exposure and incubation on histidine-deficient medium, the rate of reversion (back mutation) to prototrophy is determined by counting the number of colonies that are seen on the histidine-deficient medium.

Although testing of chemicals for mutagenesis in bacteria has been performed for a long time, two new features are included in the Ames test that make it a powerful tool. The first is that the strain of *S. typhimurium* used here lacks DNA repair enzymes, which prevents the correction of DNA injury. The second feature of the test is the incorporation of mammalian liver enzyme preparations with the chemical agent. The latter is significant because there is evidence that liver enzymes convert many noncarcinogenic chemical agents to carcinogenic ones.

There are two ways to perform the Ames test. The method illustrated in figure 76.1 is a **spot test** that is widely used for screening purposes. The other method is the **plate incorporation test**, which is used for quantitative analysis of the mutagenic effectiveness of compounds. Our concern here will be with the spot test; however, since the concentration of the liver extract is very critical, we will omit using it in our test. The test, as performed here, will work well without it.

Success in performing the spot test requires considerable attention to careful measurements and timing. It is for this reason that students will work in pairs to perform the test.

Note in figure 76.1 that 0.1 ml of *S. typhimurium* is first added to a small tube that contains 2 ml of top agar that is held at 45° C. This top agar contains 0.6% agar, 0.5% NaCl, and a *trace* of histidine and biotin. The histidine allows the bacteria to go through several rounds of cell division, which is essential for mutagenesis to occur. Since the histidine deletion extends through the biotin gene, biotin is also needed. This early growth of cells produces a faint background lawn that is barely visible to the naked eye.

Before pouring the top agar over the glucose–minimal salts agar, the tube must be vortexed at slow speed for 3 seconds and poured quickly to get even distribution. The addition of the bacteria, vortexing, and pouring must be accomplished in 20 seconds. Failure to move quickly enough will cause stippling of the top agar.

There are two ways that one can use to apply the chemical agent to the top agar: a filter paper disk may be used, or the chemical can be applied directly to the center of the plate without a disk. The procedure shown in figure 76.1 involves using a disk.

Note the unusual way in which a filter paper disk is impregnated in figure 76.1. To get it to stand on edge it must be put in position with sterile forceps and pressed in slightly to hold it upright. Just the right amount of the chemical agent is then added with a Pasteur pipette to the upper edge of the disk to completely saturate it without making it dripping wet; then the disk is lowered onto the top agar.

Once the test reagent is deposited on the top agar, the plate is incubated at 37° C for 48 hours. If the

Bacterial Mutagenicity and Carcinogenesis: The Ames Test • Exercise 76

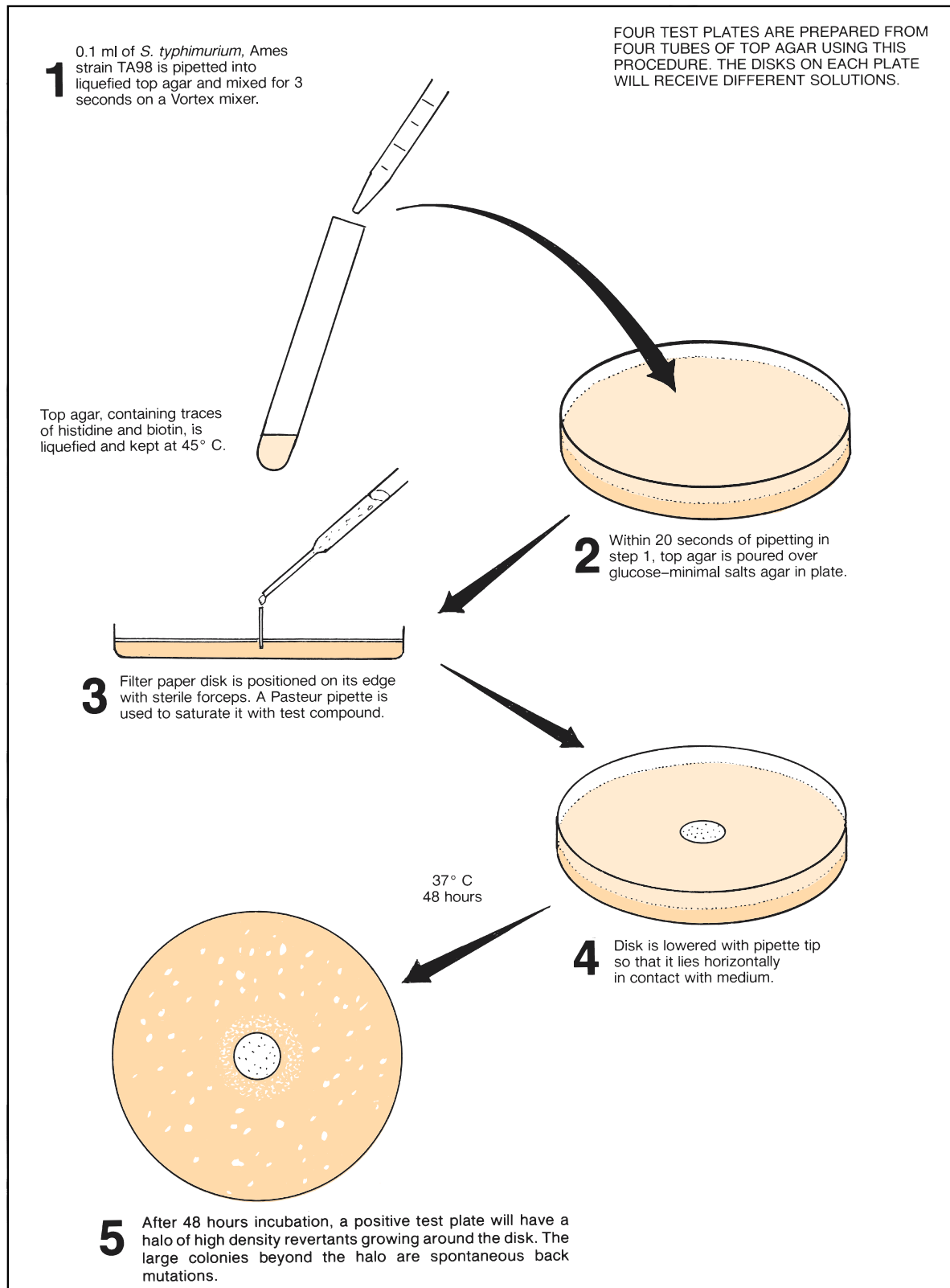


Figure 76.1 Procedure for performing a modified Ames test

Exercise 76 • Bacterial Mutagenicity and Carcinogenesis: The Ames Test

agent is mutagenic, a halo of densely packed revertant colonies will be seen around the disk. Scattered larger colonies will show up beyond the halo that represent spontaneous back mutations, not related to the test reagent.

You will be issued an unknown chemical agent to test and you will have an opportunity to test some other substance you have brought to the laboratory. In addition to these two tests you will be inoculating positive and negative test controls: thus, each pair of students will be responsible for four plates.

Keep in mind as you perform this experiment that there is a lot more to the Ames test than revealed here. While we are using only one tester strain of *S. typhimurium*, there are several others that are used in routine testing. The additional strains are needed to accommodate different kinds of chemical compounds. While one chemical agent may be mutagenic on one tester strain, it may produce a negative result on another strain. Also, keep in mind that we are not taking advantage of using the liver extract.

FIRST PERIOD (Inoculations)

Materials:*per pair of students:*

- 4 plates of glucose–minimal salts agar
(30 ml per plate)
- 4 tubes of top agar (2 ml per tube)
- tube of sterile water
- Vortex mixer

sterile Pasteur pipettes, forceps
serological pipettes (1 ml size)
filter paper disks, sterile in Petri dish
test reagents:

- 4-NOPD (10 µg/ml) solution*
- tube of unknown possible carcinogen
substance from home for testing

culture of *S. typhimurium*, Ames strain, TA98 in
trypticase soy broth
*4-nitro-o-phenylenediamine

1. Working with your laboratory partner, label the bottoms of four glucose–minimal salts agar plates as follows: POSITIVE CONTROL, NEGATIVE CONTROL, UNKNOWN, and OPTIONAL.
2. Liquefy four tubes of top agar and cool to 45° C.
3. With a 1 ml serological pipette, inoculate a tube of top agar with 0.1 ml of *S. typhimurium*.
4. Thoroughly mix the organisms into the top agar by vortexing (slow speed) for 3 seconds, or

rolling the tube between the palms of both hands. Pour the contents onto the positive control plate of glucose–minimal salts agar. The agar plate must be at room temperature. **Work rapidly to achieve pipetting, mixing, and spreading in 20 seconds.**

5. Repeat steps 3 and 4 for each of the other three tubes of top agar.
6. With sterile forceps place a disk on its edge near the center of the **positive control plate**. Sterilize the forceps by dipping in alcohol and flaming.
7. With a sterile Pasteur pipette, deposit just enough 4-NOPD on the upper edge of the disk to saturate it; then, push over the disk with the pipette tip onto the agar so that it lies flat.
8. Insert a sterile disk on the **negative control plate** in the same manner as above. Moisten this disk with sterile water, and reposition it flat on the agar surface. Be sure to use a fresh Pasteur pipette.
9. Place a disk on the **unknown plate**, and, using the same procedures, infiltrate it with your unknown, and position it flat on the agar.
10. On the fourth plate (**optional**) deposit a drop of your unknown from home. If the test substance from home is crystalline, place a few crystals directly on the top agar of the optional plate in its center. Liquid substances should be handled in same manner as above.
11. Incubate all four plates for 48 hours at 37° C.

SECOND PERIOD (Evaluation)

Examine all four plates. You should have a pronounced halo of revertant colonies around the disk on the positive control plate and no, or very few, revertants on the negative control plate. The presence of a few scattered revertants on the negative control plate is due to spontaneous back mutations, which always occur. Examine the areas beyond the halo to see if you can detect a faint lawn of bacterial growth.

CAUTION

Since much of the glassware in this experiment contains carcinogens, do not dispose of any of it in the usual manner. Your instructor will indicate how this glassware is to be handled.

Record your results on the Laboratory Report and answer all the questions.

PART 14 Medical Microbiology and Immunology

Although many of the exercises up to this point in this manual pertain in some way to medical microbiology, they also have applications that are nonmedical. The exercises of this unit, however, are primarily medical or dental in nature.

Medical (clinical) microbiology is primarily concerned with the isolation and identification of pathogenic organisms. Naturally, the techniques for studying each type of organism are different. A complete coverage of this field of microbiology is very extensive, encompassing the *Mycobacteriaceae*, *Brucellaceae*, *Enterobacteriaceae*, *Corynebacteriaceae*, *Micrococcaceae*, *ad infinitum*. It is not possible to explore all of these groups in such a short period of time; however, this course would be incomplete if it did not include some of the routine procedures that are used in the identification of some of the more common pathogens.

Exercise 77 in this unit differs from the other 13 exercises in that it pertains to the spread of disease (epidemiology) rather than to specific microorganisms. Its primary function is to provide an understanding of some of the tools used by public health epidemiologists to determine the sources of infection in the disease transmission cycle.

Since the most frequently encountered pathogenic bacteria are the gram-positive pyogenic cocci and the intestinal organisms, Exercises 78, 79, and 80 have been devoted to the study of those bacteria. The exercise that provides the greatest amount of depth is Exercise 79 (The Streptococci). To provide assistance in the identification of streptococci, it has been necessary to provide supplementary information in Appendix E.

Four exercises (82, 83, 84, and 85) are related to various applications of the agglutination reaction to serological testing. Two of these exercises pertain to slide tests and two of them are tube tests. It is anticipated that the instructor will select those tests from this group that fit time and budget limitations.

Exercises 87, 88, and 89 cover some of the basic hematological tests that might be included in a microbiology laboratory. The last exercise (90) pertains to an old test that has been revived pertaining to caries susceptibility.

77

A Synthetic Epidemic

A disease caused by microorganisms that enter the body and multiply in the tissues at the expense of the host is said to be an **infectious disease**. Infectious diseases that are transmissible to other persons are considered to be **communicable**. The transfer of communicable infectious agents between individuals can be accomplished by direct contact, such as in handshaking, kissing, and sexual intercourse, or they can be spread indirectly through food, water, objects, animals, and so on.

Epidemiology is the study of how, when, where, what, and who are involved in the spread and distribution of diseases in human populations. An epidemiologist is, in a sense, a medical detective who searches out the sources of infection so that the transmission cycle can be broken.

Whether an epidemic actually exists is determined by the epidemiologist by comparing the number of new cases with previous records. If the number of newly reported cases in a given period of time in a specific area is excessive, an **epidemic** is considered to be in progress. If the disease spreads to one or more continents, a **pandemic** is occurring.

In this experiment we will have an opportunity to approximate, in several ways, the work of the epidemiologist. Each member of the class will take part in the spread of a “synthetic infection.” The mode of transmission will be handshaking. For obvious safety reasons, the agent of transmission will not be a pathogen.

Two different approaches to this experiment are given: procedures A and B. In procedure A a white powder is used. In Procedure B two non-pathogens (*Micrococcus luteus* and *Serratia marcescens*) will be used. The advantage of procedure A is that it can be completed in one laboratory session. Procedure B, on the other hand, is more realistic in that viable organisms are used; however, it involves two periods. Your instructor will indicate which procedure is to be followed.

PROCEDURE A

In this experiment each student will be given a numbered container of white powder. Only one member in the class will be given a powder that is to be consid-

ered the infectious agent. The other members will be issued a transmissible agent that is considered noninfectious. After each student has spread the powder on his or her hands, all members of the class will engage in two rounds of handshaking, directed by the instructor. A record of the handshaking contacts will be recorded on a chart similar to the one on the Laboratory Report. After each round of handshaking, the hands will be rubbed on blotting paper so that a chemical test can be applied to it to determine the presence or absence of the infectious agent.

Once all the data are compiled, an attempt will be made to determine two things: (1) the original source of the infection, and (2) who the carriers are. The type of data analysis used in this experiment is similar to the procedure that an epidemiologist would employ. Proceed as follows:

Materials:

- 1 numbered container of white powder*
- 1 piece of white blotting paper
- spray bottles of “developer solution”*

Preliminaries

1. After assembling your materials, write your name and unknown number at the top of your sheet of blotting paper. In addition, draw a line down the middle, top to bottom, and label the left side ROUND 1 and the right side ROUND 2.
2. Wash and dry your hands thoroughly.
3. Moisten the right hand with water and prepare it with the agent by thoroughly coating it with the white powder, especially on the palm surface. This step is similar to the contamination that would occur to one’s hand if it were sneezed into during a cold.

IMPORTANT: Once the hand has been prepared do not rest it on the tabletop or allow it to touch any other object.

***Instructor:** To prevent students from preguessing the outcome of this experiment, the compositions of the powders and developer solution are known only to the instructor. The *Instructor’s Handbook* provides this information.

A Synthetic Epidemic • Exercise 77

Round 1

1. On the cue of the instructor, you will begin the first round of handshaking. Your instructor will inform you when it is your turn to shake hands with someone. You may shake with anyone, but it is best not to shake your neighbor's hand. *Be sure to use only your treated hand, and avoid extracurricular glad-handing.*
2. In each round of handshaking you will be selected by the instructor *only once* for handshaking; however, due to the randomness of selection by the handshakers, it is possible that you may be selected as the "shakee" several times.
3. After every member of the class has shaken someone's hand, you need to assess just who might have picked up the "microbe." To accomplish this, wipe your fingers and palm of the contaminated hand on the left side of your blotting paper. Press fairly hard, but don't tear the surface.
IMPORTANT: Don't allow your hand to touch any other object. A second round of handshaking follows.

Round 2

1. On the cue of your instructor, shake hands with another person. Avoid contact with any other objects.
2. Once the second handshaking episode is finished, rub the fingers and palm of the contaminated hand on the right side of the blotting paper.
CAUTION: Keep your contaminated hand off the left side of the blotting paper.

Chemical Identification

1. To determine who has been "infected" we will now spray the developer solution on the handprints of both rounds. One at a time, each student, with the help of the instructor, will spray his or her blotting paper with developer solution.
2. Color interpretation is as follows:
Blue:—positive for infectious agent
Brown or yellow:—negative

Tabulation of Results

1. Tabulate the results on the chalkboard, using a table similar to the one on the Laboratory Report.
2. Once all results have been recorded, proceed to determine the originator of the epidemic. The easiest way to determine this is to put together a flowchart of shaking.
3. Identify those persons that test positive. You will be working backward with the kind of informa-

tion an epidemiologist has to work with (contacts and infections). Eventually, a pattern will emerge that shows which person started the epidemic.

4. Complete the Laboratory Report.

PROCEDURE B

In this experiment each student will be given a piece of hard candy that has had a drop of *Micrococcus luteus* or *Serratia marcescens* applied to it. Only one person in the class will receive candy with *S. marcescens*, the presumed pathogen. All others will receive *M. luteus*.

After each student has handled the piece of candy with a glove-covered right hand, he or she will shake hands (glove to glove) with another student as directed by the instructor. A record will be kept of who takes part in each contact. Two rounds of handshaking will take place. After each round, a plate of trypticase soy agar will be streaked.

After incubating the plates, a tabulation will be made for the presence or absence of *S. marcescens* on the plates. From the data collected, an attempt will be made to determine two things: (1) the original source of the infection and (2) who the carriers are. The type of data analysis used in this experiment is similar to the procedure that an epidemiologist would employ. Proceed as follows:

CAUTION

Although the pathogenicity of *S. marcescens* is considered to be relatively low, avoid allowing any skin contact during this experiment.

Materials:

sterile rubber surgical gloves (1 per student)
hard candy contaminated with *M. luteus*
hard candy contaminated with *S. marcescens*
sterile swabs (2 per student)
TSA plates (1 per student)

Preliminaries

1. Draw a line down the middle of the bottom of a TSA plate, dividing it into two halves. Label one half ROUND 1 and the other ROUND 2.
2. Put a sterile rubber glove on your right hand. Avoid contaminating the palm surface.
3. Grasp the piece of candy in your gloved hand, rolling it around the surface of your palm. Discard the candy into a beaker of disinfectant set aside for disposal. You are now ready to do the first-round handshake.

Exercise 77 • A Synthetic Epidemic

Round 1

1. *On the cue of your instructor*, select someone to shake hands with. You may shake with anyone, but it is best not to shake hands with your neighbor.
2. In each round of handshaking you will be selected by the instructor *only once* for handshaking; however, due to the randomness of selection by the handshakers, it is possible that you may be selected as the “shakee” several times. The instructor or a recorder will record the initials of the shaker and shakee each time.
3. After you have shaken someone’s hand, swab the surface of your palm and transfer the organisms to the side of your plate designated as ROUND 1. Discard this swab into the appropriate container for disposal.

Round 2

1. Again, on the cue of your instructor, select someone at random to shake hands with. Be sure not to contaminate your gloved hand by touching something else.
2. With a fresh swab, swab the palm of your hand and transfer the organisms to the side of your

plate designated as ROUND 2. Make sure that your initials and the initials of the shakee are recorded by the instructor or recorder.

3. Incubate the TSA plate at room temperature for 48 hours.

Tabulation and Analysis

1. After 48 hours’ incubation look for typical red *S. marcescens* colonies on your Petri plate. If such colonies are present, record them as positive on your Laboratory Report chart and on the chart on the chalkboard.
2. Fill out the chart on your Laboratory Report with all the information from the chart on the chalkboard.
3. Identify those persons that test positive. You will be working backwards with the kind of information an epidemiologist has to work with (contacts and infections). Eventually a pattern will emerge that shows which person started the epidemic.

LABORATORY REPORT

Complete the Laboratory Report for this exercise.

The Staphylococci: Isolation and Identification

78

Often in conjunction with streptococci, the staphylococci cause abscesses, boils, carbuncles, osteomyelitis, and fatal septicemias. Collectively, the staphylococci and streptococci are referred to as the pyogenic (pus-forming) gram-positive cocci. Originally isolated from pus in wounds, the staphylococci were subsequently demonstrated to be normal inhabitants of the nasal membranes, the hair follicles, the skin, and the perineum of healthy individuals. The fact that 90% of hospital personnel are carriers of staphylococci portends serious epidemiological problems, especially since most strains are penicillin-resistant.

The **staphylococci** are gram-positive spherical bacteria that divide in more than one plane to form irregular clusters of cells. They are listed in section 12, volume 2, of *Bergey's Manual of Systematic Bacteriology*. The genus *Staphylococcus* is grouped with three other genera in family Micrococcaceae:

SECTION 12 GRAM-POSITIVE COCCI

Family I	Micrococcaceae
Genus I	<i>Micrococcus</i>
Genus II	<i>Stomatococcus</i>
Genus III	<i>Planococcus</i>
Genus IV	<i>Staphylococcus</i>
Family II	Deinococcaceae
Genus I	<i>Deinococcus</i>
Genus II	<i>Streptococcus</i>

Although the staphylococci make up a coherent phylogenetic group, they have very little in common with the streptococci except for their basic similarities of being gram-positive, non-spore-forming cocci. Note that *Bergey's Manual* puts these two genera into separate families due to their inherent differences.

Of the 19 species of staphylococci listed in *Bergey's Manual*, the most important ones are *S. aureus*, *S. epidermidis*, and *S. saprophyticus*. The single most significant characteristic that separates these species is the ability or inability of these organisms to coagulate plasma: only *S. aureus* has this ability; the other two are coagulase-negative.

Although *S. aureus* has, historically, been considered to be the only significant pathogen of the three, the others do cause infections. Some cerebrospinal fluid infections (2), prosthetic joint infec-

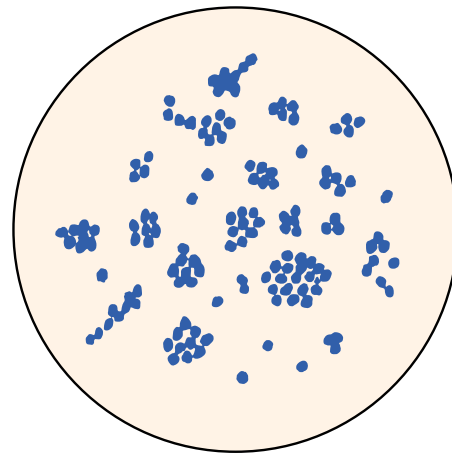


Figure 78.1 Staphylococci

tions (3), and vascular graft infections (1) have been shown to be due to coagulase-negative staphylococci. Numbers in parentheses designate references at the end of this exercise.

Our concern in this exercise will pertain exclusively to the differentiation of only three species of staphylococci. If other species are encountered, the student may wish to use the API Staph-Ident miniaturized test strip system (Exercise 55).

In this experiment we will attempt to isolate staphylococci from (1) the nose, (2) a fomite, and (3) an "unknown-control." The unknown-control will be a mixture containing staphylococci, streptococci, and some other contaminants. If the nasal membranes and fomite prove to be negative, the unknown-control will yield positive results, providing all inoculations and tests are performed correctly.

Since *S. aureus* is by far the most significant pathogen in this group, most of our concern here will be with this organism. It is for this reason that the characteristics of only this pathogen will be outlined next.

Staphylococcus aureus cells are 0.8 to 1.0 μm in diameter and may occur singly, in pairs, or as clusters. Colonies of *S. aureus* on trypticase soy agar or blood agar are opaque, 1 to 3 mm in diameter, and yellow, orange, or white. They are salt-tolerant, growing well

Exercise 78 • The Staphylococci: Isolation and Identification

on media containing 10% sodium chloride. Virtually all strains are coagulase-positive. Mannitol is fermented aerobically to produce acid. Alpha toxin is produced that causes a wide zone of clear (beta-type) hemolysis on blood agar; in rabbits it causes local necrosis and death.

The other two species lack alpha toxin (do not exhibit hemolysis) and are coagulase-negative. Mannitol is fermented to produce acid (no gas) by all strains of *S. aureus* and most strains of *S. saprophyticus*. Table 78.1 lists the principal characteristics that differentiate these three species of staphylococcus.

Table 78.1 Differentiation of three species of staphylococci

	<i>S. aureus</i>	<i>S. epidermidis</i>	<i>S. saprophyticus</i>
Alpha toxin	+	—	—
Mannitol (acid only)	+	—	(+)
Coagulase	+	—	—
Biotin for growth	—	+	NS
Novobiocin	S	S	R

Note: NS = not significant; S = sensitive; R = resistant;
(+) = mostly positive

To determine the incidence of carriers in our classroom, as well as the incidence of the organism on common fomites, we will follow the procedure illustrated in figure 78.2. Results of class findings will be tabulated on the chalkboard so that all members of the class can record data required on the Laboratory Report. The characteristics we will look for in our isolates will be (1) beta-type hemolysis (alpha toxin), (2) mannitol fermentation, and (3) coagulase production. Organisms found to be positive for these three characteristics will be *presumed* to be *S. aureus*. Final confirmation will be made with additional tests. Proceed as follows:

FIRST PERIOD

(Specimen Collection)

Note in figure 78.2 that swabs that have been applied to the nasal membranes and fomites will be placed in tubes of enrichment medium containing 10% NaCl (*m*-staphylococcus broth). Since your unknown-control will lack a swab, initial inoculations from this culture will have to be done with a loop.

Materials:

- 1 tube containing numbered unknown-control
- 3 tubes of *m*-staphylococcus broth
- 2 sterile cotton swabs

1. Label the three tubes of *m*-staphylococcus broth NOSE, FOMITE, and the number of your unknown-control.
2. Inoculate the appropriate tube of *m*-staphylococcus broth with one or two loopfuls of your unknown-control.
3. After moistening one of the swabs by immersing partially into the “nose” tube of broth, swab the nasal membrane just inside your nostril. A small amount of moisture on the swab will enhance the pickup of organisms. Place this swab into the “nose” tube.
4. Swab the surface of a fomite with the other swab that has been similarly moistened and deposit this swab in the “fomite” tube.
The fomite you select may be a coin, drinking glass, telephone mouthpiece, or any other item that you might think of.
5. Incubate these tubes of broth for 4 to 24 hours at 37° C.

SECOND PERIOD

(Primary Isolation Procedure)

Two kinds of media will be streaked for primary isolation: mannitol salt agar and staphylococcus medium 110.

Mannitol salt agar (MSA) contains mannitol, 7.5% sodium chloride, and phenol red indicator. The NaCl inhibits organisms other than staphylococci. If the mannitol is fermented to produce acid, the phenol red in the medium changes color from red to yellow.

Staphylococcus medium 110 (SM110) also contains NaCl and mannitol, but it lacks phenol red. Its advantage over MSA is that it favors colony pigmentation by different strains of *S. aureus*. Since this medium lacks phenol red, no color change takes place as mannitol is fermented.

Materials:

- 3 culture tubes from last period
- 2 Petri plates of MSA
- 2 Petri plates of SM110

1. Label the bottoms of the MSA and SM110 plates as shown in figure 78.2. Note that to minimize the number of plates required, it will be necessary to make half-plate inoculations for the nose and fomite. The unknown-control will be inoculated on separate plates.
2. Quadrant streak the MSA and SM110 plates with the unknown control.
3. Inoculate a portion of the nose side of each plate with the swab from the nose tube; then, with a

The Staphylococci: Isolation and Identification • Exercise 78

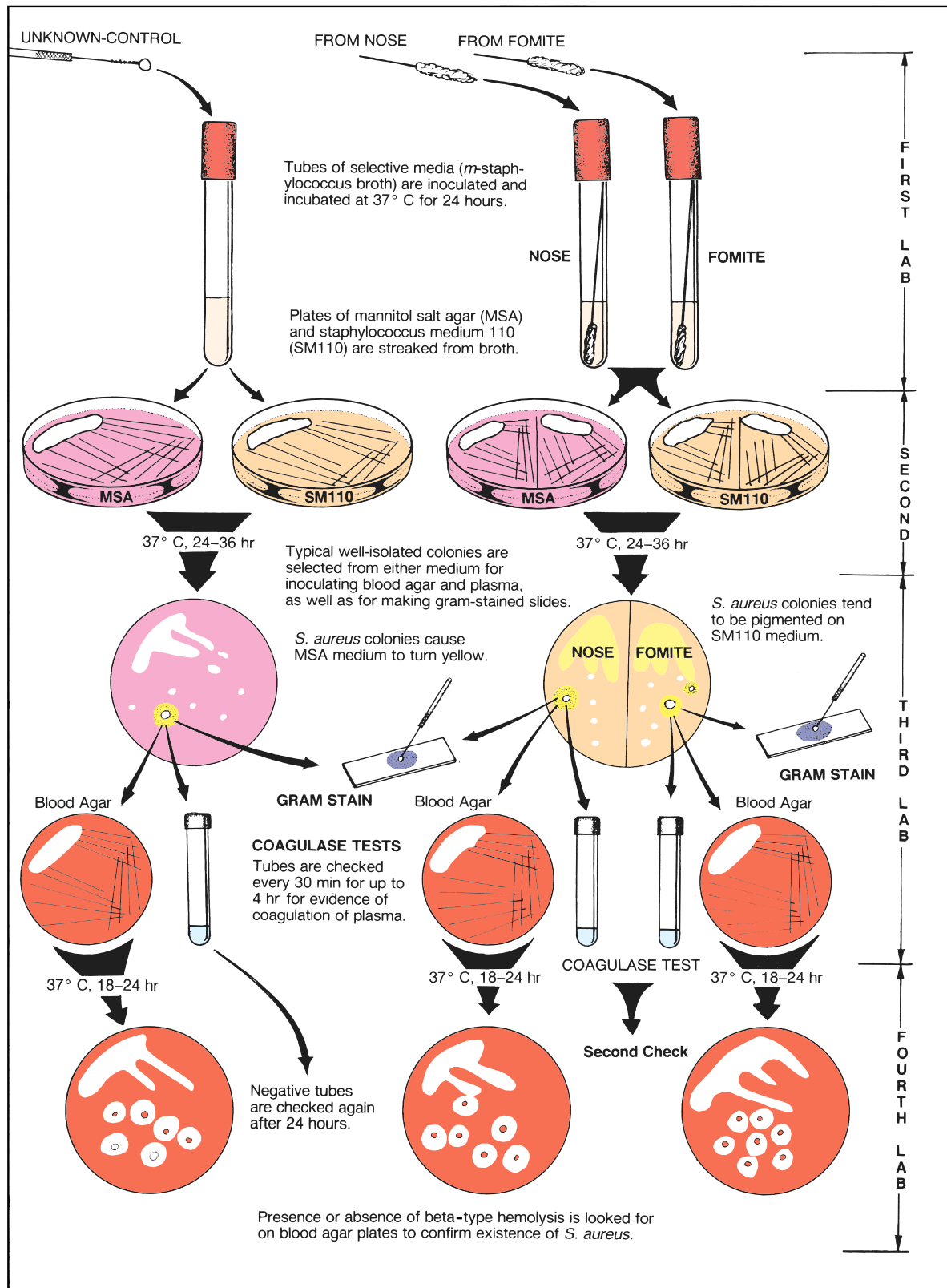


Figure 78.2 Procedure for presumptive identification of staphylococci

Exercise 78 • The Staphylococci: Isolation and Identification

- sterile loop, streak out the organisms on the remainder of the agar on that half of each plate. The swabbed areas will provide massive growth; the streaked-out areas should yield good colony isolation.
- Repeat step 3 to inoculate the other half of each agar plate with the swab from the fomite tube.
 - Incubate the plates aerobically at 37° C for 24 to 36 hours.
 - Select staphylococcus-like colonies from the MSA and SM110 plates from the nose and fomites for streaking out on another blood agar plate. Use half-plate streaking methods, if necessary.
 - Incubate the blood agar plates at 37° C for 18 to 24 hours. *Don't leave plates in incubator longer than 24 hours.* Overincubation will cause blood degeneration.

THIRD PERIOD

(Plate Evaluations and Coagulase/DNase Tests)

During this period we will perform the following tasks: (1) evaluate the plates from the previous period, (2) inoculate blood agar plates, (3) make gram-stained slides, and (4) perform coagulase and/or DNase tests on organisms from selected colonies. Proceed as follows:

Materials:

MSA and SM110 plates from previous period
2 blood agar plates
serological tubes containing 0.5 ml of 1:4 saline dilution of rabbit or human plasma (one tube for each isolate)
Petri plates of DNase agar
gram-staining kit

Evaluation of Plates

- Examine the mannitol salt agar plates. Has the phenol red in the medium surrounding any of the colonies turned yellow?
If this color change exists, it can be presumed that you have isolated a strain of *S. aureus*. Record your results on the Laboratory Report and chalkboard. (Your instructor may wish to substitute a copy of the chart from the Laboratory Report to be filled out at the demonstration table.)
- Examine the plates of SM110. The presence of growth here indicates that the organisms are salt-tolerant. Note color of the colonies (white, yellow, or orange).
- Record your observations of these plates on the Laboratory Report and chalkboard.

Blood Agar Inoculations

- Label the bottom of one blood agar plate with your unknown-control number, and streak out the organisms from a staph-like colony.

Coagulase Tests

The fact that 97% of the strains of *S. aureus* have proven to be coagulase-positive and that the other two species are *always* coagulase-negative makes the coagulase test an excellent definitive test for confirming identification of *S. aureus*.

The procedure is simple. It involves inoculating a small tube of plasma with several loopfuls of the organism and incubating it in a 37° C water bath for several hours. If the plasma coagulates, the organism is coagulase-positive. Coagulation may occur in 30 minutes or several hours later. *Any degree of coagulation, from a loose clot suspended in plasma to a solid immovable clot, is considered to be a positive result, even if it takes 24 hours to occur.*

It should be emphasized that this test is valid only for gram-positive, staphylococcus-like bacteria, because some gram-negative rods, such as *Pseudomonas*, can cause a false-positive reaction. The mechanism of clotting in such organisms is not due to coagulase. Proceed as follows:

- Label the plasma tubes NOSE, FOMITE, or UNKNOWN, depending on which of your plates have staph-like colonies.
- With a wire loop, inoculate the appropriate tube of plasma with organisms from one or more colonies on SM110 or MSA. Use several loopfuls. Success is more rapid with a heavy inoculation. If positive colonies are present on both nose and fomite sides, be sure to inoculate a separate tube for each side.
- Place the tubes in a 37° C water bath.
- Check for solidification of the plasma every 30 minutes for the remainder of the period. Note in figure 78.3 that solidification may be complete, as in the lower tube, or show up as a semisolid ball, as seen in the middle tube.

Any cultures that are negative at the end of the period will be left in the water bath. At 24 hours your instructor will remove them from the water bath and place them in the refrigerator, so that you can evaluate them in the next laboratory period.

- Record your results on the Laboratory Report.

The Staphylococci: Isolation and Identification • Exercise 78

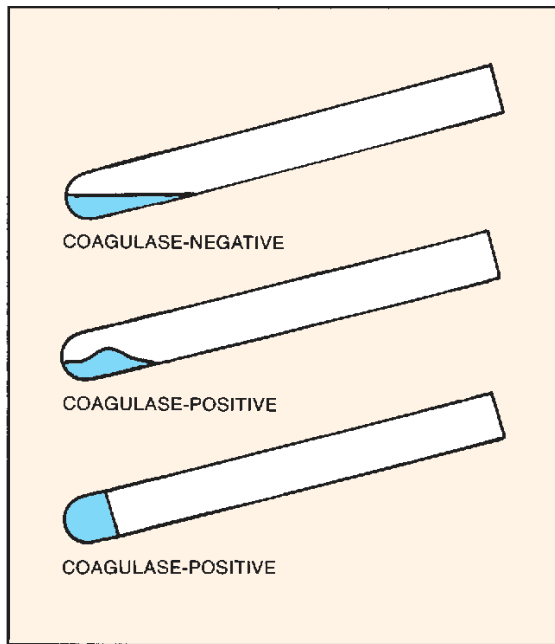


Figure 78.3 Coagulase test results: one negative and two positive tests

DNase Test

The fact that coagulase-positive bacteria are also able to hydrolyze DNA makes the DNase test a reliable means of confirming *S. aureus* identification. The following procedure can be used to determine if a staphy-like organism can hydrolyze DNA.

1. Heavily streak the organism on a plate of DNase test agar. One plate can be used for several test cultures by making short streaks about 1 inch long.
2. Incubate for 18–24 hours at 35° C.

Gram-Stained Slides

While your tubes of plasma are incubating in the water bath, prepare gram-stained slides from the same colonies that were used for the blood agar plates and coagulase tests.

Examine the slides under oil immersion lens and draw the organisms in the appropriate areas on the Laboratory Report.

FOURTH PERIOD (Confirmation)

During this period we will make final assessment of all tests and perform any other confirmatory tests that might be available to us.

Materials:

coagulase tubes from previous tests
blood agar plates from previous period
DNase test agar plates from previous period
0.1N HCl

1. Examine any coagulase tubes that were carried over from the last laboratory period that were negative at the end of that period. Record your results on the Laboratory Report.
2. Examine the colonies on your blood agar plates. Look for clear (beta-type) hemolysis around the colonies. The presence of alpha toxin is a definitive characteristic of *S. aureus*. Record your results on the Laboratory Report.
3. Look for zones of clearing near the streaks on the DNase agar plate. If none is seen, develop by flooding the plate with 0.1N HCl. The acid will render the hydrolyzed areas somewhat opaque.
4. Record your results on the chart on the chalkboard or chart on demonstration table. If an instructor-supplied tabulation chart is used, the instructor will have copies made of it to be supplied to each student.

FURTHER TESTING

In addition to using the API Staph-Ident miniaturized test strip system (Exercise 55) to confirm your identification of staphylococci, you may wish to use the latex agglutination slide test described in Exercise 83. Your instructor will inform you as to the availability of these materials and the desirability of proceeding further.

LABORATORY REPORT

After recording your results on the chalkboard (or on chart on demonstration table), complete the chart on the Laboratory Report and answer all the questions.

LITERATURE CITED

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2. Schoenbaum, S. C., P. Gardner, and J. Shillito. 1975. Infections in cerebrospinal shunts: Epidemiology, clinical manifestations, and therapy. *J. Infect. Dis.* 131:543–52.
3. Wilson, P. D., Jr., E. A. Salvati, P. Aglietti, and L. J. Kutner. 1973. The problem of infection in endoprosthetic surgery of the hip joint. *Clin. Orthop. Relat. Res.* 96:213–21.

79

The Streptococci: Isolation and Identification

The streptococci differ from the staphylococci in that they are arranged primarily in chains rather than in clusters. In addition to causing many mixed infections with staphylococci, the streptococci can also, separately, cause diseases such as pneumonia, meningitis, endocarditis, pharyngitis, erysipelas, and glomerulonephritis.

Several species of streptococci are normal inhabitants of the pharynx. They can also be isolated from surfaces of the teeth, the saliva, skin, colon, rectum, and vagina.

The streptococci of greatest medical significance are *S. pyogenes*, *S. agalactiae*, and *S. pneumoniae*. Of lesser importance are *S. faecalis*, *S. faecium*, and *S. bovis*. Appendix E describes in greater detail the characteristics and significance of these and other streptococcal species.

The purpose of this exercise is twofold: (1) to learn about standard procedures for isolating streptococci from the pharynx and (2) to learn how to differentiate between the most significant medically important streptococci.

Figure 79.2 illustrates the overall procedure to be followed in the pursuit of the above two goals. Note that blood agar is used to separate the streptococci into two groups on the basis of the type of hemolysis they produce on blood agar. Those organisms that produce alpha hemolysis on blood agar can be differentiated by four tests. Those that produce beta-type hemolysis can be differentiated with the CAMP test and three other tests. The procedure outlined here is, primarily, designed to achieve *presumptive identification* of seven groups of streptococci. A few extra tests are usually required to confirm identification.

To broaden the application of these tests you may be given two or three unknown cultures of streptococci to be identified along with the pharyngeal isolates. *If unknowns are to be used, they will not be issued until physiological media are to be inoculated.*

FIRST PERIOD

(Making a Streak-Stab Agar Plate)

During this period a plate of blood agar is swabbed and streaked in a special way to determine the type of hemolytic bacteria that are present in the pharynx.

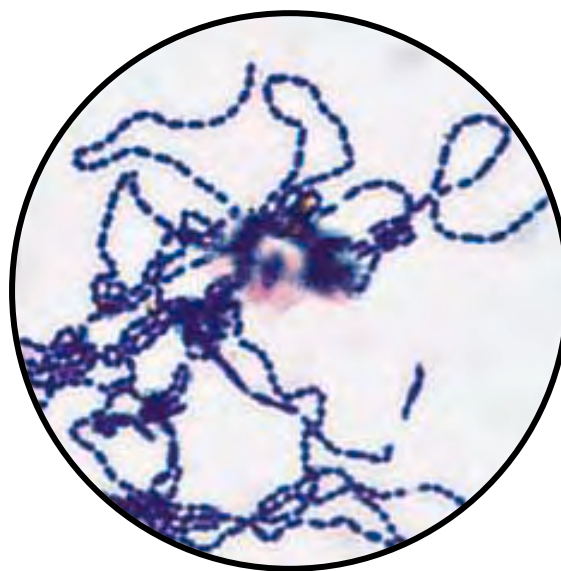


Figure 79.1 Streptococci

Before making such a streak plate, however, clinicians prefer to use a tube of enrichment broth (TSB) or a selective medium of TSB with a little crystal violet added to it (TSBCV). Media of this type are usually incubated at 37° C for 24 hours. This is particularly useful if the number of organisms might be low or if the swab cannot be applied to blood agar immediately. *Although this enrichment/selective step has been omitted here, it should be understood that the procedure is routine.*

Since swabbing one's own throat properly can be difficult, it will be necessary for you to work with your laboratory partner to swab each other's throats. Once your throat has been swabbed, you will proceed to use the swab to streak and stab your own agar plate according to a special procedure shown in figure 79.3.

Materials:

- 1 tongue depressor
- 1 sterile cotton swab
- inoculating loop
- 1 blood agar plate

1. With the subject's head tilted back and the tongue held down with the tongue depressor, rub the back

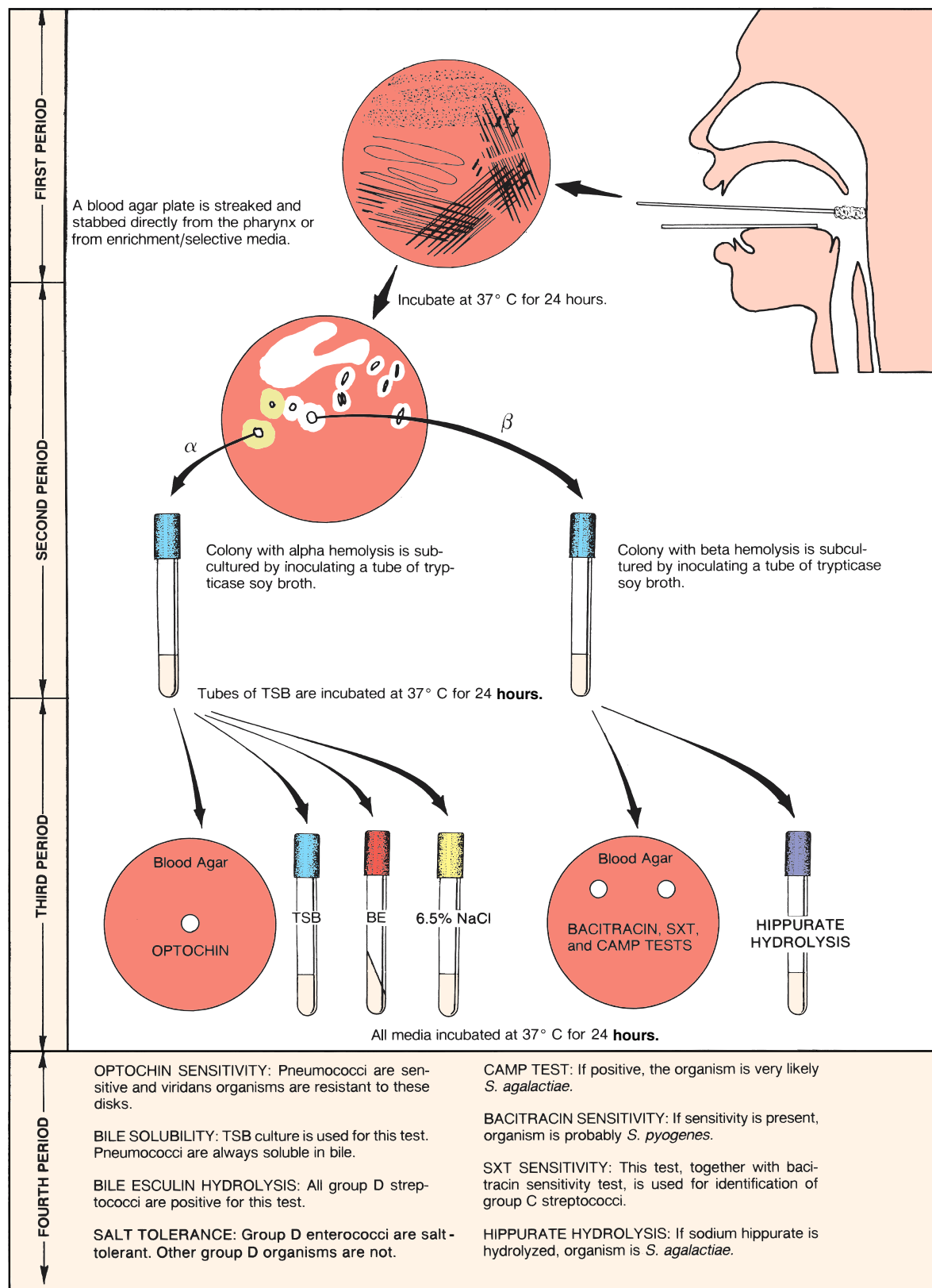


Figure 79.2 Media inoculations for the presumptive identification of streptococci

Exercise 79 • The Streptococci: Isolation and Identification

surface of the pharynx up and down with the sterile swab.

Also, look for white patches in the tonsillar area. Avoid touching the cheeks and tongue.

2. Since streptococcal hemolysis is most accurately analyzed when the colonies develop anaerobically beneath the surface of the agar, it will be necessary to use a streak-stab technique as shown in figure 79.3. The essential steps are as follows:
 - Roll the swab over an area approximating one-fifth of the surface. The entire surface of the swab should contact the agar.
 - With a wire loop, streak out three areas as shown to thin out the organisms.
 - Stab the loop into the agar to the bottom of the plate at an angle perpendicular to the surface to make a clean cut without ragged edges.
 - Be sure to make one set of stabs in an unstreaked area so that streptococcal hemolysis will be easier to interpret with a microscope.

CAUTION

Dispose of swabs and tongue depressors in beaker of disinfectant.

3. Incubate the plate aerobically at 37° C for 24 hours. *Do not incubate longer than 24 hours.*

SECOND PERIOD

(Analysis and Subculturing)

During this period, two things must be accomplished: first, the type of hemolysis must be correctly determined and, second, well-isolated colonies must be selected for making subcultures. The importance of proper subculturing cannot be overemphasized: without a pure culture, future tests are certain to fail. Proceed as follows:

Materials:

blood agar plate from previous period
tubes of TSB (one for each different type of colony)
dissecting microscope

1. Look for isolated colonies that have alpha or beta hemolysis surrounding them. Streptococcal colonies are characteristically very small.
2. Do any of the stabs appear to exhibit hemolysis? Examine these hemolytic zones near the stabs under 60× magnification with a dissecting microscope.
3. Consult figure 79.4 to analyze the type of hemolysis. Note that the illustrations on the left side indicate what the colonies would look like if they were submerged under a layer of blood agar (two-layer pour plate). The illustrations on the right indicate the nature of hemolysis around stabs on streak-stab plates. Although this illustration is very diagrammatic, it reveals the microscopic differences between three kinds of hemolysis: alpha, alpha-prime, and beta.

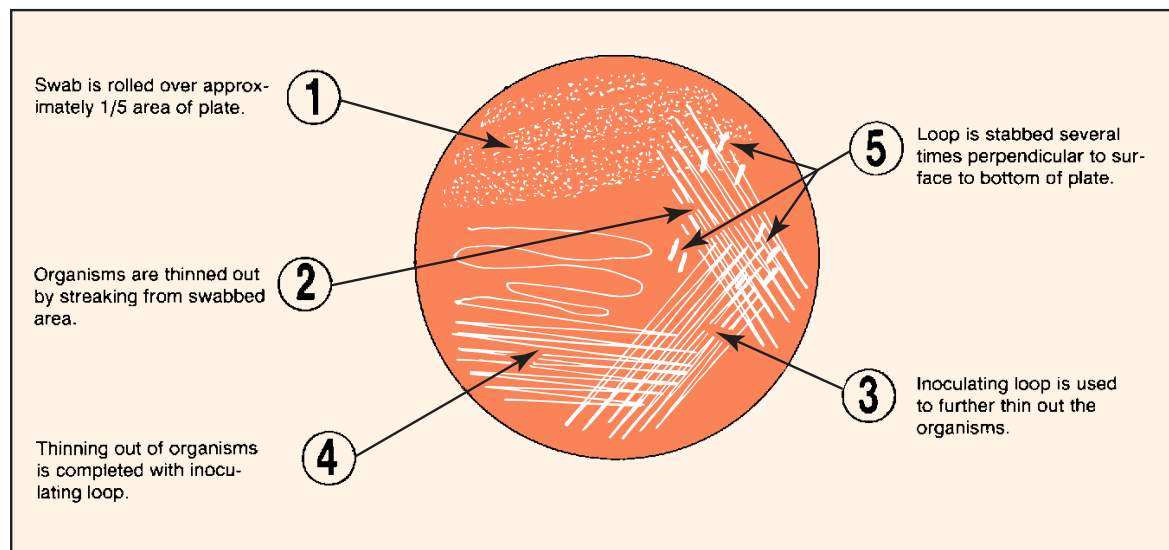


Figure 79.3 Streak-stab procedure for blood agar inoculations

The Streptococci: Isolation and Identification • Exercise 79

Only those stabs that are completely free of red blood cells in the hemolytic area are considered to be **beta hemolytic**. The chance of isolating a colony of this type from your own throat is very slim, for the beta hemolytic streptococci are the most serious pathogens.

If some red blood cells are seen dispersed throughout the hemolytic zone, the organism is classified as **alpha-prime hemolytic**. Viridans streptococci often fall in this category.

4. Record your observations on the Laboratory Report.
5. Select well-isolated colonies that exhibit hemolysis (alpha, beta, or both) for inoculating tubes of TSB. Be sure to label the tubes ALPHA or BETA. Whether or not the organism is alpha or beta is crucial in identification.

Since the chances of isolating beta hemolytic streptococci from the pharynx are usually quite slim, notify your instructor if you think you have isolated one.

6. Incubate the tubes at 37° C for 24 hours.
7. **Important:** At some time prior to the next laboratory session, review the material in Appendix E that pertains to this exercise.

THIRD PERIOD

(Inoculations for Physiological Tests)

Presumptive identification of the various groups of streptococci is based on seven or eight physiological tests. Table 79.1 on page 267 reveals how they perform on these tests. Note that groups A, B, and C are all beta hemolytic; a few enterococci are also beta hemolytic. The remainder are all alpha hemolytic or nonhemolytic.

Since each of the physiological tests is specific for differentiating only two or three groups, it is not desirable to do all the tests on all unknowns. For economy and preciseness, only four tests that are mentioned for the third period in figure 79.2 should be performed on an isolate or unknown.

Before any inoculations are made, however, it is desirable to do a purity check on each TSB culture from the previous period. To accomplish this it will be necessary to make a gram-stained slide of each of the cultures.

If unknowns are to be issued, they will be given to you at this time. They will be tested along with your pharyngeal isolates. The only information that will be given to you about each unknown is its hemolytic type so that you will be able to determine what physiological tests to perform on each one. Proceed as follows:

Gram-Stained Slides (Purity Check)

Materials:

TSB cultures from previous period
gram-staining kit

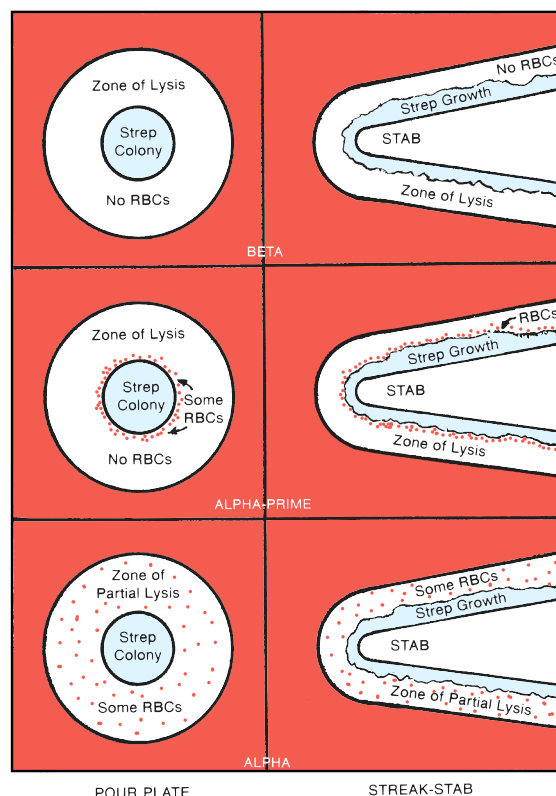


Figure 79.4 Comparison of hemolysis types as seen on pour plates and streak-stab plates

1. Make a gram-stained slide from each of the pharyngeal isolates and examine them under oil immersion lens. Do they appear to be pure cultures?
2. Draw the organisms in the appropriate circles on the Laboratory Report.

Beta-Type Inoculations

Use the following procedure to perform tests on each isolate that has beta-type hemolysis:

Materials:

for each isolate:

- 1 blood agar plate
- 1 tube of sodium hippurate broth
- 1 bacitracin differential disk
- 1 SXT sensitivity disk
- 1 broth culture of *S. aureus*
- dispenser or forceps for transferring disks

1. Label a blood agar plate and a tube of sodium hippurate broth with proper identification information of each isolate and unknown to be tested.
2. Follow the procedure outlined in figure 79.5 to inoculate each blood agar plate with the isolate (or unknown) and *S. aureus*.

Exercise 79 • The Streptococci: Isolation and Identification

Note that a streak of the unknown is brought down perpendicular to the *S. aureus* streak, keeping the two organisms about 1 cm apart.

3. With forceps or dispenser, place one bacitracin differential disk and one SXT disk on the heavily streaked area at points shown in figure 79.5. Press down on each disk slightly.
4. Inoculate one tube of sodium hippurate broth for each isolate or unknown.
5. Incubate the blood agar plates at 37° C, aerobically, for 24 hours, and the hippurate broth tubes at 35° C, aerobically, for 24 hours. If the hippurate broths prove to be negative or weakly positive at 24 hours, they should be given more time to see if they change.

place one optochin (Taxo P) disk in the center of each area. Press down slightly on each disk to secure it to the medium.

3. Inoculate one tube each of TSB, BE, and 6.5% NaCl broth with each test organism.
4. Incubate all media at 35°–37° C as follows:

Blood agar plates: 24 hours in a candle jar

6.5% NaCl broths: 24, 48, and 72 hours

Bile esculin slants: 48 hours

Trypticase soy broths: 24 hours

Note: While the blood agar plates should be incubated in a candle jar or CO₂ incubator, the remaining cultures can be incubated aerobically.

FOURTH PERIOD**(Evaluation of Physiological Tests)**

Once all of the inoculated media have been incubated for 24 hours, you are ready to examine the plates and tubes and add test reagents to some of the cultures. Some of the tests will also have to be checked at 48 and 72 hours.

After you have assembled all the plates and tubes from the last period, examine the blood agar plates first that were double-streaked with the unknowns and *S. aureus*. Note that the second, third, and fourth tests listed in table 79.1 can be read from these plates. Proceed as follows:

CAMP Reaction

If you have an unknown that produces an enlarged arrowhead-shaped hemolytic zone at the juncture where the unknown meets the *S. aureus* streak, as seen

Alpha-Type Inoculations

As shown in figure 79.2, four inoculations will be made for each isolate or unknown that is alpha hemolytic.

Materials:

- 1 blood agar plate (for up to 4 unknowns)
- 1 6.5% sodium chloride broth
- 1 trypticase soy broth (TSB)
- 1 bile esculin (BE) slant
- 1 optochin (Taxo P) disk
- candle jar setup or CO₂ incubator

1. Mark the bottom of a blood agar plate to divide it into halves, thirds, or quarters, depending on the number of alpha hemolytic organisms to be tested. Label each space with the code number of each test organism.
2. Completely streak over each area of the blood agar plate with the appropriate test organism, and

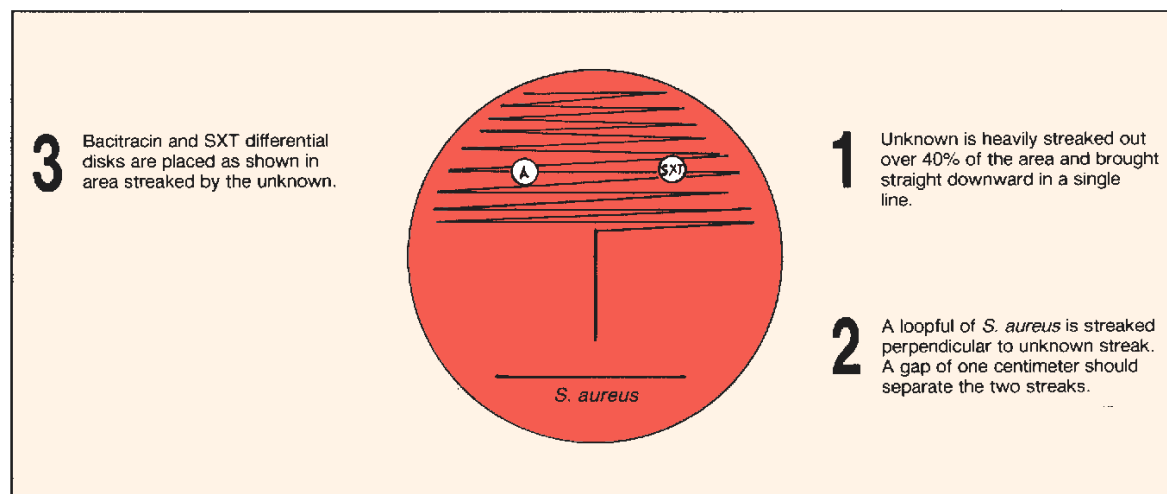


Figure 79.5 Blood agar inoculation technique for the CAMP, bacitracin, and SXT tests

The Streptococci: Isolation and Identification • Exercise 79

in figure 79.6, the organism is *S. agalactiae*. This phenomenon is due to what is called the *CAMP factor*. The only problem that can arise from this test is that if the plate is incubated anaerobically, a positive CAMP reaction can occur on *S. pyogenes* inoculated plates.

Record the CAMP reactions for each of your isolates or unknowns on the Laboratory Report.

Bacitracin Susceptibility

Any size zone of inhibition seen around the bacitracin disks should be considered to be a positive test result. Note in table 79.1 that *S. pyogenes* is positive for this characteristic.

This test has two limitations: (1) the disks must be of the *differential type*, not sensitivity type, and (2) the test should not be applied to alpha hemolytic streptococci. Reasons: Sensitivity disks have too high a concentration of the antibiotic, and many alpha hemolytic streptococci are sensitive to these disks.

Record the results of this test on table under D of your Laboratory Report.

SXT Sensitivity Test

The disks used in this test contain 1.25 mg of trimethoprim and 27.75 mg of sulfamethoxazole (SXT). The purpose of this test is to distinguish

Table 79.1 Physiological Tests for Streptococcal Differentiation

GROUP	Type of Hemolysis	Bacitracin Susceptibility	CAMP Reaction or Hippurate Hydrolysis	SXT Sensitivity	Bile Esculin Hydrolysis	Tolerance to 6.5% NaCl	Optochin Susceptibility	Bile Solubility
Group A <i>S. pyogenes</i>	beta	+	—	R	—	—	—	—
Group B <i>S. agalactiae</i>	beta	—*	+	R	—	±	—	—
Group C <i>S. equi</i> <i>S. equisimilis</i> <i>S. zooepidemicus</i>	beta	—*	—	S	—	—	—	—
Group D** (enterococci) <i>S. faecalis</i> <i>S. faecium</i> etc.	alpha beta none	—	—	R	+	+	—	—
Group D** (nonenterococci) <i>S. bovis</i> etc.	alpha none	—	—	R/S	+	—	—	—
Viridans <i>S. mitis</i> <i>S. salivarius</i> <i>S. mutans</i> etc.	alpha none	—*	—*	S	—	—	—	—
Pneumococci <i>S. pneumoniae</i>	alpha	±	—		—	—	+	+

Note: R = resistant; S = sensitive; blank = not significant.

*Exceptions occur occasionally.

**See comments on pp. 457 and 458 concerning correct genus.

Exercise 79 • The Streptococci: Isolation and Identification



Figure 79.6 Note positive SXT disk on right, negative bacitracin disk on left, and positive CAMP reaction (arrowhead). Organism: *S. agalactiae*.

groups A and B from other beta hemolytic streptococci. Note in table 79.1 that both groups A and B are uniformly resistant to SXT.

If a beta hemolytic streptococcus proves to be bacitracin-resistant and SXT-susceptible, it is classified as being a **non-group-A or -B beta hemolytic streptococcus**. This means that the organism is probably a species within group C. *Keep in mind that an occasional group A streptococcal strain is susceptible to both bacitracin and SXT disks.* One must always remember that exceptions to most tests do occur; that is why this identification procedure leads us only to *presumptive* conclusions.

Record any zone of inhibition (resistance) as positive for this test.

Hippurate Hydrolysis

Note in table 79.1 that hippurate hydrolysis and the CAMP test are grouped together as positive tests for *S. agalactiae*. If an organism is positive for both tests, or either one, one can assume with almost 100% certainty that the organism is *S. agalactiae*.

Proceed as follows to determine which of your isolates are able to hydrolyze sodium hippurate:

Materials:

serological test tubes
serological pipettes (1 ml size)
ferric chloride reagent
centrifuge

1. Centrifuge the culture for 3 to 5 minutes.
2. Pipette 0.2 ml of the supernatant and 0.8 ml of ferric chloride reagent into an empty serological test tube. Mix well.

3. Look for a **heavy precipitate** to form. If the precipitate forms and persists for 10 minutes or longer, the test is positive. If the culture proves to be weakly positive, incubate the culture for another 24 hours and repeat the test.
4. Record your results on the Laboratory Report.

Bile Esculin Hydrolysis

This is the best physiological test that we have for the identification of group D streptococci. Both enterococcal and nonenterococcal species of group D are able to hydrolyze esculin in the agar slant, causing the slant to blacken.

A positive BE test tells us that we have a group D streptococcus; differentiation of the two types of group D streptococci depends on the salt-tolerance test.

Examine the BE agar slants, looking for **blackening of the slant**, as illustrated in figure 79.7. If less than half of the slant is blackened, or if no blackening occurs within 24 to 48 hours, the test is negative.

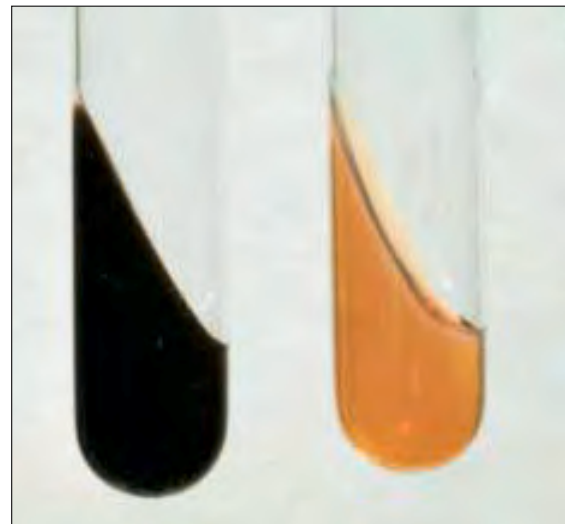


Figure 79.7 Positive bile esculin hydrolysis on left; negative on right

Salt Tolerance (6.5% NaCl)

All enterococci of group D produce heavy growth in 6.5% NaCl broth. As indicated in table 79.1, none of the nonenterococci, group D, grow in this medium. This test, then, provides us with a good method for differentiating the two types of group D streptococci.

A positive result shows up as turbidity within 72 hours. A color change of **purple to yellow** may also be present. If the tube is negative at 24 hours, incubate it and check it again at 48 and 72 hours. *If the organ-*

The Streptococci: Isolation and Identification • Exercise 79

ism is salt-tolerant and BE-positive, it is considered to be an enterococcus. Parenthetically, it should be added here that approximately 80% of group B streptococci will grow in this medium.

Optochin Susceptibility

Optochin susceptibility is used for differentiation of the alpha hemolytic viridans streptococci from the pneumococci. The pneumococci are sensitive to these disks; the viridans organisms are resistant.

Materials:

blood agar plates with optochin disks
plastic metric ruler

1. Measure the diameters of zones of inhibition that surround each disk, evaluating whether the zones are large enough to be considered positive. The standards are as follows:
 - For 6 mm diameter disks, the zone must be at least 14 mm diameter to be considered positive.
 - For 10 mm diameter disks, the zone must be at least 16 mm diameter to be considered positive.
2. Record your results on the Laboratory Report.

Bile Solubility

If an alpha hemolytic streptococcal organism is soluble in bile and positive on the optochin test, presumptive evidence indicates that the isolate is *S. pneumoniae*. Perform the bile solubility test on each of your alpha hemolytic isolates as follows:

Materials:

2 empty serological tubes (per test)
dropping bottle of phenol red indicator

dropping bottle of 0.05N NaOH
TSB culture of unknown
2% bile solution (sodium desoxycholate)
bottle of normal saline solution
2 serological pipettes (1 ml size)
water bath (37° C)

1. Mark one empty serological tube BILE and the other SALINE. Into their respective tubes, pipette 0.5 ml of 2% bile and 0.5 ml of saline.
2. Shake the TSB unknown culture to suspend the organisms and pipette 0.5 ml of the culture into each tube.
3. Add one or two drops of phenol red indicator to each tube and adjust the pH to 7.0 by adding drops of 0.05N NaOH.
4. Place both tubes in a 37° C water bath and examine periodically for 2 hours. If the turbidity clears in the bile tube, it indicates that the cells have disintegrated and the organism is *S. pneumoniae*. Compare the tubes side by side.
5. Record your results on the Laboratory Report.

FINAL CONFIRMATION

All the laboratory procedures performed so far lead us to presumptive identification. To confirm these conclusions it is necessary to perform serological tests on each of the unknowns. If commercial kits are available for such tests, they should be used to complete the identification procedures.

LABORATORY REPORT

Complete the Laboratory Report for this exercise.

80

Gram-Negative Intestinal Pathogens

The enteric pathogens of prime medical concern are the **salmonella** and **shigella**. They cause enteric fevers, food poisoning, and bacillary dysentery. *Salmonella typhi*, which causes typhoid fever, is by far the most significant pathogen of the salmonella group. In addition to the typhoid organism, there are 10 other distinct salmonella species and over 2,200 serotypes. The shigella, which are the prime causes of human dysentery, comprise four species and many serotypes. *Serotypes* within genera are organisms of similar biochemical characteristics that can most easily be differentiated by serological typing.

Routine testing for the presence of these pathogens is a function of public health laboratories at various governmental levels. The isolation of these pathogenic enterics from feces is complicated by the fact that the colon contains a diverse population of bacteria. Species of such genera as *Escherichia*, *Proteus*, *Enterobacter*, *Pseudomonas*, and *Clostridium* exist in large numbers; hence it is necessary to use media that are differential and selective to favor the growth of the pathogens.

Figure 80.1 is a separation outline that is the basis for the series of tests that are used to demonstrate the

presence of salmonella or shigella in a patient's blood, urine, or feces. Note that lactose fermentation separates the salmonella and shigella from most of the other Enterobacteriaceae. Final differentiation of the two enteric pathogens from *Proteus* relies on motility, hydrogen sulfide production, and urea hydrolysis. The differentiation information of the positive lactose fermenters on the left side of the separation outline is provided here mainly for comparative references that can be used for the identification of other unknown enterics.

The procedural diagram in figure 80.2 on the opposite page reveals how we will apply these facts in the identification of an unknown salmonella or shigella. The entire process will involve four laboratory periods.

In this experiment you will be given a mixed culture containing a coliform, *Proteus*, and a salmonella or shigella. The pathogens will be of the less dangerous types, but their presence will, naturally, demand utmost caution in handling. Your problem will be to isolate the pathogen from the mixed culture and make a genus identification. There are five steps that are used to prove the presence of these pathogens in a

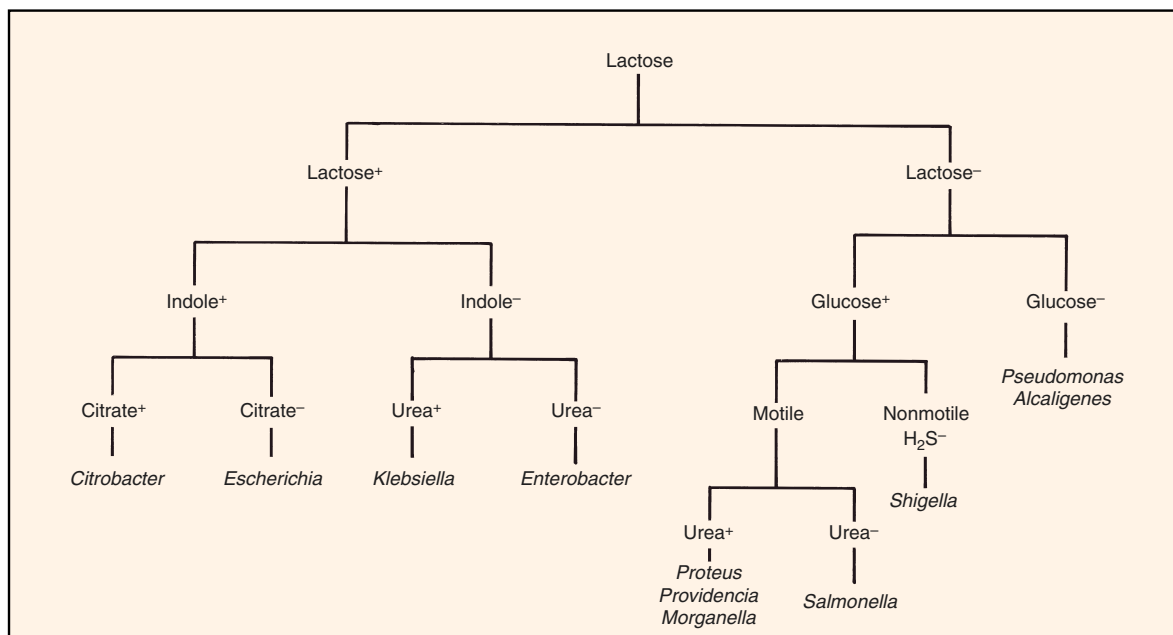


Figure 80.1 Separation outline of Enterobacteriaceae

Gram-Negative Intestinal Pathogens • Exercise 80

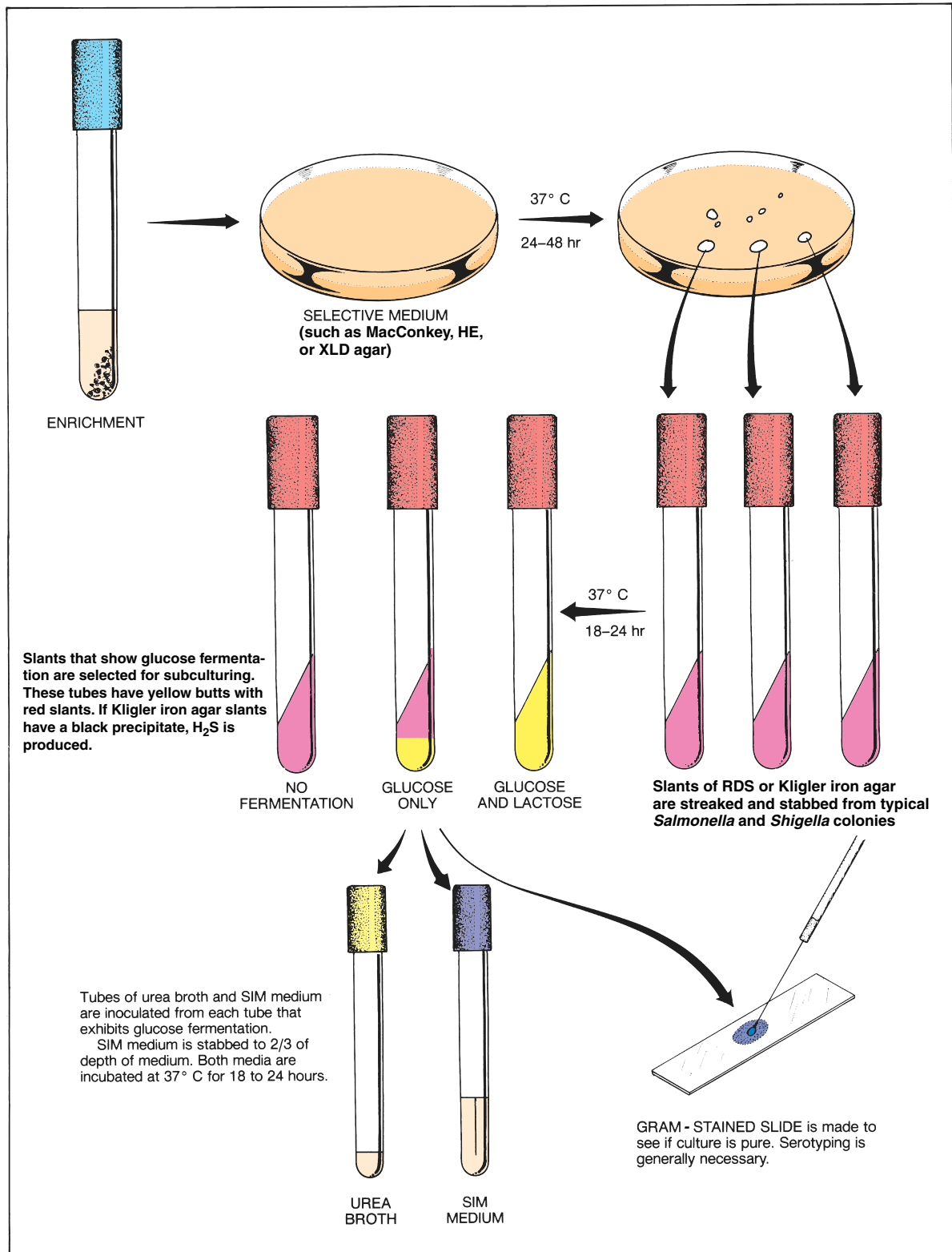


Figure 80.2 Isolation and presumptive identification of *Salmonella* and *Shigella*

Exercise 80 • Gram-Negative Intestinal Pathogens

stool sample: (1) enrichment, (2) isolation, (3) fermentation tests, (4) final physiological tests, and (5) serotyping.

Enrichment

There are two enrichment media that are most frequently used to inhibit the nonpathogens and favor the growth of pathogenic enterics. They are selenite F and gram-negative (GN) broths. While most salmonella grow unrestricted in these two media, some of the shigella are inhibited to some extent in selenite F broth; thus, for shigella isolation, GN broth is preferred. In many cases, stool samples are plated directly on isolation media.

In actual practice, 1 to 5 grams of feces are placed in 10 ml of enrichment broth. In addition, plates of various kinds of selective media are inoculated directly. The broths are usually incubated for 4 to 6 hours.

Since we are not using stool samples in this exercise, the enrichment procedure is omitted. Instead, you will streak the isolation media directly from the unknown broth.

FIRST PERIOD
(Isolation)

There are several excellent selective differential media that have been developed for the isolation of these pathogens. Various inhibiting agents such as brilliant green, bismuth sulfite, sodium desoxycholate, and sodium citrate are included in them. For *Salmonella typhi*, bismuth sulfite agar appears to be the best medium. Colonies of *S. typhi* on this medium appear black due to the reduction of sulfite to sulfide.

Other widely used media are MacConkey agar, Hektoen Enteric agar (HE), and Xylose Lysine Desoxycholate (XLD) agar. These media may contain bile salts and/or sodium desoxycholate to inhibit gram-positive bacteria. To inhibit coliforms and other nonenterics, they may contain a citrate. All of them contain lactose and a dye so that if an organism is a lactose fermenter, its colony will take on a color characteristic of the dye present.

Since the enrichment procedure is being omitted here, you will be issued an unknown broth culture with a pathogenic enteric. Your instructor will indicate which selective media will be used. Proceed as follows to inoculate the selective media with your unknown mixture:

Materials:

- unknown culture (mixture of a coliform, *Proteus*, and a salmonella or shigella)
- 1 or more Petri plates of different selective media: MacConkey, Hektoen Enteric (HE), or Xylose Lysine Desoxycholate (XLD) agar

1. Label each plate with your name and unknown number.
2. With a loop, streak each plate with your unknown in a manner that will produce good isolation.
3. Incubate the plates at 37° C for 24 to 48 hours.

SECOND PERIOD
(Fermentation Tests)

As stated above, the fermentation characteristic that separates the SS pathogens from the coliforms is their *inability to ferment lactose*. Once we have isolated colonies on differential media that look like salmonella or shigella, the next step is to determine whether the isolates can ferment lactose. All media for this purpose contain at least two sugars, glucose and lactose. Some contain a third sugar, sucrose. They also contain phenol red to indicate when fermentation occurs. Russell Double Sugar (RDS) agar is one of the simpler media that works well. Kligler iron agar may also be used. It is similar to RDS with the addition of iron salts for detection of H₂S. Your instructor will indicate which one will be used.

Proceed as follows to inoculate three slants from colonies on the selective media that look like either salmonella or shigella. The reason for using three slants is that the you may have difficulty distinguishing *Proteus* from the SS pathogens. By inoculating three tubes from different colonies, you will be increasing your chances of success.

Materials:

- 3 agar slants (RDS or Kligler iron)
- streak plates from first period

1. Label the three slants with your name and the number of your unknown.
2. Look for isolated colonies that look like salmonella or shigella organisms. The characteristics to look for on each medium are as follows:
 - **MacConkey agar**—*Salmonella*, *Shigella* and other non-lactose-fermenting species produce smooth, colorless colonies. Coliforms that ferment lactose produce reddish, mucoid, or dark-centered colonies.
 - **Hektoen Enteric (HE) agar**—*Salmonella* and *Shigella* colonies are greenish-blue. Some species of *Salmonella* will have greenish-blue colonies with black centers due to H₂S production. Coliform colonies are salmon to orange and may have a bile precipitate.
 - **Xylose Lysine Desoxycholate (XLD) agar**—although most *Salmonella* produce red colonies with black centers, a few may produce red colonies that lack black centers. *Shigella* colonies are red. Coliform colonies are yellow.

Some *Pseudomonas* produce false-positive red colonies.

3. With a straight wire, inoculate the three agar slants from separate SS-appearing colonies. Use the streak-stab technique. When streaking the surface of the slant before stabbing, move the wire over the entire surface for good coverage.
4. Incubate the slants at 37° C for 18 to 24 hours. Longer incubation time may cause alkaline reversion. Even refrigeration beyond this time may cause reversion.

Alkaline reversion is a condition in which the medium turns yellow during the first part of the incubation period and then changes to red later due to increased alkalinity.

THIRD PERIOD

(Slant Evaluations and Final Inoculations)

During this period you will inoculate tubes of SIM medium and urea broth with organisms from the slants of the previous period. Examination of the separation outline in figure 80.1 reveals that the final step in the differentiation of the SS pathogens is to determine whether a non-lactose-fermenter can do three things: (1) exhibit motility, (2) produce hydrogen sulfide, and (3) produce urease. You will also be making a gram-stained slide to perform a purity check. If miniaturized multitest media are available, they can also be inoculated at this time.

Materials:

RDS or Kligler iron agar slants from previous period
1 tube of SIM medium for each positive slant
1 tube of urea broth for each positive slant
miniaturized multitest media such as API 20E or Enterotube II (optional)

1. Examine the slants from the previous period and **select those tubes that have a yellow butt with a red slant**. These tubes contain organisms that ferment only glucose (non-lactose-fermenters). If you used Kligler's iron agar, a black precipitate in the medium will indicate that the organism is a producer of H₂S.
Note in figure 80.1 that slants that are completely yellow are able to ferment lactose as well as glucose. Tubes that are completely red are either nonfermenters or examples of alkaline reversion. Ignore those tubes.
2. With a loop, inoculate one tube of urea broth from each slant that has a yellow butt and red slant (non-lactose-fermenter).

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3. With a straight wire, stab one tube of SIM medium from each of the same agar slants. Stab in the center to two-thirds of depth of medium.
4. Incubate these tubes at 37° C for 18 to 24 hours.
5. Make gram-stained slides from the same slants and confirm the presence of gram-negative rods.
6. If miniaturized multitest media are available, such as API 20E or Enterotube II, inoculate and incubate for evaluation in the next period. Consult Exercises 52 and 53 for instructions.
7. Refrigerate the positive RDS and Kligler iron slants for future use, if needed.

FOURTH PERIOD

(Final Evaluation)

During this last period the tubes of SIM medium, urea broth, and any miniaturized multitest media from the last period will be evaluated. Serotyping can also be performed, if desired.

Materials:

tubes of urea broth and SIM medium from previous period
Kovacs' reagent and chloroform
5 ml pipettes
miniaturized multitest media from previous period
serological testing materials (optional)

1. Examine the tubes of SIM medium, checking for motility and H₂S production. If you see **cloudiness** spreading from the point of inoculation, the organism is motile. A **black precipitate** will be evidence of H₂S production.
2. Test for indole production by pipetting 2 ml of chloroform into each SIM tube and then adding 2 ml of Kovacs' reagent. A **pink to deep red color** will form in the chloroform layer if indole is produced.
Salmonella are negative. Some *Shigella* may be positive. *Citrobacter* and *Escherichia* are positive.
3. Examine the urea broth tubes. If the medium has changed from yellow to **red** or **cerise color**, the organism is urease-positive.
4. If a miniaturized multitest media was inoculated in the last period, complete them now.
5. If time and materials are available, confirm the identification of your unknown with serological typing. Refer to Exercise 80.

LABORATORY REPORT

Record the identity of your unknown on the Laboratory Report and answer all the questions.

81

Urinary Tract Pathogens

Chronic or acute infections of the urinary tract may involve the kidneys, ureters, bladder, or urethra. Such infections may cause high blood pressure, kidney damage, uremia, or death. In some instances the infections are inapparent and may go unnoticed for some time. Most infections of this tract enter by way of the urethra; very few originate in the blood.

A multitude of organisms can cause urinary infections. The most common cause of such infections in women of childbearing age is *Escherichia coli*. In order of frequency after *E. coli* are other members of the Enterobacteriaceae, *Pseudomonas aeruginosa* and *Staphylococcus* species.

The importance of performing microbial analyses of urine on patients with urinary infections cannot be overemphasized. Some physicians tend to treat patients with antimicrobics and watch for symptomatic improvement without performing follow-up urinary tests, but this practice is not reliable. Clinical testing of urine 48 to 78 hours after the start of chemotherapy should be performed to evaluate the effectiveness of the therapy. If the antimicrobics are effective, the urine will be free of bacteria at this time.

A thorough microbial analysis of urine from a patient with urinary distress should be both quantitative and qualitative. The steps are as follows:

- Collect a urine sample as aseptically as possible.
- Do a plate count to determine the presence or absence of infection.
- Isolate the pathogen, if an infection is known to be present.
- Make a presumptive identification of the pathogen.
- Do an antimicrobial sensitivity test.

Except for antimicrobial testing, all of the above steps will be addressed in this laboratory exercise. Note that there are two parts to this experiment. The first portion pertains to doing a **plate count**. Note, also, that in figures 81.1 and 81.2 two different methods are available for making the inoculations. Your instructor will indicate which method will be used.

The second portion of this exercise is concerned with the protocol that one can follow to identify the genus of a pathogen that might be causing a urinary infection. Figure 81.3 depicts the routine that we will use to make this determination. Completion of this second portion of the exercise will yield only a **pre-**

sumptive identification of a pathogen. Further physiological tests, which are not included in this exercise, would be necessary to make species identification.

Aseptic Collection of Urine

Since the urethra in all individuals contains some bacteria, especially near its external orifice, the mere presence of bacteria in urine does not necessarily indicate that a urinary infection exists. One might assume that aseptic collection can be achieved with a catheter. However, collection by catheterization is, generally, not desirable because bacteria may be dislodged in the urethra, and there is the danger of causing an infection with this procedure.

Meaningful results, however, can be obtained with *midstream voided specimens* collected in sterile containers. For best results, the external genitalia should be cleansed with liquid soap containing chlorhexidine. Even with midstream samples, however, one can expect to find low counts of the following contaminants in normal urine: coagulase-negative staphylococci, diphtheroid bacilli, enterococci, *Proteus*, hemolytic streptococci, yeasts, and aerobic gram-positive spore-forming rods.

Specimens are most reliable when plated out immediately after collection. If bacterial tests cannot be performed immediately, refrigeration is mandatory. It must be kept in mind that urine is an ideal growth medium for many bacteria. Specimens not properly refrigerated should be considered unsatisfactory for study.

THE PLATE COUNT

Before attempting to isolate pathogens from a urine sample, one should determine, first of all, that an infection actually exists. Since normal urine always contains some bacteria, yeasts, and other organisms, we need to know if there is *an excess number of organisms* present due to an infection somewhere in the urinary tract. The best way to make this evaluation is to do a plate count to determine the number of organisms per ml that are present.

Generally speaking, if a urine sample contains 100,000 or more organisms per ml, one may assume that *significant bacteriuria* exists. In some instances, however, urine from completely normal individuals

Urinary Tract Pathogens • Exercise 81

may exceed these numbers. It is also possible for counts between 1,000 and 100,000 to be significant. Thus, it is apparent that precise evaluation of plate counts must take into consideration other factors. The clinician, aware of the effects of certain variables, will subjectively evaluate the results. Our purpose here in this experiment is not to interpret, but simply to become familiar with the basic procedures.

Proceed to inoculate two plates of trypticase soy agar with inocula from a urine sample. After 24 hours incubation, the colonies will be counted on the best plate. Your instructor will indicate whether Method A or Method B will be used.

Method A: Using Calibrated Inoculating Loops

Note in figure 81.1 that calibrated loops are used to inoculate tubes of TSA. After the poured plates have been incubated for 24 hours, counts will be made. Proceed as follows:

Materials:

First period:

- urine sample
- 1 sterile empty shake bottle
- 2 sterile Petri plates
- 2 trypticase soy agar pours
- calibrated wire loops (0.01 μm and 0.001 μm)

Second period:

- plates from previous period
- Quebec colony counter
- mechanical hand counter

1. Liquefy two TSA pours and cool to 50° C.

2. Label one Petri plate 1:100 and the other 1:1000.
3. Pour the urine into an empty sterile shake bottle, cap it tightly, and shake 25 times, as in figure 23.3, page 95.
4. With a 0.01 calibrated sterile loop, transfer 1 loopful to one of the pours, mix by rolling the tube between both palms, and pour into the 1:100 plate. Be sure to flame the neck of the tube before pouring into the plate.
5. Repeat step 4 using the 0.001 calibrated loop and the other TSA pour. Pour into the 1:1000 plate.
6. Incubate the plates at 37° C for 24 hours.
7. After incubation, select the plate that contains between 30 and 300 colonies. Count all colonies on a Quebec colony counter, using a mechanical hand counter to tally.
8. Record your count and number of organisms per ml on the Laboratory Report.

Method B: Using Pipettes

This method differs from Method A in that pipettes are used instead of calibrated loops for making the dilutions. Proceed as follows:

Materials:

First period:

- urine sample
- 1 99 ml sterile water blank
- 1 sterile empty shake bottle
- 2 sterile Petri plates
- 2 trypticase soy agar pours
- 2 1.1 ml dilution pipettes
- mechanical pipetting device

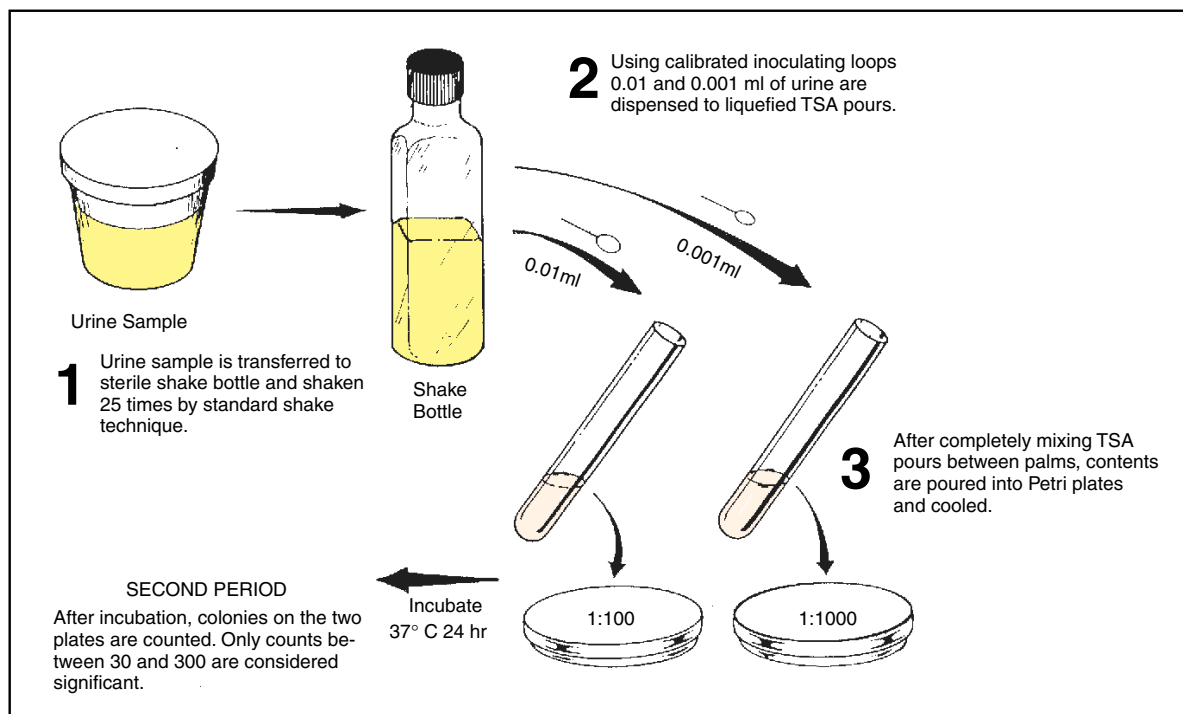


Figure 81.1 Method A procedure for doing a plate count

Exercise 81 • Urinary Tract Pathogens

Second period:

plates from previous period
Quebec colony counter
mechanical hand counter

1. Liquefy two TSA pours and cool to 50° C.
2. Label one Petri plate 1:100 and the other 1:1000.
3. Pour the urine into an empty sterile shake bottle, cap it tightly, and shake 25 times, as in figure 23.3, page 95.
4. Transfer 1 ml of the mixed urine to a 99 ml sterile water blank. Use a mechanical delivery device with the pipette.
5. Mix the water blank with 25 shakes and, with a fresh pipette, transfer 0.1 ml to the 1:1000 plate and 1.0 ml to the 1:100 plate.
6. Empty the tubes of TSA into the plates, swirl them, and let stand to cool.
7. Incubate the plates at 37° C for 24 hours.
8. After incubation, select the plate that contains between 30 and 300 colonies. Count all colonies on a Quebec colony counter, using a mechanical hand counter to tally.
9. Record your count and number of organisms per ml on the Laboratory Report.

PRESUMPTIVE IDENTIFICATION

Once it is established that an infection exists, the next step is to isolate the pathogen and identify it. Figure 81.3 illustrates the overall procedure for making a presumptive identification of the genus of a urinary pathogen.

The minimum number of laboratory periods required to arrive at a presumptive identification is two; however, if one wishes to be more explicit in identifying the unknown, a total of three or four periods will be required.

First Period

Note in figure 81.3 that two things will be done during this period with a concentrated sample of urine: (1) two microscope slides will be made for direct examination, and (2) four kinds of media will be inoculated. Proceed as follows:

Materials:

- 1 sterile centrifuge tube (with screw cap)
- 1 tube of thioglycollate medium (BBL135C)
- 1 plate of blood agar (TSA base)
- 1 plate of desoxycholate lactose agar (DLA)
- 1 plate of phenylethyl alcohol medium with blood (PEA-B)

1. Shake the sample to resuspend the organisms and decant 10 ml into a centrifuge tube. Keep the tube capped.

CAUTION

Be sure to balance centrifuge by placing a capped tube with 10 ml of water opposite your urine sample tube.

2. Centrifuge for 10 minutes at 2,000 rpm.
3. Decant all but 0.5 ml from the tube and resuspend the sediment with a sterile wire loop.

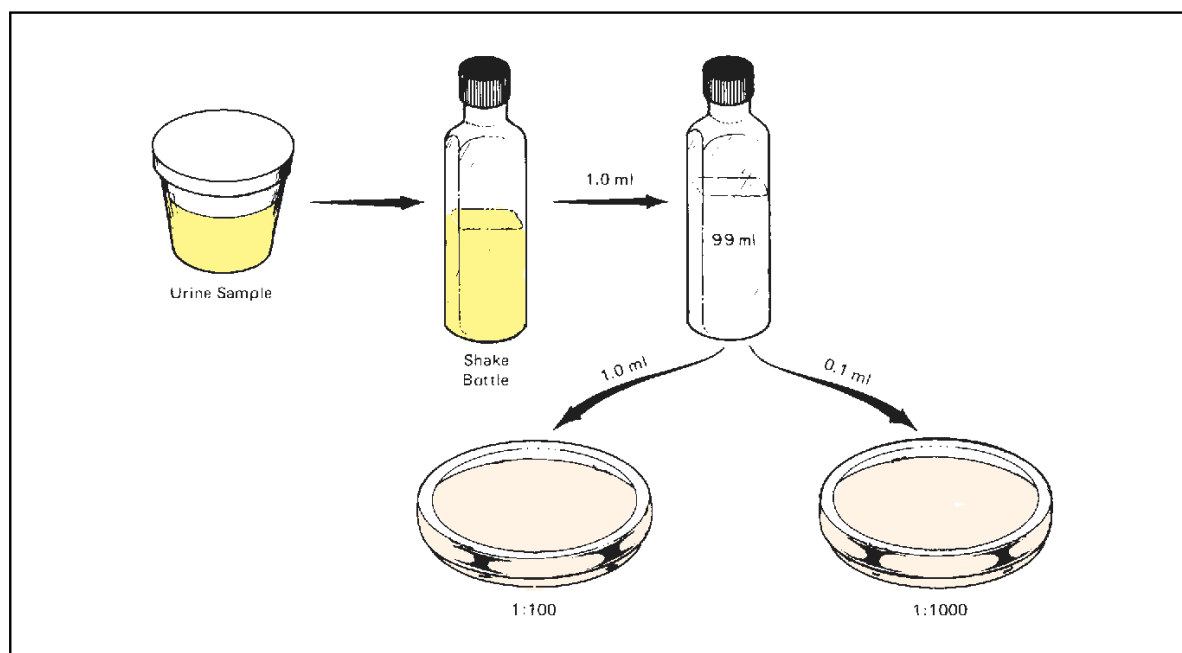


Figure 81.2 Method B procedure for doing a plate count

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4. Inoculate a tube of thioglycollate medium with a loopful of the sediment.
5. Streak out a loopful of the sediment on each of the three agar plates (blood agar, DLA, and PEA-B). Use a good isolation technique.
6. Incubate the thioglycollate tube and three plates at 37° C for 18 to 24 hours.
7. Make a wet mount slide from material in the bottom of the centrifuge tube and examine under high-dry, preferably with phase optics. Look for casts, pus cells, and other elements.

Refer to figure 81.4 for help in identifying objects that are present. Normal urine will contain an occasional leukocyte, some epithelial cells, mucus, bacteria, and crystals of various kinds.

8. Make a gram-stained slide and examine it under oil immersion. Determine the morphology and staining reaction of the predominant organism. Record your observations on the Laboratory Report.

Second Period

After the plates have been incubated for 18 to 24 hours, lay them out and evaluate them according to the characteristics of each medium.

Thioglycollate Medium This medium is inoculated to promote the growth of organisms that are not present in large numbers or are too fastidious to grow readily in nutrient broth. In the event that none of the plates produce colonies from the urine of a patient known to have a urinary infection, this tube can be used for reinoculation or to provide information pertaining to growth characteristics.

Blood Agar Practically all pathogens of the urinary tract will grow on this medium. This includes the coliforms, *Proteus*, *Pseudomonas*, *Candida*, staphylococci, streptococci, and others.

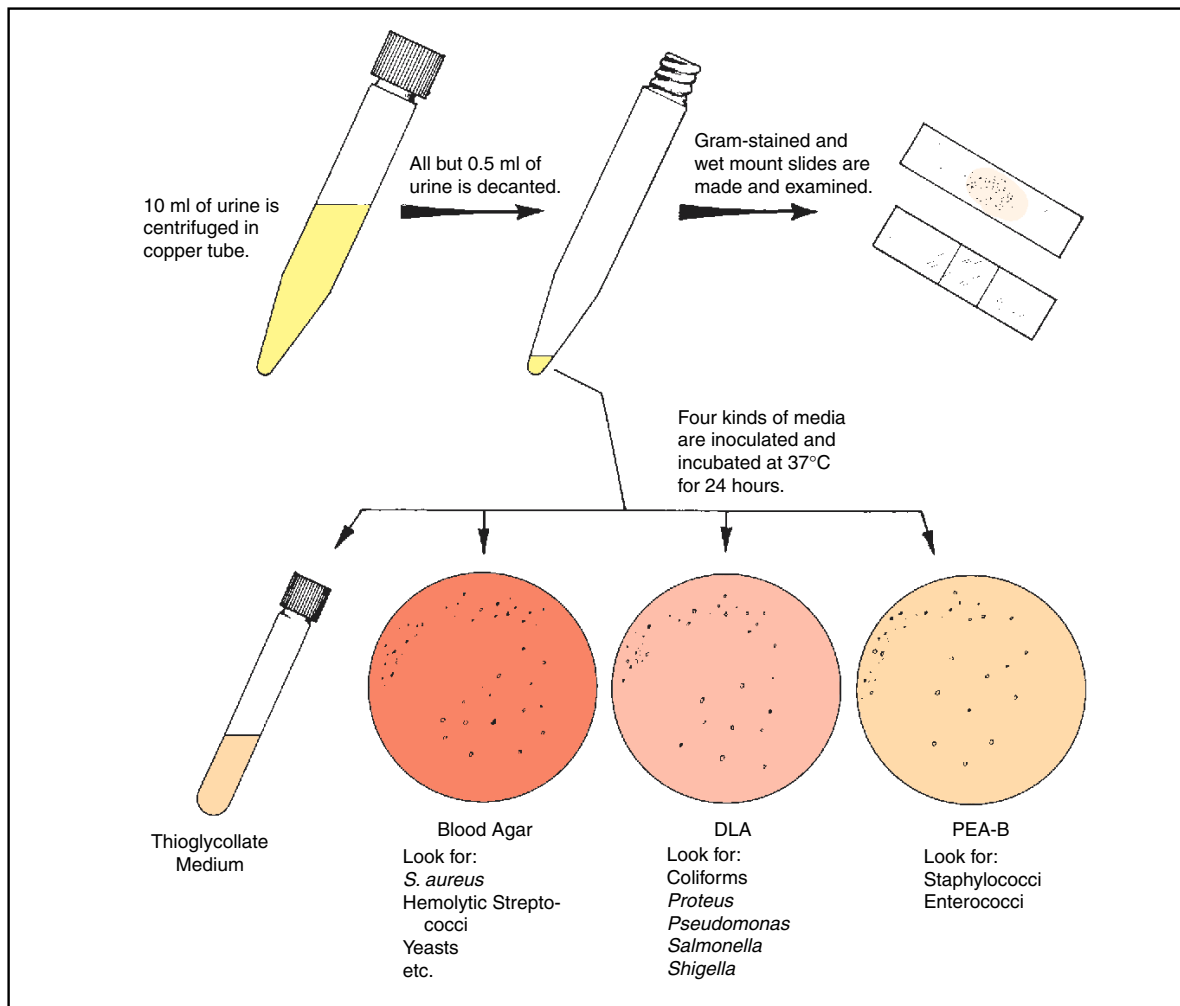


Figure 81.3 Procedure for presumptive identification of urinary pathogens

Exercise 81 • Urinary Tract Pathogens

Subculturing from this plate to trypticase soy broth can provide pure cultures of the pathogen for physiological testing or antimicrobial sensitivity testing.

The presence or absence of hemolytic activity can also be determined at this time. If the pathogen appears to be a hemolytic, gram-positive coccus, one should follow the procedures outlined in Exercises 78 and 79 for identification.

Desoxycholate Lactose Agar The presence of sodium desoxycholate and sodium citrate in this medium is inhibitory to gram-positive bacteria. Coliforms and other gram-negative bacteria grow well on it.

If the predominant organism is gram-negative, some differentiation may be made at this point. If the colonies on this medium are flat and rose-red in color, the organism is *E. coli*. *Pseudomonas* and *Proteus*, which do not ferment the lactose in the medium, produce white colonies. *Proteus* can be confirmed with the urease test, being positive for urease production.

Pseudomonas species give a positive reaction with Taxo N disks. Fermentation and additional phys-

iological testing may be necessary for species identification. Exercises 48, 49, and 80 should be consulted for further testing.

Phenylethyl Alcohol Medium This medium, to which blood has been added, is highly inhibitory to gram-negative organisms. *Proteus*, in particular, is prevented from growing on it. If considerable growth occurs on the DLA plate, and very little or no growth here, then one can assume that the disease is due to a gram-negative organism. This, of course, would be confirmed by the findings on the gram-stained slide.

The findings on this plate should be correlated with those on blood agar. If enterococci (*S. faecalis*) are suspected, a plate of Mead agar should be streaked and incubated at 37° C. Enterococci produce pink colonies on this medium.

LABORATORY REPORT

Record all findings on the Laboratory Report.

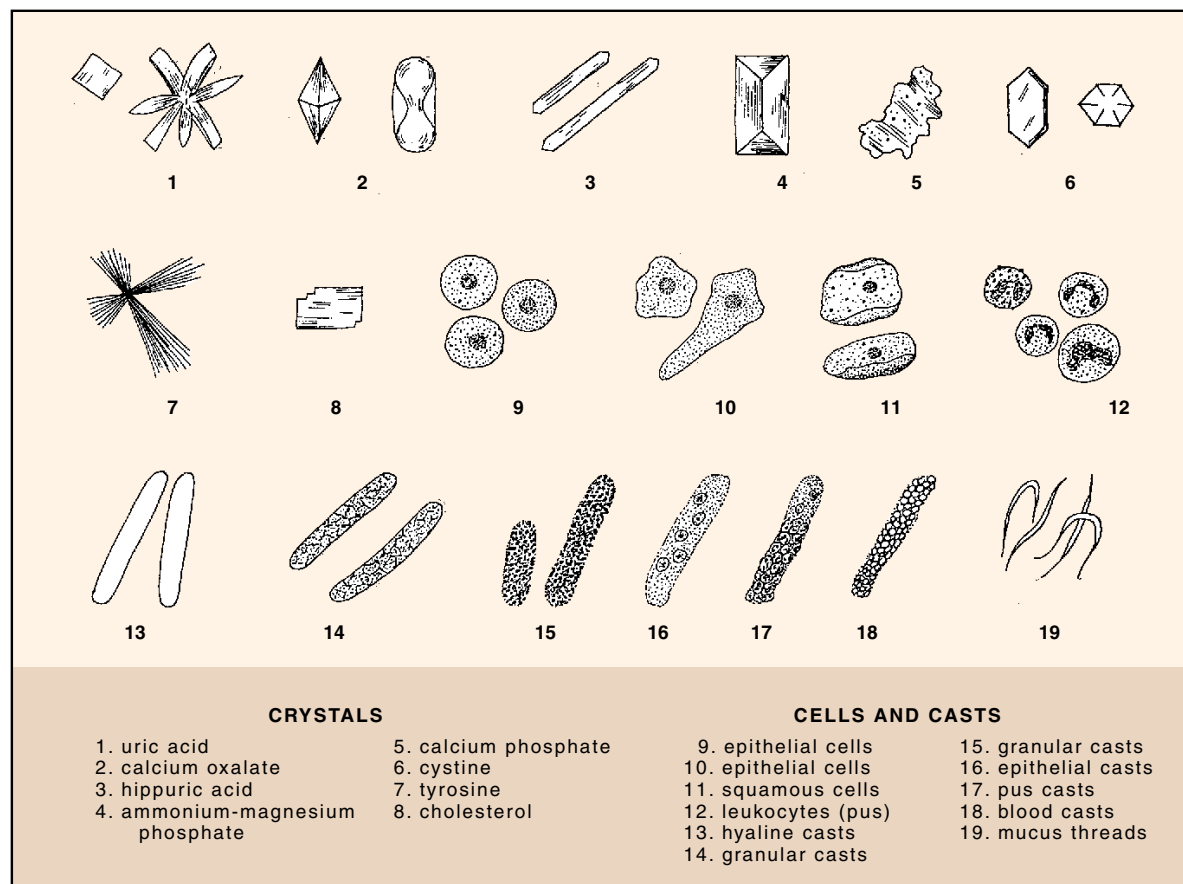


Figure 81.4 Microscopic elements in urine

Slide Agglutination Test: Serological Typing

82

Organisms of different species not only differ morphologically and physiologically, but they also differ in protein makeup. The different proteins of bacterial cells that are able to stimulate antibody production when injected into an animal are **antigens**. The antigenic structure of each species of bacteria is unique to that species and, like the fingerprint of an individual, can be used to identify the organism. Many closely related microorganisms that are identical physiologically can be differentiated only by determining their antigenic nature.

The method of determining the presence of specific antigens in a microorganism is called **serological typing** (serotyping). It consists of adding a suspension of the organisms to **antiserum**, which contains antibodies that are specific for the known antigens. If the antigens are present, the antibodies in the antiserum will combine with the antigens, causing **agglutination**, or clumping, of the bacterial cells. Serotyping is particularly useful in the identification of various organisms that cause salmonella and shigella infections. In the identification of the various serotypes of these two genera, the use of antisera is generally performed after basic biochemical tests have been utilized as in Exercise 80.

In this exercise you will be issued two unknown organisms, one of which is a salmonella. By following the procedure shown in figure 82.1, you will determine which one of the unknowns is salmonella. Note that you will use two test controls. A **negative test control** will be set up in depression A on the slide to see what the absence of agglutination looks like. The negative control is a mixture of antigen and saline (antibody is lacking). A **positive test control** will be performed in depression C with standardized antigen and antiserum to give you a typical reaction of agglutination.

Materials:

- 2 numbered unknowns per student (slant cultures of a salmonella and a coliform)
- salmonella O antigen, group B (Difco #2840-56)
- salmonella O antiserum, poly A-I (Difco #2264-47)
- depression slides or spot plates

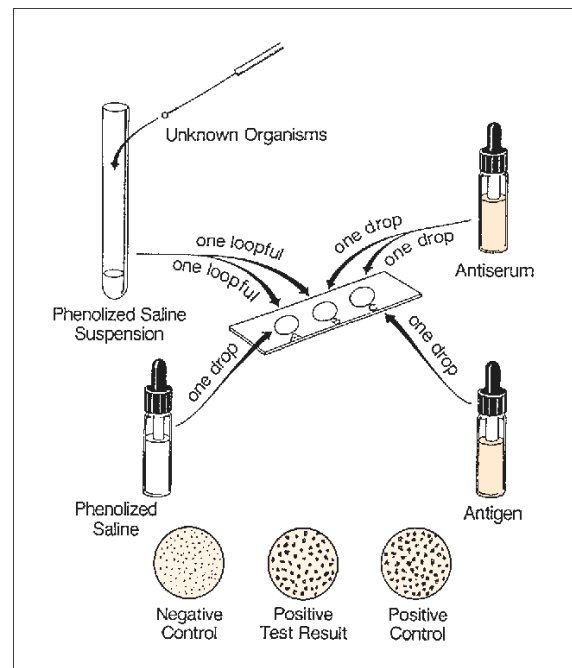


Figure 82.1 Slide agglutination technique

dropping bottle of phenolized saline solution
(0.85% sodium chloride, 0.5% phenol)
2 serological tubes per student
1-ml pipettes

CAUTION

Keep in mind that *Salmonella typhimurium* is a pathogen and can cause gastroenteritis. Be careful!

1. Label three depressions on a spot plate or depression slide **A**, **B**, and **C**, as shown in figure 82.1.
2. Make a phenolized saline suspension of each unknown in separate serological tubes by suspending one or more loopfuls of organisms in 1 ml of

Exercise 82 • Slide Agglutination Test: Serological Typing

phenolized saline. Mix the organisms sufficiently to ensure complete dispersion of clumps of bacteria. The mixture should be very turbid.

3. Transfer 1 loopful (0.05 ml) from the phenolized saline suspension of one tube to depressions A and B.
4. To depressions B and C add 1 drop of salmonella O polyvalent antiserum. To depression A, add 1 drop of phenolized saline, and to depression C, add 1 drop of salmonella O antigen, group B.
5. Mix the organisms in each depression with a clean wire loop. Do not go from one depression to the other without washing the loop first.
6. Compare the three mixtures. Agglutination should occur in depression C (positive control),

but not in depression A (negative control). If agglutination occurs in depression B, the organism is salmonella.

7. Repeat this process on another slide for the other organism.

CAUTION

Deposit all slides and serological tubes in container of disinfectant provided by the instructor.

LABORATORY REPORT

Record your results on the first portion of Laboratory Report 82, 83.

Slide Agglutination (Latex) Test: For *S. aureus* Identification

83

Many manufacturers of reagents for slide agglutination tests utilize polystyrene latex particles as carriers for the antibody particles. By adsorbing reactive antibody units to these particles, an agglutination reaction results that occurs rapidly and is much easier to see than ordinary precipitin type reactions that might be used to demonstrate the presence of a soluble antigen.

In this exercise we will use reagents manufactured by Difco Laboratories to determine if a suspected staphylococcus organism produces coagulase and/or protein A. The test reagent (*Difco Staph Reagent*) is a suspension of yellow latex particles sensitized with antibodies for coagulase and protein A. Reagents are also included to provide positive and negative controls in the test. Instead of using depression slides or spot plates, Difco provides disposable cards with eight black circles printed on them for performing the test. As indicated in figure 83.1, only three circles are used when performing the test on one unknown. The additional circles are provided for testing five additional unknowns at the same time. The black background of the cards facilitates rapid interpretation by providing good contrast for the yellow clumps that form.

There are two versions of this test: direct and indirect. The procedure for the direct method is illustrated in figure 83.1. The indirect method differs in that saline is used to suspend the organism being tested.

It should be pointed out that the reliability correlation between this test for coagulase and the tube test (page 260) is very high. Studies reveal that a reliability correlation of over 97% exists. Proceed as follows to perform this test.

forming the test. As indicated in figure 83.1, only three circles are used when performing the test on one unknown. The additional circles are provided for testing five additional unknowns at the same time. The black background of the cards facilitates rapid interpretation by providing good contrast for the yellow clumps that form.

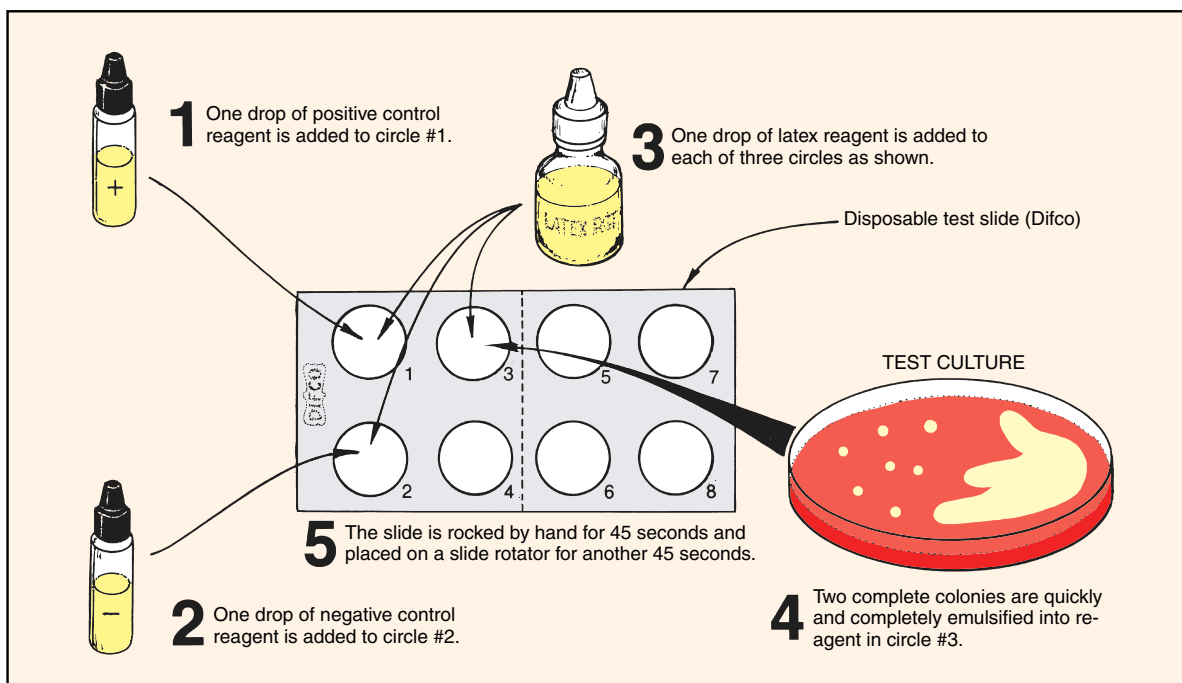


Figure 83.1 Slide agglutination test (direct method) for the presence of coagulase and/or protein A

Exercise 83 • Slide Agglutination (Latex) Test: For *S. aureus* Identification**Materials:**

plate culture of staphylococcus-like organism
(trypticase soy agar plus blood)
Difco Staph Latex Test kit #3850-32-7, which
consists of:
bottle of Bacto Staph Latex Reagent
bottle of Bacto Staph Positive Control
bottle of Bacto Staph Negative Control
bottle of Bacto Normal Saline Reagent
disposable test slides (black circle cards)
mixing sticks (minimum of 3)
slide rotator

DIRECT METHOD

If the direct method is to be used, as illustrated in figure 83.1, follow this procedure:

1. Place 1 drop of Bacto Staph Positive Control reagent onto circle #1.
2. Place 1 drop of Bacto Staph Negative Control reagent on circle #2.
3. Place 1 drop of Bacto Staph Latex Reagent onto circles #1, #2, and #3.
4. Using a sterile inoculating needle or loop, quickly and completely emulsify *two isolated colonies* from the culture to be tested into the drop of Staph Latex Reagent in circle #3.

Also, emulsify the Staph Latex Reagent in the positive and negative controls in circles #1 and #2 using separate mixing sticks supplied in the kit.

All mixing in these three circles should be done quickly to minimize drying of the latex on the slide and to avoid extended reaction times for the first cultures emulsified.

5. Rock the slide by hand for 45 seconds.
6. Place the slide on a slide rotator capable of providing 110 to 120 rpm and rotate it for another 45 seconds.
7. Read the results immediately, according to the descriptions provided in the table at right. If agglutination occurs before 45 seconds, the results may be read at that time. *The slide should be read at normal reading distance under ambient light.*

INDIRECT METHOD

The only differences between the direct and indirect methods pertain to the amount of inoculum and the use of saline to emulsify the unknown being tested. Proceed as follows:

1. Place 1 drop of Bacto Staph Positive Control reagent onto test circle #1.
2. Place 1 drop of Bacto Staph Negative Control onto circle #2.
3. Place 1 drop of Bacto Normal Saline Reagent onto circle #3.
4. Using a sterile inoculating needle or loop, completely emulsify *four isolated colonies* from the culture to be tested into the circle containing the drop of saline (circle #3).
5. Add 1 drop of Bacto Staph Latex Reagent to each of the three circles.
6. Quickly mix the contents of each circle, using individual mixing sticks.
7. Rock the slide by hand for 45 seconds.
8. Place the slide on a slide rotator capable of providing 110 to 120 rpm and rotate it for another 45 seconds.
9. Read the results immediately according to the descriptions provided in the table below. If agglutination occurs before 45 seconds, the results may be read at that time. *The slide should be read at normal reading distance under ambient light.*

POSITIVE REACTIONS

- | | |
|-----|---|
| 4 + | Large to small clumps of aggregated yellow latex beads; clear background |
| 3 + | Large to small clumps of aggregated yellow latex beads; slightly cloudy background |
| 2 + | Medium to small but clearly visible clumps of aggregated yellow latex beads; moderately cloudy background |
| 1 + | Fine clumps of aggregated yellow latex beads; cloudy background |

NEGATIVE REACTIONS

- | | |
|---|--|
| + | Smooth cloudy suspension; particulate grainy appearance that cannot be identified as agglutination |
| – | Smooth, cloudy suspension; free of agglutination or particles |

LABORATORY REPORT

Record your results on the last portion of Laboratory Report 82, 83.

Tube Agglutination Test: The Heterophile Antibody Test

84

Infectious mononucleosis (IM) is a benign disease, occurring principally in individuals in the 13 to 25 year age group. It is caused by the Epstein-Barr virus (EBV), a herpesvirus, that is one of the most ubiquitous viruses in humans. Studies have shown that the virus can be isolated from saliva of patients with IM, as well as from some healthy, asymptomatic individuals. Between 80% and 90% of all adults possess antibodies for EBV.

The disease is characterized by a sudden onset of fever, sore throat, and pronounced enlargement of the cervical lymph nodes. There is also moderate leukocytosis with a marked increase in the number of lymphocytes (50% to 90%).

The serological test for IM takes advantage of an unusual property: the antibodies produced against the EBV coincidentally agglutinate sheep red blood cells. This is an example of a **heterophile antigen**—a sub-

stance isolated from a living form that stimulates the production of antibodies capable of reacting with tissues of other organisms. The antibodies are referred to as **heterophile antibodies**.

This test is performed by adding a suspension of sheep red blood cells to dilutions of inactivated patient's serum and incubating the tubes overnight in the refrigerator. Figure 84.1 illustrates the overall procedure. Agglutination titers of 320 or higher are considered significant. Titers of 40,960 have been obtained.

Proceed as follows to perform this test on a sample of test serum:

FIRST PERIOD

Materials:

test-tube rack (Wasserman type) with 10 clean serological tubes

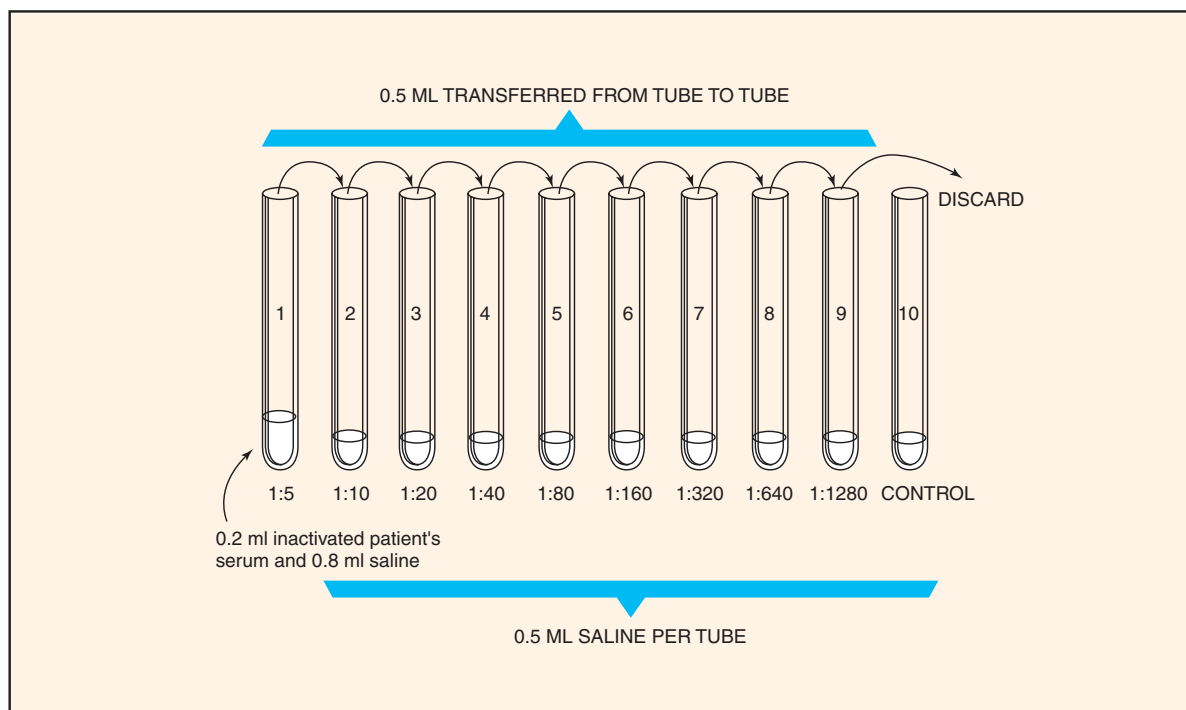


Figure 84.1 Procedure for setting up heterophile antibody test

Exercise 84 • Tube Agglutination Test: The Heterophile Antibody Test

bottle of saline solution (0.85% NaCl), clear or filtered

1 ml pipettes

5 ml pipettes

2% suspension of sheep red blood cells

patient's serum (known to be positive)

1. Place the test serum in a 56° C water bath for 30 minutes to inactivate the complement.
2. Set up a row of 10 serological tubes in the front row of a test-tube rack and number them from 1 to 10 (left to right) with a marking pencil.
3. Into tube 1, pipette 0.8 ml of physiological saline.
4. Dispense 0.5 ml of physiological saline to tubes 2 through 10. Use a 5 ml pipette.
5. With a 1 ml pipette add 0.2 ml of the inactivated serum to tube 1. Mix the contents of this tube by drawing into the pipette and expelling about five times.
6. Transfer 0.5 ml from tube 1 to tube 2, mix five times, and transfer 0.5 ml from tube 2 to tube 3, etc., through the ninth tube. *Discard 0.5 ml from the ninth tube after mixing.* Tube 10 is the **control**.

7. Add 0.2 ml of 2% sheep red blood cells to all tubes (1 through 10) and shake the tubes. Final dilutions of the serum are shown in figure 84.1.
8. Allow the rack of tubes to stand at room temperature for 1 hour, then transfer the tubes to a small wire basket, and place in a refrigerator to remain overnight.

SECOND PERIOD

Set up the tubes in a tube rack in order of dilution and compare each tube with the control by holding the tubes overhead and looking up at the bottoms of the tubes. Nonagglutinated cells will tumble to the bottom of the tube and form a small button (as in control tube). Agglutinated cells will form a more-amorphous "blanket."

The **titer** should be recorded as the reciprocal of the last tube in the series that shows positive agglutination.

LABORATORY REPORT

Complete the first portion of Laboratory Report 84, 85.

Tube Agglutination Test: The Widal Test

85

A tube test for determining the quantity of agglutinating antibodies, or **agglutinins**, in the serum of a patient with typhoid fever was described by Grunbaum and Widal in 1896. This technique is still in use today and has been adapted to many other diseases as well.

The procedure involves adding a suspension of dead typhoid bacterial cells to a series of tubes containing the patient's serum, which has been diluted out to various concentrations. After the tubes have been incubated for 30 minutes at 37° C, they are centrifuged and examined to note the amount of agglutination that has occurred.

The reciprocal of the highest dilution at which agglutination is seen is designated as the **antibody titer** of the patient's serum. For example, if the highest dilution at which agglutination occurs is 1:320, the titer is 320 antibody units per milliliter of serum. Naturally, the higher the titer, the greater is the antibody response of the individual to the disease.

This technique can be used clinically to determine whether a patient with typhoidlike symptoms

actually has the disease. If successive daily tests on a patient's serum reveal no antibody titer, or a low titer that does not increase from day to day, it can be assumed that some other disease is present. On the other hand, if one sees a daily increase in the titer, it can be assumed that a typhoid infection does exist. Since the treatment of typhoid fever requires powerful antibiotics that are not widely used on other similar diseases, it is very important to diagnose this disease early to begin the proper form of chemotherapy as soon as possible.

In this exercise you will be given a sample of blood serum that is known to contain antibodies for the typhoid organism. By using the Widal tube agglutination method, you will determine the antibody titer.

Materials:

- test-tube rack (Wassermann type) with 10 clean serological tubes
- bottle of saline solution (0.85%), clear or filtered

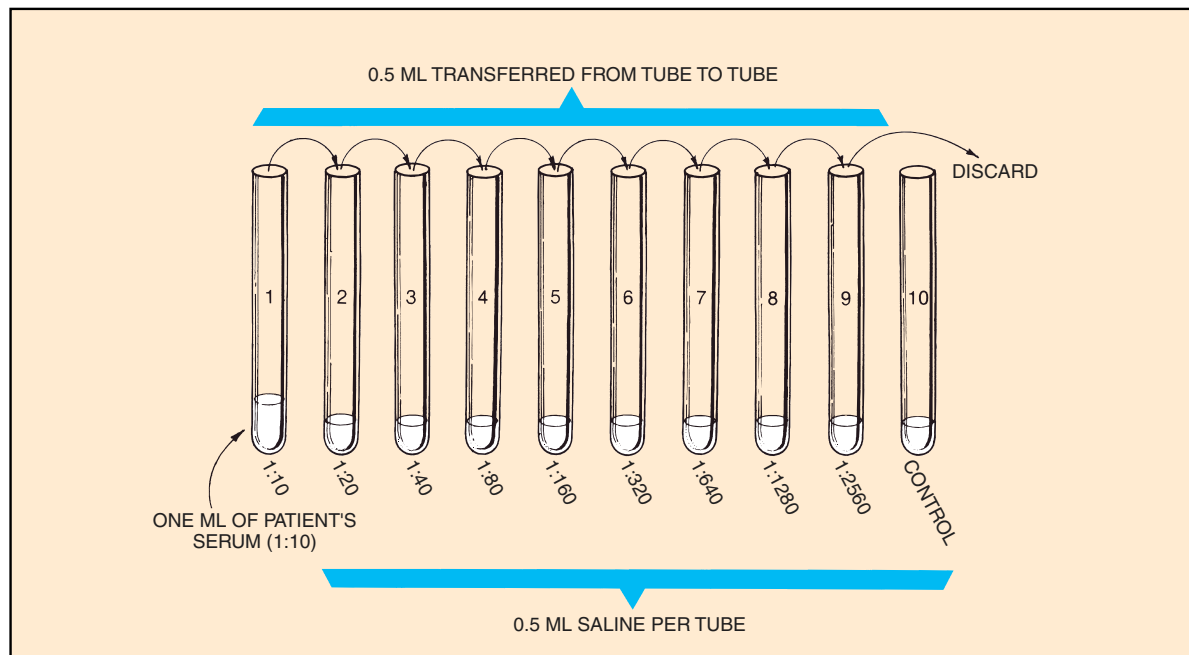


Figure 85.1 Procedure for dilution of serum

Exercise 85 • Tube Agglutination Test: The Widal Test

1 ml pipettes
5 ml pipettes
water bath at 37° C
centrifuges
antigen (1:10 dilution) *Salmonella typhi* “O”
patient’s serum (1:10 dilution), known
positive “O”

1. Dilute the patient’s serum as shown in figure 85.1. Follow this routine:
 - a. Set up 10 clean serological test tubes in the front row of a test-tube rack and number them from 1 to 10 (left to right) with a marking pencil.
 - b. Into tube 1 pipette 1 ml of the patient’s serum (1:10 dilution). *For convenience, the instructor may wish to dispense this material to each student.*
 - c. With a 5 ml pipette, dispense 0.5 ml of saline to each of the remaining nine tubes.
 - d. With a 1 ml pipette, transfer 0.5 ml of the serum from tube 1 to tube 2. Mix the serum and saline in tube 2 by carefully drawing the liquid up into the pipette and discharging it slowly back down into the tube a minimum of three times.
 - e. Repeat this process by transferring 0.5 ml from tube 2 to 3, tube 3 to 4, 4 to 5, etc. *When you*

*get to tube 9, discard the 0.5 ml drawn from it instead of adding it to tube 10; thus, tube 10 will contain only saline and can be used as a **negative test control** for comparing with the other tubes.*

2. With a fresh 5 ml pipette, transfer 0.5 ml of antigen to each tube. Shake the rack to completely mix the antigen and diluted serum.
3. Place the rack in a water bath at 37° C for 30 minutes.
4. Centrifuge all tubes for 3 minutes at 2,000 rpm. (If time permits, 7 minutes centrifugation is preferable.)
5. Examine each tube for agglutination and record the titer as the reciprocal of the highest dilution in which agglutination is seen.

When examining each tube, jar it first by rapping the side of the tube with a snap of the finger to suspend the clumps of agglutinated cells. Hold it up against the light of a desk lamp in the manner shown in figure 85.2. Do not look directly into the light. The reflection of the light off the particles is best seen against a dark background. Compare each tube with tube 10, which is your negative test control.

6. Record your results on the last portion of Laboratory Report 84, 85.

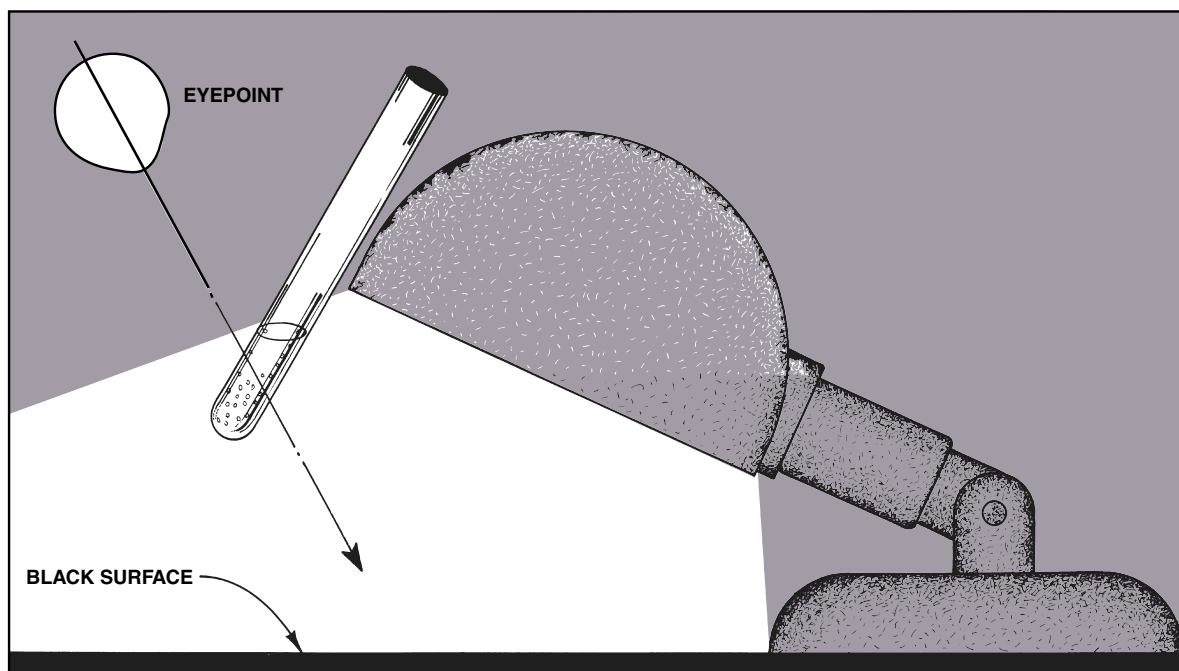


Figure 85.2 Agglutination is more readily seen when the tube is examined against a black surface

Phage Typing

86

The host specificity of bacteriophage is such that it is possible to delineate different strains of individual species of bacteria on the basis of their susceptibility to various kinds of bacteriophage. In epidemiological studies, where it is important to discover the source of a specific infection, determining the phage type of the causative organism can be an important tool in solving the riddle. For example, if it can be shown that the phage type of *S. typhi* in a patient with typhoid fever is the same as the phage type of an isolate from a suspected carrier, chances are excellent that the two cases are epidemiologically related. Since all bacteria are probably parasitized by bacteriophages, it is theoretically possible, through research, to classify each species into strains or groups according to their phage type susceptibility. This has been done for *Staphylococcus aureus*, *Salmonella typhi*, and several other pathogens. The following table illustrates the lytic groups of *S. aureus* as proposed by M. T. Parker.

Lytic Group	Phages in Group
I	29 52 52A 79 80
II	3A 3B 3C 55 71
III	6 7 42E 47 53 54 75 77 83A
IV	42D
not allotted	81 187

In bacteriophage typing, a suspension of the organism to be typed is swabbed over an agar surface. The bottom of the plate is marked off in squares and labeled to indicate which phage types are going to be used. To the organisms on the surface, a small drop of each phage type is added to their respective squares. After incubation, the plate is examined to see which

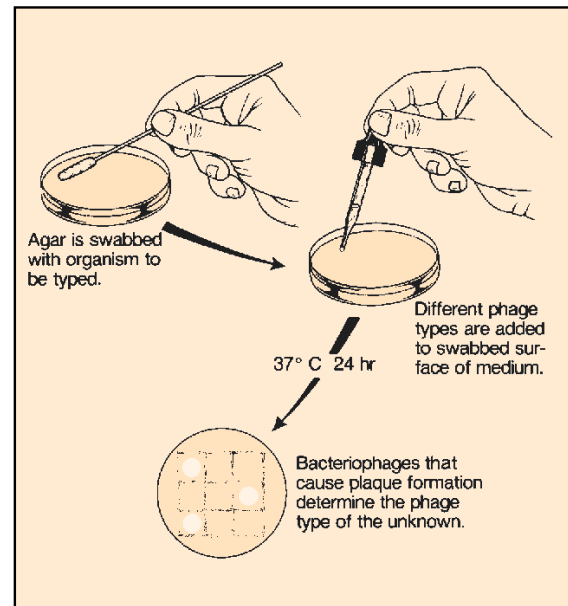


Figure 86.1 Bacteriophage typing

phages were able to lyse the organisms. This is the procedure to be used in this exercise. See figure 86.1.

Materials:

1 Petri plate of tryptone yeast extract agar
bacteriophage cultures (available types)
nutrient broth cultures of *S. aureus* with swabs

1. Mark the bottom of a plate of tryptone yeast extract agar with as many squares as there are phage types to be used. Label each square with the phage type numbers.
2. Swab the entire surface of the agar with the organisms.
3. Deposit 1 drop of each phage in its respective square.
4. Incubate the plate at 37° C for 24 hours and record the lytic group and phage type of the culture.
5. Record your results on the Laboratory Report.

87

White Blood Cell Study: The Differential WBC Count

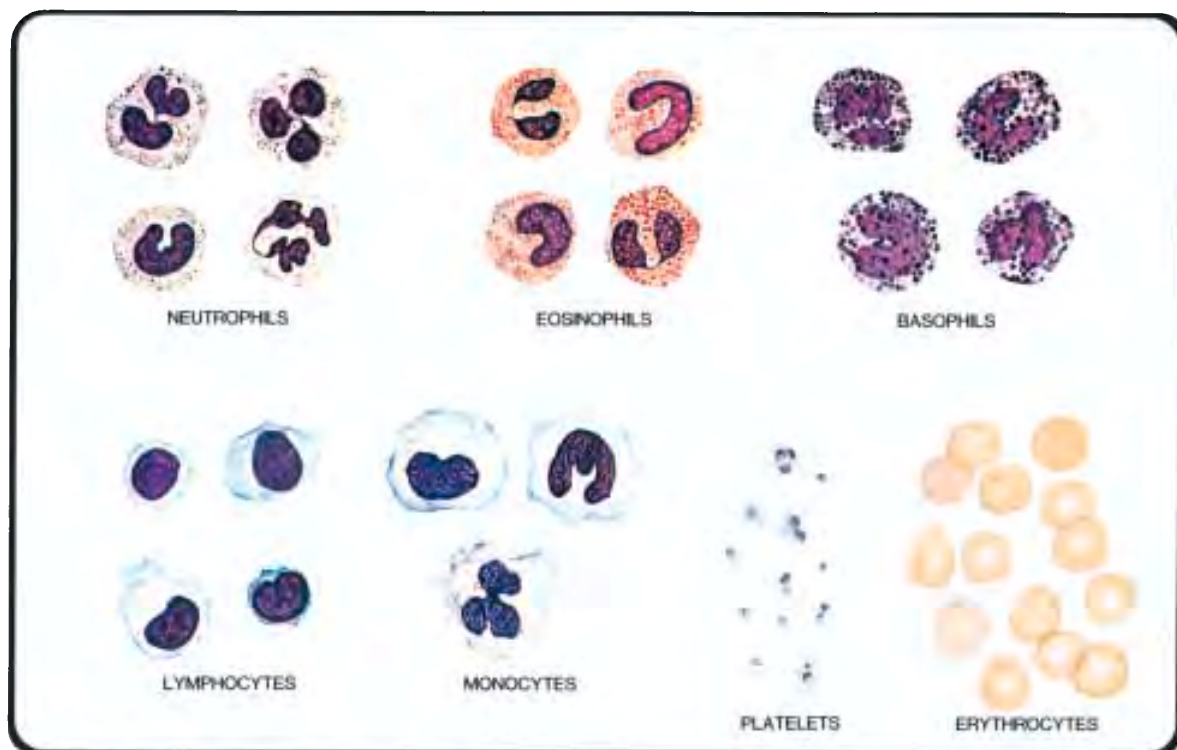
In 1883, at the Pasteur Institute in Paris, Metchnikoff published a paper proposing the **phagocytic theory of immunity**. On the basis of his studies performed on transparent starfish larvae, he postulated that amoeboid cells in the tissue fluid and blood of all animals are the major guardians of health against bacterial infection. He designated the large phagocytic cells of the blood as *macrophages* and the smaller ones as *microphages*. Today, Metchnikoff's macrophages are known as monocytes and his microphages as neutrophils or polymorphonuclear leukocytes.

Figure 87.1 illustrates the five types of leukocytes that are normally seen in the blood. Blood platelets and erythrocytes also are shown to present a complete picture of all formed elements in the blood. When observed as living cells under the microscope, they appear as refractile, colorless structures. As shown here,

however, they reflect the dyes that are imparted by Wright's stain.

In this exercise we will do a study of the white blood cells in human blood. This study may be made from a prepared stained microscope slide or from a slide made from your own blood. By scanning an entire slide and counting the various types, you will have an opportunity to encounter most, if not all, types. The erythrocytes and blood platelets will be ignored.

Figures 87.1 and 87.2 will be used to identify the various types of cells. Figure 87.3 illustrates the procedure for preparing a slide stained with Wright's stain. The relative percentages of each type will be determined after a total of 100 white blood cells have been identified. This method of white blood cell enumeration is called a **differential WBC count**.



K.P. Talaro

Figure 87.1 Formed elements of blood

White Blood Cell Study: The Differential WBC Count • Exercise 87

As you proceed with this count, it will become obvious that the neutrophils are most abundant (50%–70%). The next most prominent cells are the lymphocytes (20%–30%). Monocytes comprise about 2%–6%; eosinophils, 1%–5%; and basophils, less than 1%.

A normal white blood cell count is between 5,000 and 10,000 white cells per cubic millimeter. Elevated white blood cell counts are referred to as *leukocytosis*; counts of 30,000 or 40,000 represent marked leukocytosis. When counts fall considerably below 5,000, *leukopenia* is said to exist. Both conditions can have grave implications.

The value of a differential count is immeasurable in the diagnosis of infectious diseases. High neutrophil counts, or *neutrophilia*, often signal localized infections, such as appendicitis or abscesses in some other part of the body. *Neutropenia*, a condition in

which there is a marked decrease in the numbers of neutrophils, occurs in typhoid fever, undulant fever, and influenza. *Eosinophilia* may indicate allergic conditions or invasions by parasitic roundworms such as *Trichinella spiralis*, the “pork worm.” Counts of eosinophils may rise to as high as 50% in cases of trichinosis. High lymphocyte counts, or *lymphocytosis*, are present in whooping cough and some viral infections. Increased numbers of monocytes, or *monocytosis*, may indicate the presence of the Epstein-Barr virus, which causes infectious mononucleosis.

Note in the materials list that items needed for making a slide (option B) are listed separately. If a prepared slide (option A) is to be used, ignore the instructions under the heading “Preparation of Slide,” and proceed to the heading “Performing the Cell Count.” Your instructor will indicate which option will be used. Proceed as follows:

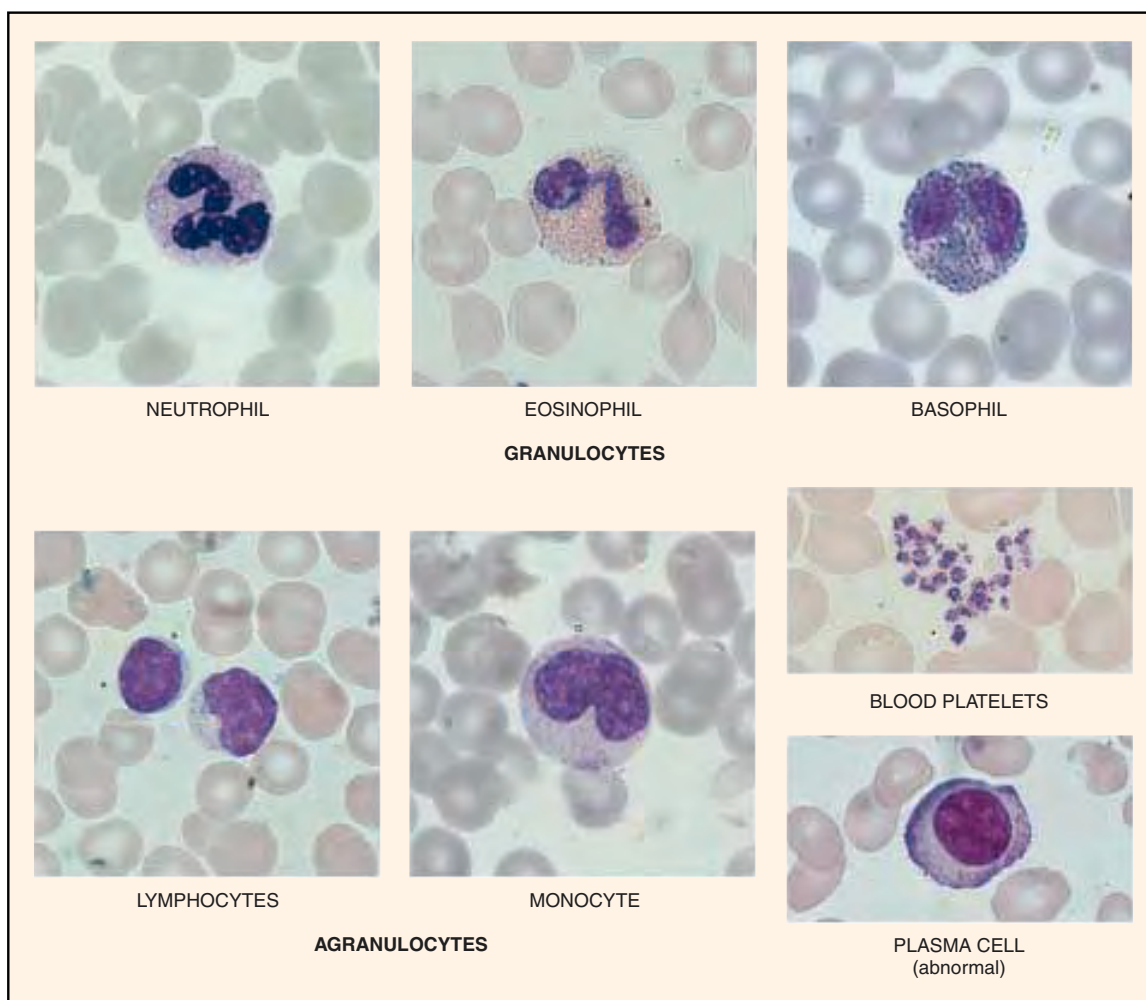


Figure 87.2 Photomicrographs of formed elements in blood

Exercise 87 • White Blood Cell Study: The Differential WBC Count

PRECAUTIONS

When working with blood observe the following precautions:

1. Always disinfect the finger with alcohol prior to piercing it.
2. Use sterile disposable lancets only one time.
3. Dispose of used lancets by placing them into a beaker of disinfectant.
4. Avoid skin contact with blood of other students. Wear disposable latex gloves.
5. Disinfect finger with alcohol after blood has been taken.

Materials:

prepared blood slide (option A):

stained with Wright's or Giemsa's stains

for staining a blood smear (option B):

2 or 3 clean microscope slides (should have polished edges)

sterile disposable lancets

disposable latex gloves

sterile absorbent cotton, 70% alcohol

Wright's stain, wax pencil, bibulous paper

distilled water in dropping bottle

PREPARATION OF SLIDE

Figure 87.3 illustrates the procedure that will be used to make a stained slide of a blood smear. The most difficult step in making such a slide is getting a good spread of the blood, which is thick at one end and thin at the other end. If done properly, the smear will have a gradient of cellular density that will make it possible to choose an area that is ideal for study. The angle at which the spreading slide is held in making the smear will determine the thickness of the smear. It may be necessary for you to make more than one slide to get an ideal one.

1. Clean three or four slides with soap and water. Handle them with care to avoid getting their flat surfaces soiled by your fingers. Although only two slides may be used, it is often necessary to repeat the spreading process, thus the extra slides.
2. Scrub the middle finger with 70% alcohol and stick it with a lancet. Put a drop of blood on the slide $\frac{3}{4}$ " from one end and spread with another slide in the manner illustrated in figure 87.3.

Note that the blood is dragged over the slide, not pushed. Do not pull the slide over the smear a second time. If you don't get an even smear the

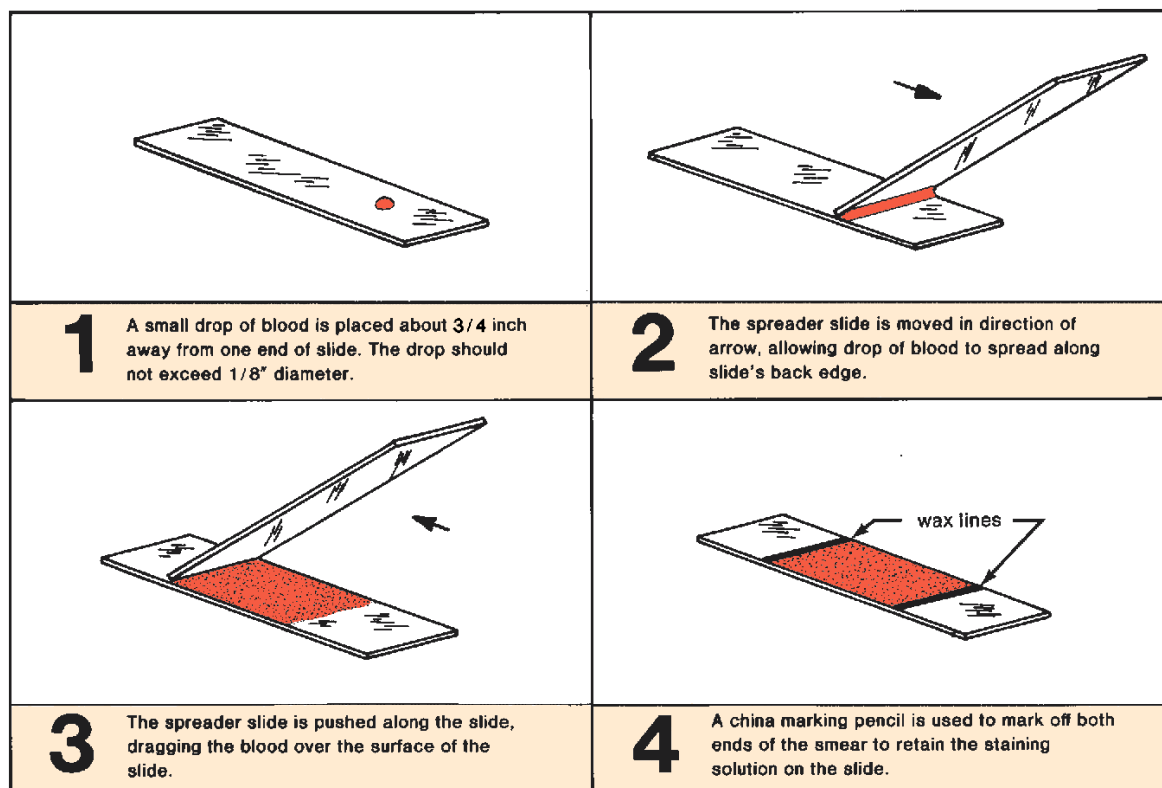


Figure 87.3 Smear preparation technique for making a stained blood slide

White Blood Cell Study: The Differential WBC Count • **Exercise 87**

first time, repeat the process on a fresh clean slide. To get a smear that will be the proper thickness, hold the spreading slide at an angle somewhat greater than 45°.

3. Draw a line on each side of the smear with a wax pencil to confine the stain that is to be added. (Note: This step is helpful for beginners, and usually omitted by professionals.)
4. Cover the film with Wright's stain, *counting the drops* as you add them. Stain for **4 minutes** and then add the same number of drops of distilled water to the stain and let stand for another **10 minutes**. Blow gently on the mixture every few minutes to keep the solutions mixed.
5. Gently wash off the slide under running water for 30 seconds and shake off the excess. Blot dry with bibulous paper.

PERFORMING THE CELL COUNT

Whether you are using a prepared slide or one that you have just stained, the procedure is essentially the same. Although the high-dry objective can be used for the count, the oil immersion lens is much better. Differentiation of some cells is difficult with high-dry optics. Proceed as follows:

1. Scan the slide with the low-power objective to find an area where cell distribution is best. A good area is one in which the cells are not jammed together or scattered too far apart.
2. Systematically scan the slide, following the pathway indicated below in figure 87.4. As each leukocyte is encountered, identify it, using figures 87.1 and 87.2 for reference.

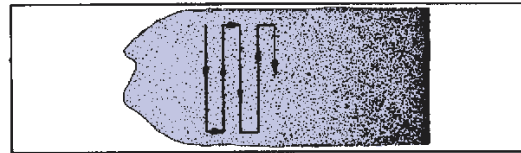


Figure 87.4 Path to follow when seeking cells

3. Tabulate your count on the Laboratory Report sheet according to the instructions there. It is best to remove the Lab Report sheet from the back of the manual for this identification and tabulation procedure.

LABORATORY REPORT

Complete the first portion of Laboratory Report 87–89.

88

Total WBC Count

Although the differential white blood cell count provides us with the relative percentages of leukocytes, it alone cannot reveal the true picture of the extent of an infection. For a more complete picture, one must also know the total number of WBCs per cubic millimeter of blood.

Although the number of leukocytes may vary with the time of day, exercise, and other factors, a range of 5,000 to 9,000 WBCs per cubic millimeter is considered normal. If an individual were to have an abnormally high neutrophil percentage, and a total count of, say, 17,000 WBCs, the presence of an infection of some sort would be highly probable.

To determine the number of leukocytes in a cubic millimeter of blood, one must dilute the blood and count the WBCs on a specialized slide called a **hemacytometer**. Figure 88.2 shows the sequence of steps in performing for this count.

Note in illustration 1 of figure 88.2 that blood is drawn up into a special pipette and then diluted in the pipette with a weak acid solution (illustration 2). After shaking the pipette (illustration 3) to mix the acid and blood, a small amount of diluted blood is allowed to flow under the cover glass of the hemacytometer (illustration 4). The count of white blood cells is then made with the low-power microscope objective.

PREPARATION OF HEMACYTOMETER

Working with your laboratory partner, assist each other to prepare a “charged” hemacytometer as follows:

PRECAUTIONS

Review the precautions that are stated in the previous exercise and use a mechanical suction device as shown in figure 88.1.

Materials:

hemacytometer and cover glass
WBC diluting pipette
WBC diluting fluid
mechanical hand counter
mechanical suction device
pipette cleaning solutions
cotton, alcohol, lancets, clean cloth



Figure 88.1 Using a mechanical suction device to draw up fluids into pipette

1. Wash the hemacytometer and cover glass with soap and water, rinse well, and dry with a clean cloth or Kimwipes.
2. Produce a free flow of blood, wipe away the first drop and draw the blood up into the diluting pipette to the 0.5 mark. See illustration 1 of figure 88.2. If the blood happens to go a little above the mark, the volume can be reduced to the mark by placing the pipette tip on a piece of blotting paper. If the blood goes substantially past the 0.5 mark, discharge the blood into a beaker of disinfectant, wash the pipette in the four cleansing solutions (illustration 6, figure 88.2), and start over. The ideal way is to draw up the blood exactly to the mark on the first attempt. To clean the pipette, rinse it first with acid, then water, alcohol, and finally with acetone.
3. As shown in illustration 2, figure 88.2, draw the WBC diluting fluid up into the pipette until it reaches the 11.0 mark.

Total WBC Count • Exercise 88

4. Place your thumb over the tip of the pipette, and place your third finger over the other end (illustration 3, figure 88.2).
5. Mix the blood and diluting fluid in the pipette for **2–3 minutes** by holding it as shown in illustration 3, figure 88.2. The pipette should be held *parallel to the tabletop* and moved through a 90° arc, with the wrist held rigidly.
6. Discharge one-third of the bulb fluid from the pipette by allowing it to drop onto a piece of paper toweling.
7. While holding the pipette as shown in illustration 1, figure 88.2, deposit a tiny drop on the polished surface of the counting chamber next to the edge of the cover glass. **Do not let the tip of the pipette touch the polished surface for more than an instant.** If it is left there too long, the chamber will overflow.

A properly filled chamber will have diluted blood filling only the space between the cover glass and counting chamber. No fluid should run down into the moat.

8. Charge the other side if the first side was overfilled.

PERFORMING THE COUNT

Place the hemacytometer on the microscope stage and bring the grid lines into focus under the **low-power** (10×) objective. Use the coarse adjustment knob and reduce the lighting somewhat to make both the cells and lines visible.

Locate one of the large “W” (white) areas shown in illustration 5, figure 88.2. One of these areas should fill up the entire field. Since the diluting fluid contains acid, all red blood cells have been destroyed; only the leukocytes will show up as very small dots.

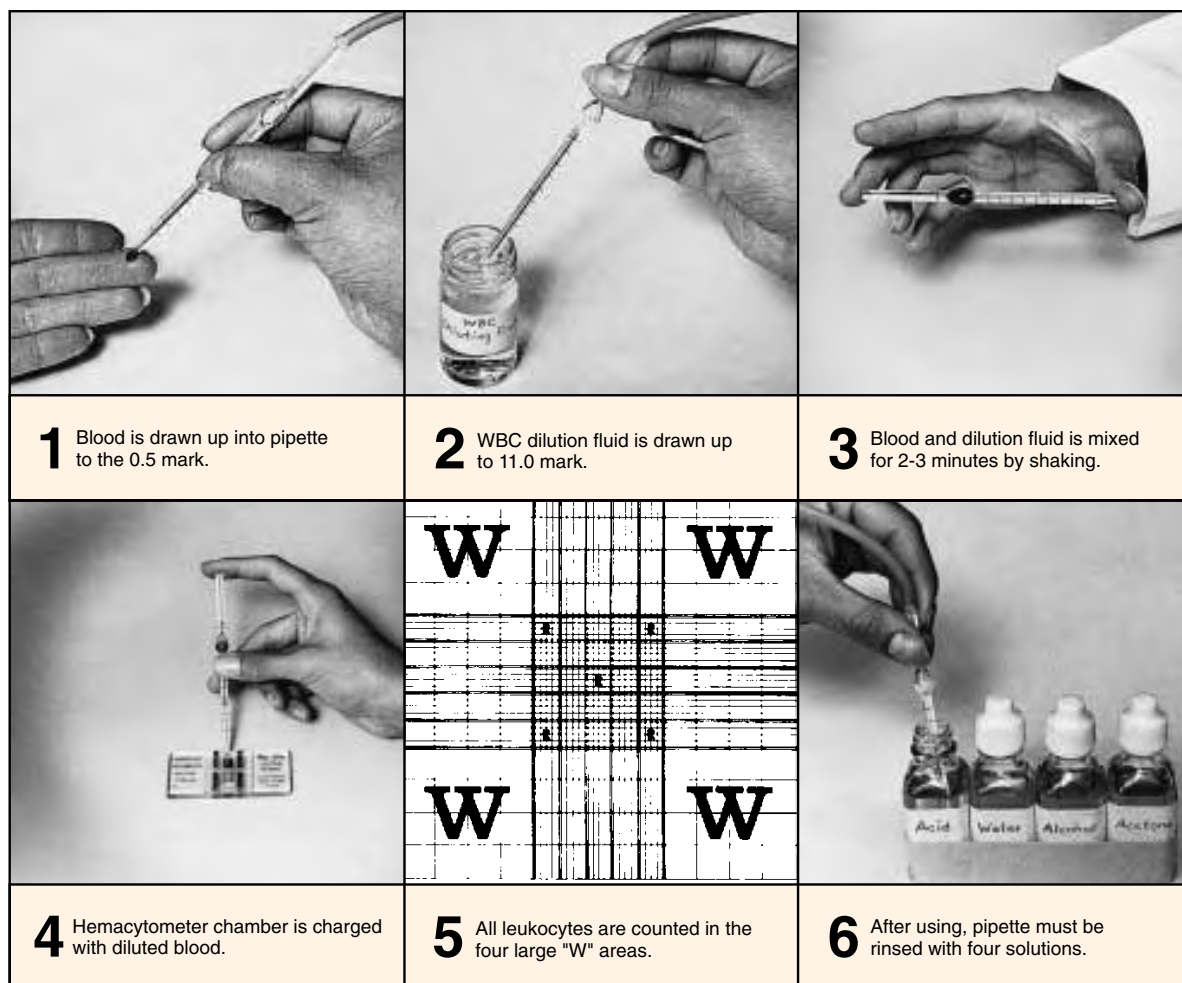


Figure 88.2 Procedure for charging a hemacytometer and doing a total white blood cell count

Total WBC Count • Exercise 88

Do the cells seem to be evenly distributed? If not, charge the other half of the counting chamber after further mixing. If the other chamber had been previously charged unsuccessfully by overflooding, wash off the hemacytometer and cover glass, shake the pipette for 2–3 minutes, and recharge it.

Count all the cells in the four “W” areas, using a mechanical hand counter. To avoid overcounting of cells at the boundaries, **count the cells that touch the lines on the left and top sides only**. Cells that touch the boundary lines on the right and bottom sides should be ignored. This applies to the boundaries of each entire “W” area.

Discharge the contents of the pipette and rinse it out by sequentially flushing with the following fluids:

acid, water, alcohol, and acetone. See illustration 6, figure 88.2.

CALCULATIONS

To determine the number of leukocytes per cubic millimeter, multiply the total number of cells counted in the four “W” areas by 50. The factor of 50 is the product of the volume correction factor and dilution factor, or

$$2.5 \times 20 = 50$$

LABORATORY REPORT

Answer the questions that pertain to this experiment on Laboratory Report 87–89.

89

Blood Grouping

Exercises 82 through 85 illustrate three uses of agglutination tests as related to (1) the identification of serological types, (2) species identification (*S. aureus*), and (3) disease identification (infectious mononucleosis and typhoid fever). The typing of blood is another example of a medical procedure that relies on this useful phenomenon.

The procedure for blood typing was developed by Karl Landsteiner around 1900. He is credited with having discovered that human blood types can be separated into four groups on the basis of two antigens that are present on the surface of red blood cells. These antigens are designated as A and B. The four groups (types) are A, B, AB, and O. The last group type O, which is characterized by the absence of A or B antigens, is the most common type in the United States (45% of the population). Type A is next in frequency, found in 39% of the population. The incidences of types B and AB are 12% and 4%, respectively.

Blood typing is performed with antisera containing high titers of anti-A and anti-B antibodies. The test may be performed by either slide or tube methods. In both instances, a drop of each kind of antiserum is added to separate samples of saline suspension of red blood cells. Figure 89.1 illustrates the slide technique. If agglutination occurs only in the suspension to which the anti-A serum was added, the blood is type A. If agglutination occurs only in the anti-B mixture, the blood is type B. Agglutination in both samples indicates that the blood is type AB. The absence of agglutination indicates that the blood is type O.

Between 1900 and 1940 a great deal of research was done to uncover the presence of other antigens in human red blood cells. Finally, in 1940, Landsteiner and Wiener reported that rabbit sera containing antibodies against the red blood cells of the rhesus monkey would agglutinate the red blood cells of 5% of white humans. This antigen in humans, which was first designated as the **Rh factor** (in due respect to the rhesus monkey), was later found to exist as six antigens: C, c, D, d, E, and e. Of these six antigens, the D factor is responsible for the Rh-positive condition and is found in 85% of caucasians, 94% of blacks, and 99% of orientals.

Typing blood for the Rh factor can also be performed by both tube and slide methods, but there are certain differences in the two techniques. First of all, the antibodies in the typing sera are of the incomplete albumin variety, which *will not agglutinate human red cells when they are diluted with saline*. Therefore, it is necessary to use whole blood or dilute the cells with plasma. Another difference is that the test *must be performed at higher temperatures: 37° C for tube test, 45° C for the slide test*.

In this exercise, two separate slide methods are presented for typing blood. If only the Landsteiner ABO groups are to be determined, the first method may be preferable. If Rh typing is to be included, the second method, which utilizes a slide warmer, will be followed. The availability of materials will determine which method is to be used.

PRECAUTIONS

Review the precautionary comments highlighted on page 290.

ABO BLOOD TYPING

Materials:

small vial (10 mm dia × 50 mm long)
disposable lancets (B-D Microlance,
Serasharp, etc.)
70% alcohol and cotton
china marking pencil
microscope slides
typing sera (anti-A and anti-B)
applicators or toothpicks
saline solution (0.85% NaCl)
1 ml pipettes
disposable latex gloves

1. Mark a slide down the middle with a marking pencil, dividing the slide into two halves as shown in figure 89.1. Write “anti-A” on the left side and “anti-B” on the right side.
2. Pipette 1 ml of saline solution into a small vial or test tube.
3. Scrub the middle finger with a piece of cotton saturated with 70% alcohol and pierce it with a

Blood Grouping • Exercise 89

sterile disposable lancet. Allow 2 or 3 drops of blood to mix with the saline by holding the finger over the end of the vial and washing it with the saline by inverting the tube several times.

4. Place a drop of this red cell suspension on each side of the slide.
5. Add a drop of anti-A serum to the left side of the slide and a drop of anti-B serum to the right side.

Do not contaminate the tips of the serum pipettes with the material on the slide.

6. After mixing each side of the slide with separate applicators or toothpicks, look for agglutination. The slide should be held about 6" above an illuminated white background and rocked gently for 2 or 3 minutes. Record your results on the Laboratory Report as of 3 minutes.

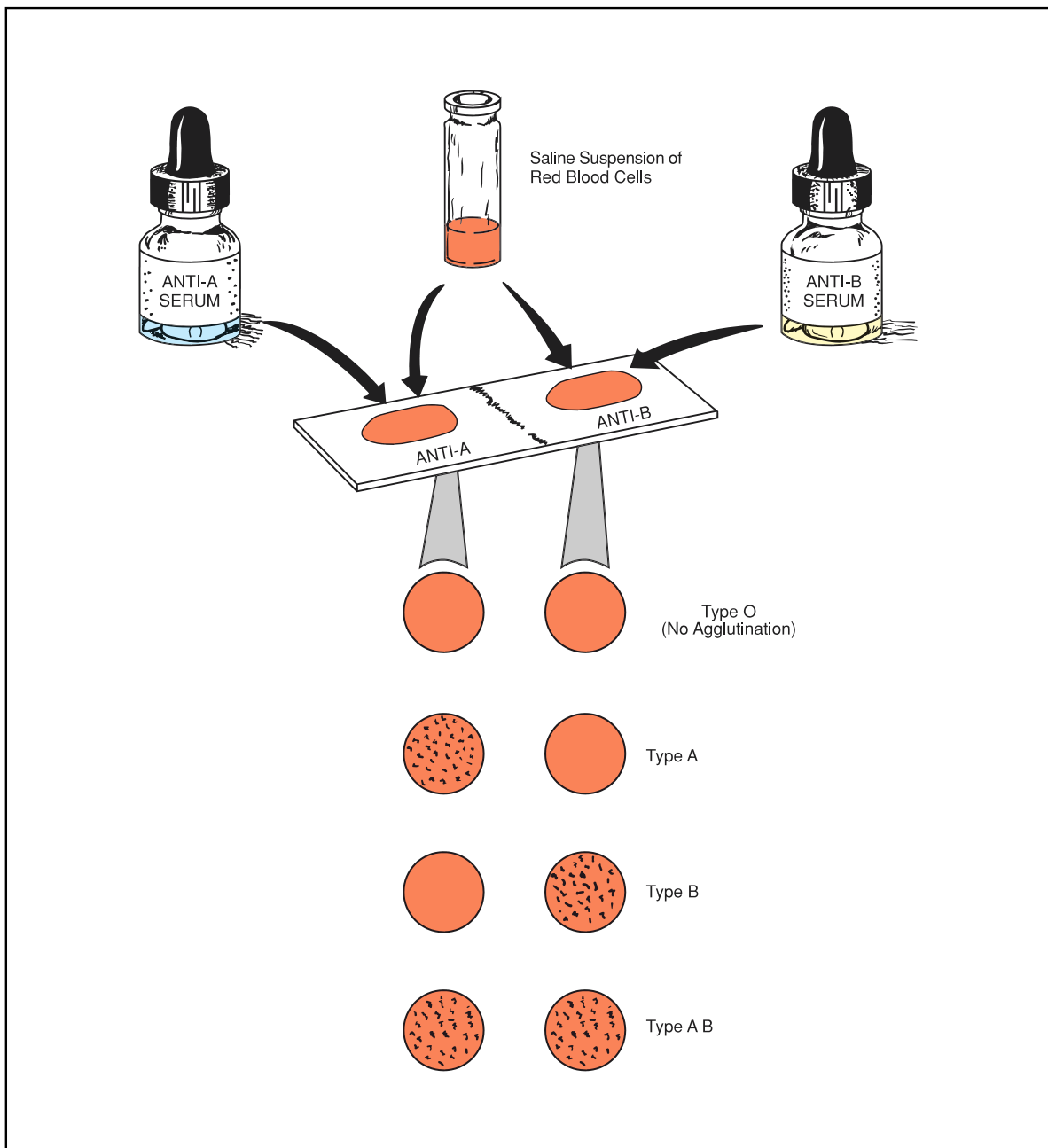


Figure 89.1 Typing of ABO blood groups

Exercise 89 • Blood Grouping**COMBINED ABO AND RH TYPING**

As stated, Rh typing must be performed with heat on blood that has not been diluted with saline. A warming box such as the one in figure 89.2 is essential in this procedure. In performing this test, two factors are of considerable importance: first, only a small amount of blood must be used (a drop of about 3 mm diameter on the slide) and, second, proper agitation must be executed. The agglutination that occurs in this antibody-antigen reaction results in finer clumps; therefore, closer examination is essential. If the agitation is not properly performed, agglutination may not be as apparent as it should be.

In this combined method we will use whole blood for the ABO typing as well as for the Rh typing. Although this method works satisfactorily as a classroom demonstration for the ABO groups, it is *not as reliable* as the previous method in which saline and room temperature are used. *This method is not recommended for clinical situations.*

Materials:

slide warming box with a special marked slide
anti-A, anti-B, and anti-D typing sera
applicators or toothpicks

70% alcohol and cotton
disposable sterile lancets

1. Scrub the middle finger with a piece of cotton saturated with 70% alcohol and pierce it with a sterile disposable lancet. Place a small drop of blood in each of three squares on the marked slides on the warming box.
2. To get the proper proportion of serum to blood, do not use a drop larger than 3 mm diameter on the slide.
3. Add a drop of anti-D serum to the blood in the anti-D square, mix with a toothpick, and note the time. **Only 2 minutes should be allowed for agglutination.**
4. Add a drop of anti-B serum to the anti-B square and a drop of anti-A serum to the anti-A square. Mix the sera and blood in both squares with *separate* fresh toothpicks.
5. Agitate the mixtures on the slide by slowly rocking the box back and forth on its pivot. At the end of 2 minutes, examine the anti-D square carefully for agglutination. If no agglutination is apparent, consider the blood to be Rh-negative. By this time the ABO type can also be determined.
5. Record your results on Laboratory Report 87–89.

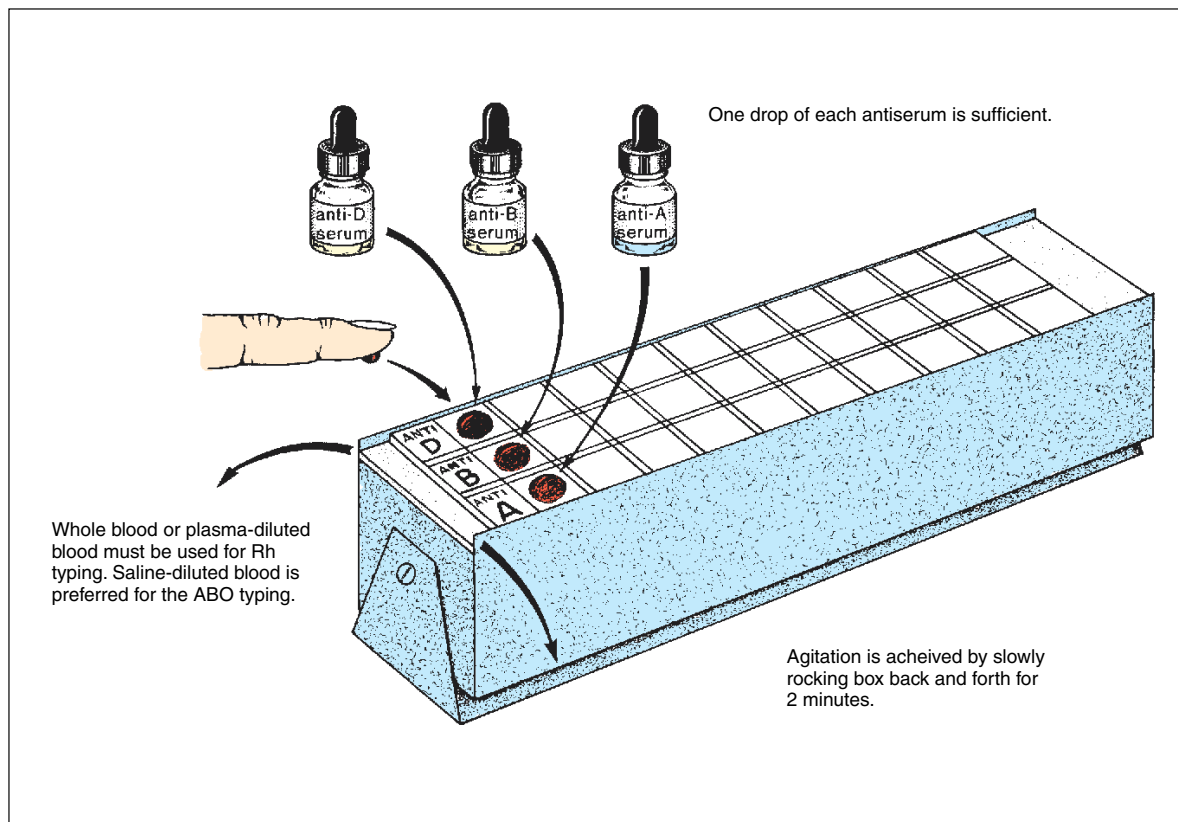


Figure 89.2 Blood typing with warming box

The Snyder Caries Susceptibility Test

90

The degradation of enamel and dentin in the formation of tooth decay (dental caries) occurs as a result of the production of lactic acid by bacteria (*Streptococcus mutans* and others) in the presence of high levels of sucrose. Of the various methods that have been devised to determine one's susceptibility to tooth decay, M. L. Snyder's caries susceptibility test is a relatively simple test that has been shown to have a fairly high reliability correlation.

This method relies on the rapidity of organisms in saliva to lower the pH in a medium that contains 2% dextrose (Snyder test agar). Since decalcification of enamel begins at a pH of 5.5, and progresses rapidly as the pH is lowered to 4.4 and less, the demonstration of pH lowering becomes evidence of susceptibility to caries.

To indicate the presence of acid production in the medium, the indicator bromocresol green is incorporated in it. This indicator is green at pH 4.8 and becomes yellow at pH 4.4, remaining yellow below 4.4.

Figure 90.1 illustrates the procedure that is used in the Snyder caries susceptibility test. Note that 0.2 ml of saliva is added to a tube of liquefied Snyder test agar (50° C) and mixed well by rotating the tube between the palms of both hands. After the medium has

solidified, the tube is incubated at 37° C for a period of 24–72 hours. If the medium turns yellow in 24–48 hours, the individual is said to be susceptible to caries.

Although we will be performing this test only once, it should be noted that test reliability is enhanced by performing the test on three consecutive days at the same time each day. If the test is performed right after toothbrushing, it is not as reliable as if 2 or 3 hours have elapsed after brushing. Proceed as follows to perform this test:

Materials:

- 1 tube of Snyder test agar (5 ml in 15 mm dia tube)
- 1 30 ml sterile beaker
- 1 piece of paraffin (1/4" × 1/4" × 1/8")
- 1 ml pipette
- 1 gummed label

1. Liquefy a tube of Snyder test agar and cool it to 50° C.
2. After allowing a piece of paraffin to soften under the tongue for a few minutes, start chewing it. Chew it for **3 minutes**, moving it from one side of the

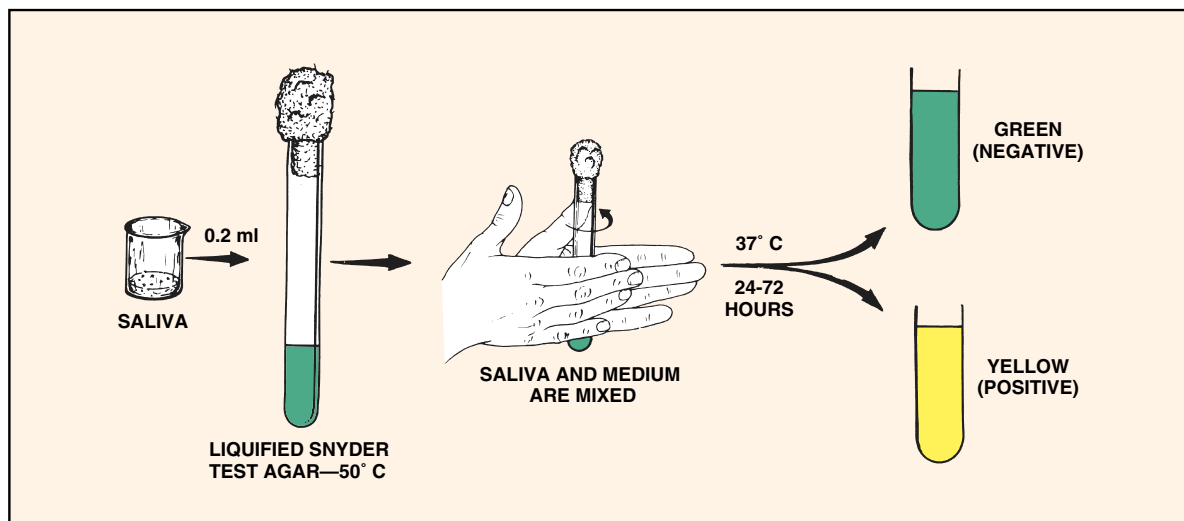


Figure 90.1 Snyder caries susceptibility test

Exercise 90 • The Snyder Caries Susceptibility Test

- mouth to the other. *Do not swallow the saliva.* As it accumulates, deposit it in the small sterile beaker.
- Vigorously shake the sample in the beaker from side to side for 30 seconds to disperse the organisms.
 - With a 1 ml pipette transfer 0.2 ml of saliva to the tube of agar. Do not allow the pipette to touch the side of the tube or agar.
 - Before the medium solidifies, mix the contents of the tube by rotating the tube vigorously between the palms of the hands.
 - Write your name on a gummed label and attach it to the tube.
 - Incubate the tube at 37° C. Examine the tube every 24 hours to see if the bromcresol green indicator has changed to yellow. If it has, the test is positive. The degree of caries susceptibility is determined from the table below.
 - Record your results on the Laboratory Report.

Caries Susceptibility	Medium turns yellow in:		
	24 hours	48 hours	72 hours
<i>Marked</i>	<i>Positive</i>		
<i>Moderate</i>	<i>Negative</i>	<i>Positive</i>	
<i>Slight</i>	<i>Negative</i>	<i>Negative</i>	<i>Positive</i>
<i>Negative</i>	<i>Negative</i>	<i>Negative</i>	<i>Negative</i>

LABORATORY REPORT

1, 2

Student: _____

Desk No.: _____ Section: _____

Ex. 1 Brightfield Microscopy

A. Completion Questions

Record the answers to the following questions in the column at the right.

1. List three fluids that may be used for cleaning lenses.
2. How can one greatly increase the bulb life on a microscope lamp if voltage is variable?
3. What characteristic of a microscope enables one to switch from one objective to another without altering the focus?
4. What effect (*increase* or *decrease*) does closing the diaphragm have on the following?
 - a. Image brightness
 - b. Image contrast
 - c. Resolution
5. In general, at what position should the condenser be kept?
6. Express the maximum resolution of the compound microscope in terms of micrometers (μm).
7. If you are getting $225\times$ magnification with a $45\times$ high-dry objective, what is the power of the eyepiece?
8. What is the magnification of objects observed through a $100\times$ oil immersion objective with a $7.5\times$ eyepiece?
9. Immersion oil must have the same refractive index as _____ to be of any value.
10. Substage filters should be of a _____ color to get the maximum resolution of the optical system.

B. True-False

Record these statements as *true* or *false* in the answer column.

1. Eyepieces are of such simple construction that almost anyone can safely disassemble them for cleaning.
2. Lenses can be safely cleaned with almost any kind of tissue or cloth.
3. When swinging the oil immersion objective into position after using high-dry, one should always increase the distance between the lens and slide to prevent damaging the oil immersion lens.
4. Instead of starting first with the oil immersion lens, it is best to use one of the lower magnifications first, and then swing the oil immersion into position.
5. The $45\times$ and $100\times$ objectives have shorter working distances than the $10\times$ objective.

Answers

Completion

- 1.a. _____
- b. _____
- c. _____
2. _____
3. _____
- 4.a. _____
- b. _____
- c. _____
5. _____
6. _____
7. _____
8. _____
9. _____
10. _____

True-False

1. _____
2. _____
3. _____
4. _____
5. _____

Brightfield Microscopy

C. Multiple Choice

Select the best answer for the following statements.

1. The resolution of a microscope is increased by
 1. using blue light.
 2. stopping down the diaphragm.
 3. lowering the condenser.
 4. raising the condenser to its highest point.
 5. Both 1 and 4 are correct.
2. The magnification of an object seen through the 10× objective with a 10× ocular is
 1. ten times.
 2. twenty times.
 3. 1000 times.
 4. None of the above are correct.
3. The most commonly used ocular is
 1. 5×.
 2. 10×.
 3. 15×.
 4. 20×.
4. Microscope lenses may be cleaned with
 1. lens tissue.
 2. a soft linen handkerchief.
 3. an air syringe.
 4. Both 1 and 3 are correct.
 5. 1, 2, and 3 are correct.
5. When changing from low power to high power, it is generally necessary to
 1. lower the condenser.
 2. open the diaphragm.
 3. close the diaphragm.
 4. Both 1 and 2 are correct.
 5. Both 1 and 3 are correct.

Answers

Multiple Choice

1. _____

2. _____

3. _____

4. _____

5. _____

Ex. 2 Darkfield Microscopy

A. Questions

1. What characteristic of living bacteria makes them easier to see with a darkfield condenser than with a regular brightfield condenser?

2. If a darkfield condenser causes all light rays to bypass the objective, where does the light come from that makes an object visible in a dark field?

3. What advantage does a cardioid condenser have over a star diaphragm?

